

A COMPUTATIONAL APPROACH TO NEWS PRODUCTION

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ABSTRACT

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News organizations shape public understanding of the world by deciding what is newsworthy for their audience and by how they report on it. This dissertation develops a computational framework for studying these choices in news production at scale. Using large language models to annotate hundreds of thousands of articles from major U.S. news publishers across 2024 and 2025, I measure selection and framing at multiple levels, from broad topics and specific events to the politicians and experts quoted within these stories.

First, I introduce the Media Bias Detector, a large-scale system for collecting and labeling news articles on a variety of dimensions, showing how this data can be used to identify variations in coverage between and within publishers. I then use this framework to study political representation in news quotations during the 2024 presidential election, showing that Republicans consistently received more quotations than Democrats across publishers, topics, and time. Next, I examine how academic scholars are quoted in news coverage and find that expert sourcing is highly concentrated among elite universities and disproportionately platforms men on nearly all news topics. Finally, I develop an event clustering framework to disentangle differences in selection and framing between publishers to understand how their coverage diverges at the granular level of specific news events.

Together, these studies show that variation across news organizations is multidimensional and cannot be reduced to a single measure of bias. Instead, it emerges through a sequence of choices about what to cover, whom to platform, and how to present events. In doing so, this dissertation demonstrates how computational methods can help measure subtle editorial differences in news coverage at the scale of the modern news ecosystem.

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CHAPTER 1

INTRODUCTION

The news is not simply a record of everything that happens in the world. Producing it requires a sequence of choices to decide which stories deserve attention, which facts to emphasize and voices to include, and how these stories should be framed. These choices shape what information reaches the public and how they understand the world through it. Decades of research on agenda setting and framing have therefore shown that media organizations can influence what people think about and how they think about it (McCombs and Shaw, 1972; Entman, 1993).

Even in an era in which U.S. adults often get their news from social media and mobile apps, media organizations remain important. News websites or apps are preferred by 23% of adults versus 18% who favor social media (St. Aubin and Liedke, 2024), and most links that are shared on social media tend to originate from mainstream media sources (Wu et al., 2011). Given its reach and influence, the reliability of information produced by news organizations continues to be an important concern. In contrast to social media, mainstream media organizations typically adhere to professional journalistic standards of fact-checking and accuracy, and hence tend not to be the primary sources of what is often referred to as “misinformation” or “fake news.” Nonetheless, even factually accurate reporting can produce systematically different representations of reality through the choices involved in news production. These choices can broadly be understood as two forms of bias, selection and framing, that have long been studied by communication and media scholars (Stroud, 2008; Baum and Groeling, 2008; Chong and Druckman, 2007).

First, “selection bias” holds that media organizations actively choose which topics, events, issues, and perspectives to cover as well as how prominently or extensively to cover them (Bourgeois et al., 2018). For example, although Hillary Clinton’s use of a private email server while serving as U.S. Secretary of State was arguably a matter of public interest that ought to have been covered by the media in the lead-up to the 2016 presidential election, the appearance of ten front-page articles in The New York Times in the eight days leading up to the election represented an editorial choice to

focus on this one issue over all others (Watts and Rothschild, 2017). More broadly, the attention that media organizations pay to topics such as crime or health risks has been shown to be unrelated to, or even inversely related to, the corresponding rates expressed by objective administrative data (Sheley and Ashkins, 1981; Rothschild et al., 2023; Isch, 2025), suggesting again that the extent of coverage of these issues represents a choice by news organizations themselves rather than a reflection of external reality.

In both cases, the information contained in the stories that are published might be entirely factually accurate and may even be presented in a neutral, objective manner. Nonetheless, as theories of “agenda setting” have long held, selection bias can still distort public opinion by driving attention to some issues at the cost of others that might be equally or more important (McCombs and Shaw, 1972; Dearing and Rogers, 1996). Moreover, because the counterfactual issues are, by definition, not being paid attention to, the distorting influence of selection bias can be hard to detect. Selection also occurs within stories themselves. Journalists decide which politicians, experts, institutions, and other sources are given voice, thereby deciding whose perspectives become visible and authoritative in public discourse. Studying news selection therefore requires examining not only which stories are published, but also who is selected to speak within them.

Second, “framing bias” contends that, conditional on having selected a topic to cover, journalists and editors can exploit additional degrees of freedom in how they present the constituent facts and issues, again without necessarily saying anything that would fail a fact check (Morstatter et al., 2018). Framing bias can involve choices about the inclusion or omission of specific details, perspectives or narratives, the tone or language used, the context provided, and the inclusion or omission of background information (Entman, 1993; Groseclose and Milyo, 2005; Rothschild et al., 2023). Such decisions can lead a reader to reach a desired conclusion that, while not necessarily false, is only one of potentially many equally true—and potentially qualitatively different—conclusions.

For example, news organizations are often accused of covering elections through the frame of sporting contests (also known as “horse-race coverage”), which emphasizes the popularity of candidates, rather than through the frame of policy differences and other issues that are likely to impact voters post-

election (Watts and Rothschild, 2017; Farnsworth et al., 2007). As with selection bias, framing bias can be subtle and hard to detect from the available data. For example, “false equivalence” or “bothsidesism” is a type of framing bias in which manifestly unequal views are presented in a balanced and evenhanded manner (Haas, 2007; Shapiro, 2016). In such cases, the bias can be difficult to identify because it is actively masquerading as neutrality.

Selection and framing are conceptually distinct but closely intertwined. Two publishers can construct different news agendas through a variety of combinations of both of these tools. They can paint opposing pictures for their readers by choosing different events to cover, even if they do so seemingly neutrally. They may also choose to cover the same event but describe it using different language and quote different sources. Distinguishing between these mechanisms is important because single, static publisher-level labels are inadequate for accurately capturing how publishers differ from one another in their news coverage.

As the media landscape continues to expand and simultaneously fragment, these potential sources of media bias and their attendant consequences for public understanding of controversial issues make the development of scalable tools to measure bias more critical than ever. However, the methods that have traditionally been used to study bias were developed when news cycles were slower and media consumption was more centralized, allowing researchers to manually analyze coverage patterns across a small set of publications. For example, traditional content analysis relies on manual annotation (Budak et al., 2016; Lim et al., 2020; Mitchell et al., 2017), which has difficulty scaling even to thousands of articles. Meanwhile, computational techniques such as keyword frequency analysis and topic modeling (Gentzkow and Shapiro, 2010; Grimmer and Stewart, 2013; DiMaggio et al., 2013) scale much better, but struggle to capture the contextual nuances necessary for identifying complex forms of bias (Iyyer et al., 2014). Finally, commercial bias-rating tools such as AllSides, Ad Fontes, and Media Bias/FactCheck measure bias by assigning static, publisher-level ratings along a one-dimensional left-right spectrum (AllSides, 2025; Media Bias/Fact Check, 2025; Ad Fontes Media, 2025). Although useful, these classifications necessarily overlook within-publisher variation across different topics or events or how these patterns evolve over time, all of which are

questions of potential interest to researchers.

The central measurement challenge is therefore extracting the right data points from news articles that capture the editorial choices that produced them. Measuring selection requires identifying what different articles are about and determining when multiple publishers are covering the same underlying event. Likewise, measuring framing requires contextual judgments about the political lean of a news article, the tone in which it frames events, and the perspectives included in it. Additionally, studying the allocation of voice requires identifying quoted individuals, their affiliations, and the context in which they are cited. These kinds of subtle differences are difficult to model using simple keyword-based approaches, but at the same time are prohibitively expensive to manually annotate at the scale and speed of news production in the present day.

Recent advancements in large language models (LLMs) present new opportunities to move beyond these limitations (Vaswani et al., 2017; Radford et al., 2019). Unlike traditional natural language processing methods, LLMs can capture nuanced semantic relationships within text (Brown et al., 2020; OpenAI, 2023). By leveraging vast training corpora that encompass a wide variety of published content, these models can identify subtle variations in tone, emphasis, and narrative structure that shape media bias (Kuo et al., 2025). Although the resulting ratings remain imperfect and require careful human validation, especially as news cycles shift (Feng et al., 2023; Horych et al., 2025), carefully administered LLMs have been shown to produce labels that compare favorably with human annotations even for complex and subtle tasks (Gilardi et al., 2023; Ziems et al., 2024). Equally important, although the cost of using commercial LLMs still presents some scaling limitations, they make it feasible to rate hundreds or thousands of articles per day, enabling article-level measurement across many publishers and long periods of time.

In this dissertation, I develop a framework for studying these choices in news production at scale. Using LLMs to annotate hundreds of thousands of news articles published across 2024 and 2025, I examine media coverage at multiple levels, from broad topics and individual events to the politicians and experts whose voices are used to frame these stories. Across these analyses, I argue that the differences between news organizations cannot be reduced to a single measurement of media bias.

Instead, the differences emerge through a sequence of editorial decisions about what to cover, whom to platform, and how to present the same underlying events.

The chapters that follow address four related questions in this vein. First, how can we reliably measure complex features of news coverage at scale? Second, whose voices do news organizations amplify in political coverage and when seeking expert opinions? And finally, how much do publishers differ in the specific events they cover, and how differently do they frame common events? Together, these questions move from constructing the measurement infrastructure needed to study news production to then using it to study selection and framing at progressively more granular levels of analysis.

In Chapter 2, I introduce the Media Bias Detector (MBD)¹, a large-scale system for collecting and annotating news articles that forms the foundation for the rest of my research. The MBD scrapes the top-ranked news articles from 21 prominent news sources multiple times every day and uses GPT-4o and GPT-4o-mini to extract a variety of labels such as news topic and subtopic, article lean, tone, and type, and sentence tone, type, and focus. I conduct a manual annotation exercise on each dimension of the data to show that the MBD’s labels agree strongly with human experts and that the variance is within that of inter-annotator disagreement. Using a labeled dataset of over 140,000 news articles published during the critical 2024 election year, I then show how article-level labels can reveal variation in news coverage that is obscured by publisher-level labels.

I build on this framework in Chapter 3 to study which politicians’ voices the media amplifies when covering political events². Using quotations as a measure of media representation, I quantify disparities in attention given to Republican and Democratic politicians during the 2024 U.S. presidential election. By extracting over 110,000 quotes attributed to more than 600 politicians from over 30,000 articles, I show that Republicans received more quotes in the media than Democrats. This pattern persisted across publishers, topics, and time, leading to a consistent asymmetry that platformed

¹**Haider, S.**, Tohidi, A., Wang, J. S., Dörr, T., Rothschild, D. M., Callison-Burch, C., & Watts, D. J. (2026). “The Media Bias Detector: A Framework for Annotating and Analyzing the News at Scale.” *Science Advances* (in press).

²**Haider, S.**, Callison-Burch, C., & Watts, D. J. (2026). “Measuring political representation in news quotations during the 2024 U.S. presidential election.” *Under review at PNAS Nexus*.

Republican voices over Democratic ones. In addition to studying how often these politicians are quoted, I also analyze the nature of their quoted speech. By using LLMs to label the emotions expressed in quotations, I show that Republicans’ speech was marked by heightened levels of anger and fear, while Democrats expressed more hope, calm, unity, compassion, and motivation.

In Chapter 4, I extend the idea of voice allocation from political representation to expert authority, analyzing which scholars and institutions journalists rely on to interpret news events. By extending the aforementioned quote extraction pipeline to cover all sources in news articles, I study which individuals and institutions are platformed as experts. Among nearly 60,000 quotes attributed to individuals associated with higher education institutions, I find that a small set of elite universities account for a disproportionate share of academic voices in the news. Furthermore, I find that, while all publishers seek expert sources at the same prestigious institutions, the specific individuals within those institutions that are quoted most frequently vary widely between publishers. Finally, I document pervasive gender bias across publishers, universities, and news topics, showing that the allocation of expert authority is highly unequal even before considering what views they ultimately express.

Finally, in Chapter 5, I return to study story selection, this time at the level of individual events. I present a multi-stage event clustering approach that groups news articles according to the specific event they cover on each day. I identify more than 9,000 events across 50,000 articles in 2025 and compare which publishers cover or do not cover the same events. These event-level representations make it possible to separate the two dimensions of selection and framing decisions made by publishers which influence the stories they choose to cover and how they present them to their readers. I show that publishers vary substantially along both dimensions and that differences in framing are particularly pronounced for certain politically charged topics and events.

Together, these studies provide a computational analysis of news production as a sequence of editorial choices. They show that differences in news organizations arise gradually, from which topics and events they choose to cover, whose voices they include to accompany their reporting, and how they interpret the facts differently from others. More broadly, I demonstrate how modern computational

methods like LLMs can make these patterns measurable at the scale of the modern news ecosystem.

CHAPTER 2

A FRAMEWORK FOR ANNOTATING AND ANALYZING THE NEWS

2.1. Introduction

In this chapter, we introduce the MEDIA BIAS DETECTOR, a scalable, near-real-time system that measures elements of framing and selection bias at multiple levels, from individual articles to publisher-wide trends. The MEDIA BIAS DETECTOR continuously collects clean and comprehensive data from a set of major news publishers across the political spectrum. We also describe a unique dataset derived from the MEDIA BIAS DETECTOR and illustrate how these data can be used to analyze media coverage during a critical election year. Here, we focus on data collected during 2024 for the ten publishers included in the initial launch of the MEDIA BIAS DETECTOR; however, we note that our set of publishers has grown (currently to 21) as has the time period of our data, and we anticipate both will continue to grow over time. Finally, we make the MEDIA BIAS DETECTOR methodology and data publicly available to support academic research on media bias, and also provide an interactive online tool³ for users to explore and analyze coverage patterns dynamically (Wang et al., 2025). Figure 2.1 illustrates the core design principles of our framework and shows screenshots of our dashboard.

At the core of our methodology is a multi-stage pipeline that systematically tracks and analyzes news articles. While television and cable news also play substantial roles in shaping public discourse, we currently focus on text-based digital news where the unit of analysis is each individual news article. We capture homepage snapshots from major publishers at regular intervals, recording article content, as well as metadata on homepage placement and prominence to measure selection bias. Using LLMs, we classify articles by topic, political lean, tone, and news type. Additionally, we break our analysis down to the sentence level, capturing nuanced aspects of framing bias such as sentence type and tone.

While existing media bias ratings organizations judge media outlets as a whole, placing them on a

³<https://mediabiasdetector.seas.upenn.edu>

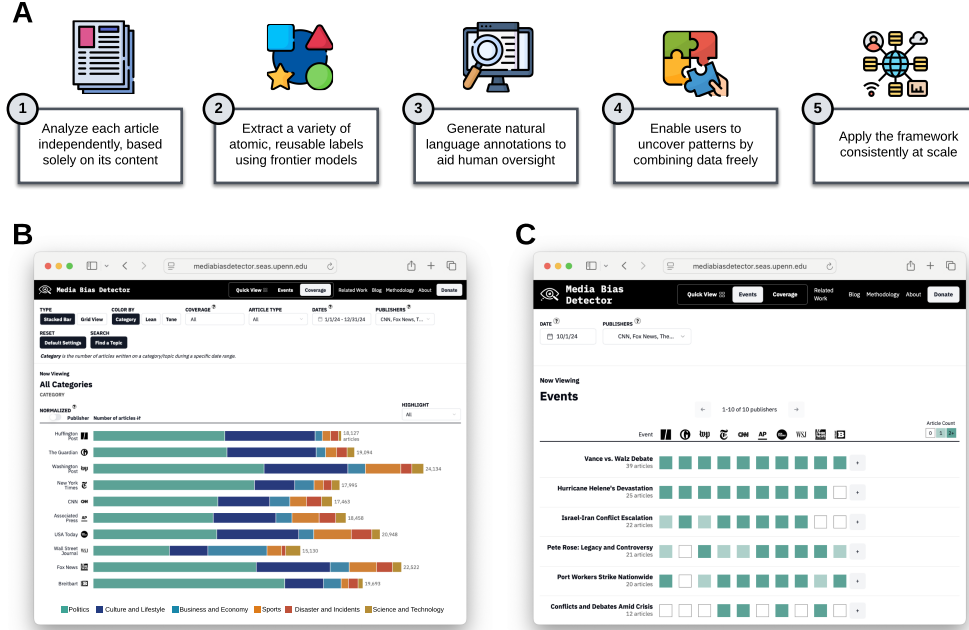


Figure 2.1: **The MEDIA BIAS DETECTOR design philosophy.** (A) The core guiding principles behind the design of our framework. We analyze each article individually to allow direct, data-driven comparisons between publishers regarding their selection and framing of news topics. By extracting simple labels using state-of-the-art LLMs, we aim to ensure a high level of reliability in our dataset. As we have a variety of data points for each article, our framework allows users to freely combine them to gain new insights into the media. (B) The coverage page gives users a long-term view of what topics the media chooses to cover. Users can set filters to narrow the time frame and click through on each news topic to compare the distribution of subtopics within it. (C) The event page, on the other hand, offers a more immediate view of the daily news cycle and shows the top news events of the past day along with the amount of attention given to them by each publisher.

discrete left-right spectrum based on the entirety of their reporting as well as any political alignment their pundits or owners may have expressed, our approach evaluates each article independently and without knowledge of the publisher or its history. To analyze articles, we use state-of-the-art LLMs like GPT-4o to extract a large set of simple labels. By focusing on labels that are tractable for current models (such as the topic or tone of a news article) and by validating them with human raters (see below), we ensure a high degree of confidence in the outputs of our framework and the resulting dataset. In addition to extracting numerical labels, we also elicit natural language analyses for each article from the model to summarize its content and aid human oversight. Finally, we allow users to freely combine our data in various ways so they can use it to answer a variety of research

questions about the media. This flexibility is facilitated by the variety of labels we extract for each news article and the number of news articles we have collected in our dataset.

To ensure accuracy and transparency, we incorporate systematic human oversight at multiple stages. Trained annotators regularly validate a random sample of the model’s outputs every day, correcting biases and ensuring consistency in the LLM’s classification. This human-in-the-loop approach balances scalability with quality control, addressing key limitations of both fully manual and fully automated methods. We also conducted an in-depth data validation exercise to benchmark the effectiveness of LLMs at complex tasks like identifying political lean and tone from news articles. Furthermore, our methodology is transparent and publicly documented, with detailed information about our LLM prompting strategies and validation procedures available to researchers who use our system.

We argue that the MEDIA BIAS DETECTOR enables new avenues of research into media bias that were previously difficult or impossible to study beyond the scale of a handful of articles. As we describe in Section 2.3, our data reveal interesting patterns in contemporary news coverage and offer relatively large-scale, quantitative evidence to earlier qualitative claims of bias. Additionally, the scale and granularity of our data enable deeper investigation of these research questions, revealing nuanced patterns and generating insights that extend beyond prior research findings. For example, in coverage of political topics we find a systematic tendency for media outlets to focus more on criticizing their ideological opposition than supporting aligned positions, extending previous research suggesting that partisan media effects are primarily driven by negative coverage of opposition candidates (Smith and Searles, 2014). In another example, we find that headlines often present issues in a manner that leans more Republican than the associated article text, complementing recent research showing that biased headlines can substantially impact readers’ ability to make inferences and recall facts from news stories (Ecker et al., 2014). In a third example, our data replicate and extend prior claims of negativity bias in the media, which can in turn shape how the public perceives the reality of the world.

The remainder of the chapter is structured as follows: First, we provide a detailed explanation of

our system for real-time analysis of media bias, which combines the processing power of LLMs with systematic human oversight. In this section, we also present evidence validating our methodology through extensive testing, demonstrating high accuracy in topic classification, and labeling article lean and tone using LLMs. Second, we use our curated dataset to explore a range of questions related to media practices and biases, showcasing the potential use cases of our dataset for gaining deeper insights into media. Finally, we discuss the practical implications of our work and outline how other researchers can build upon our transparent methodology to advance the study of media bias.

2.2. Methodology

2.2.1. System Architecture

As already noted, our multi-stage data pipeline is built upon large language models, which we utilize for both their capacity to produce deep contextual representations of text documents and their instruction-tuned ability to generate labels without requiring extensive training data.

Prior work has found that LLMs perform well in generative tasks such as summarization (Liu et al., 2024; Zhang et al., 2024), as well as discriminative tasks such as sentiment analysis and topic classification (Rathje et al., 2024; Pham et al., 2024), where these tasks can even be performed in a zero-shot setting (Chae and Davidson, 2023). These findings suggest that LLMs can be well-suited for complex content analysis tasks like media bias detection, where identifying patterns in context, tone, and subtle language cues is essential. Critically, LLMs also scale effectively to large volumes of content that would be unmanageable with manual annotation alone (Ziems et al., 2024). As a result, LLMs can competently and efficiently extract detailed information such as sentence-level labels and classifications by topic, subtopic, article type, tone, and political lean at the scale of many publishers in near-real-time (Kiss et al., 2023).

The MEDIA BIAS DETECTOR uses LLMs in two ways: first, for labeling news articles and sentences using zero-shot prompting; and second, for generating contextual embeddings to cluster articles about major events and highlight important facts within them. Figure 2.2 provides a high-level

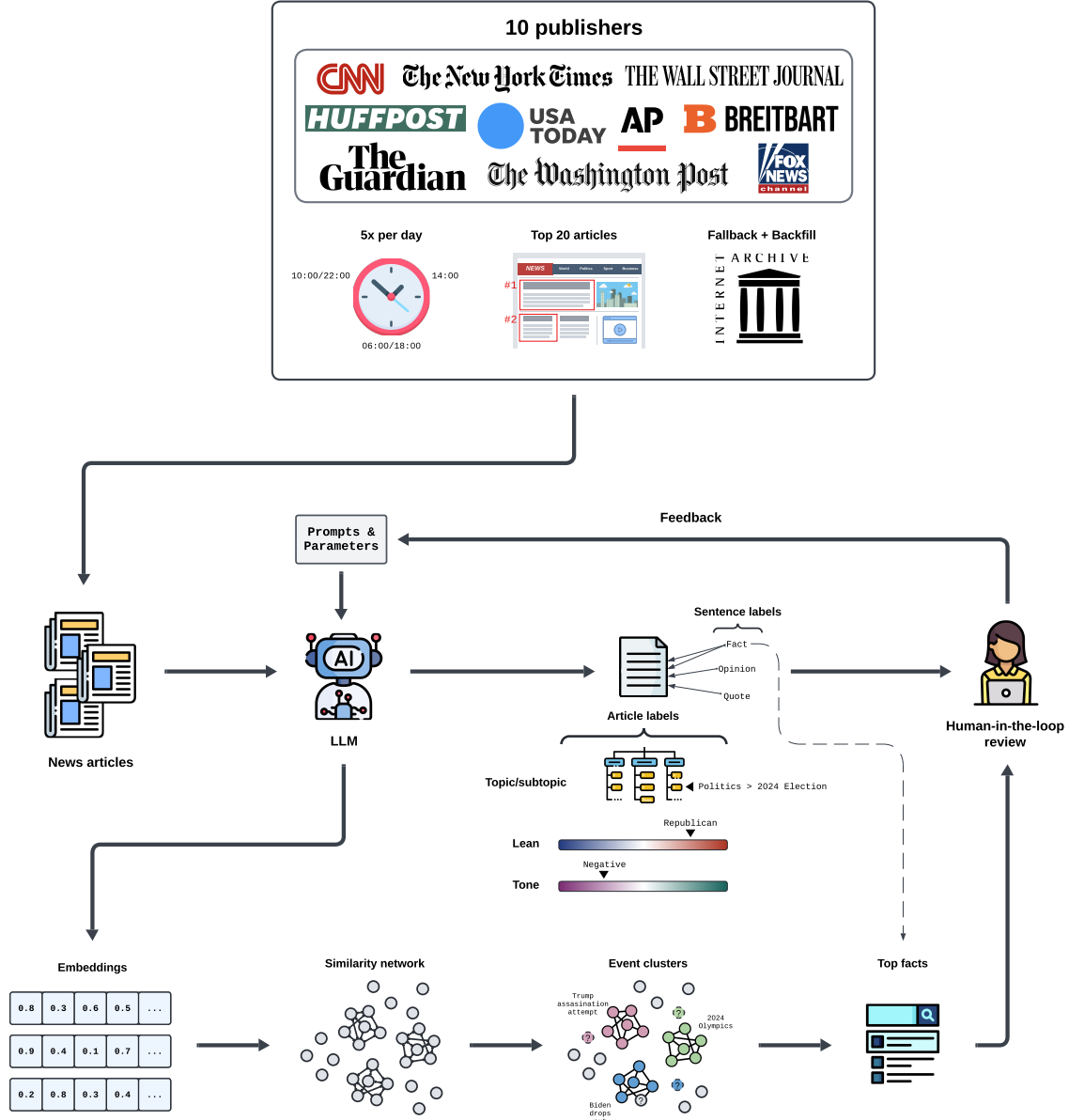


Figure 2.2: **The MEDIA BIAS DETECTOR data pipeline.** For each of the ten publishers, we take a snapshot of their homepage five times per day and scrape the top-ranking news articles. We then use LLMs to extract multiple levels of structured labels at the article level (topic, subtopic, lean, tone, type) and the sentence level (type, tone, focus). In addition, we obtain vector embeddings for all articles to cluster them based on the events they focus on. We then cluster all sentences from articles about a particular event to highlight the top facts surrounding it. Our regular human-in-the-loop process adds oversight to this framework and ensures that the generated labels are accurate.

overview of our pipeline, which comprises three primary components: data collection, data labeling, and data validation. In the following sections, we discuss each of these in detail.

2.2.2. Data Collection

Our dashboard currently covers 21 major news publishers chosen to capture a mix of audience size (see Figure 2.3), agenda-setting influence, and ideological diversity: Associated Press, Breitbart News, CNN, Fox News, HuffPost, The Guardian, The New York Times, The Wall Street Journal, The Washington Post, USA Today, Los Angeles Times, Chicago Tribune, Star Tribune, BBC, Financial Times, Reuters, Bloomberg, CNBC, New York Post, Newsmax, and The Daily Wire. Of these, the first ten were included in the initial launch in June 2024 and the remaining eleven were added in May 2025. As indicated earlier, we expect the collection of publishers will continue to increase over time, including other publication modes such as television, podcasts, and social media.

The justification for starting with a relatively small number of prominent publishers and growing the collection over time is twofold. First, our objective of characterizing bias requires us to capture all relevant text while minimizing extraneous content such as ad copy or boilerplate text that could generate spurious results. This goal is complicated by the highly variable formatting of news websites, particularly dynamic mixed-mode features such as “live” articles that increasingly serve as top stories on several publishers’ websites. Focusing on just a few publishers allows us to customize our approach to individual websites and hence obtain a level of precision that would be infeasible at the scale of hundreds or thousands of publishers. Second, media consumption, especially by elites, is highly skewed such that the handful of prominent publishers that we do cover arguably play a larger agenda-setting role than even thousands of publishers with tiny audiences.

Every day, we take a snapshot of each publisher’s homepage five times, at 6 AM, 10 AM, 2 PM, 6 PM, and 10 PM Eastern Time (ET). Then, we use headline placement on these homepage snapshots to identify the top 30 most prominent articles displayed to readers. We determine this position on the page as a combination of distance from the top, font size of the title text, and the presence and size of images accompanying the article. Currently, we disregard content that is not a news

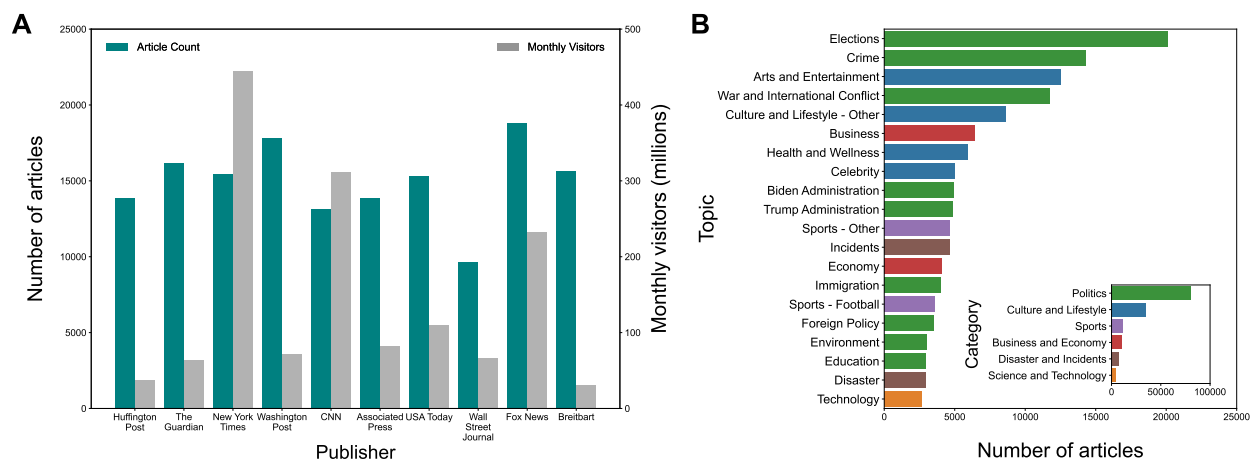


Figure 2.3: **Overall statistics of our dataset.** (A) Total number of unique articles in our dataset from each publisher in 2024, compared to their monthly traffic for May 2025 (Press Gazette, 2025). (B) The most popular news topics across this dataset, colored by category (shown in the inset). As expected in an election year, most of the news was dominated by politics, which comprises more than half of all articles in our dataset.

article, such as videos, podcasts, and photo galleries. We then scrape the text, title, and date of publication for every article in these top 30 (while we collect the data for the top 30 articles, for cost purposes the labeled data that we release on the dashboard and review in this chapter are currently restricted to the top 20 articles). We then preprocess the text to remove advertisements, boilerplate content, and superfluous text unrelated to the article. Additionally, we deduplicate articles based on normalized URLs to account for superficial variations that may cause the same article to appear under technically distinct URLs, such as trailing slashes, query parameters, fragments, hostnames, and schemes. Articles whose normalized URLs are identical are treated as duplicates, and only one copy is retained in our dataset.

Our data collection started on January 1, 2024, for an initial ten publishers (Associated Press, Breitbart News, CNN, Fox News, HuffPost, The Guardian, The New York Times, The Wall Street Journal, The Washington Post, USA Today) and an additional eleven publishers (Los Angeles Times, Chicago Tribune, Star Tribune, BBC, Financial Times, Reuters, Bloomberg, CNBC, New York Post, Newsmax, and The Daily Wire) were added in May 2025. We note, however, that the illustrative results presented in this chapter are based only on the initial ten publishers for the

period January 1 to December 31, 2024, during which time we collected over 140,000 unique news articles. Figure 2.3A shows the total number of articles per publisher, along with their most recent monthly traffic statistics (Press Gazette, 2025). Because all publications are being scraped at the same times, for the same number of articles, the differences in the number of unique articles reflect the speed with which they rotate their content.

2.2.3. Data Labeling

Measuring media bias is an inherently complex task even for humans, in part due to the lack of a universally accepted set of standard metrics (Spinde et al., 2023). Unlike outright falsehoods, which can be fact-checked against verifiable sources (Lazer et al., 2018; Vosoughi et al., 2018; Pennycook and Rand, 2021), bias is inherently subjective and difficult to quantify (Park et al., 2021). Readers often perceive articles as “biased” on the grounds that they disagree with them, but in doing so, they are (usually implicitly) treating their own subjective opinions as the “truth” (Pronin et al., 2004). In many cases of interest, such as contentious social and political issues on which subjective views vary widely, there is no *objective* or even *intersubjective* ground truth for measuring bias in the news. As noted earlier, for example, even ostensibly neutral articles can be perceived as biased on the grounds that they ought not to have been neutral. Unfortunately, however, in such cases the appropriate lack of neutrality is highly subjective. A second challenge is that the distinction between selection and framing bias is often blurry in practice. Consider, for example, a journalist who—intentionally or unintentionally—writes a negative story about the state of the economy by emphasizing facts about high inflation while omitting facts about high employment and wage increases. From one perspective, this example looks like framing bias: having selected the topic of the economy, the journalist is framing the story around inflation rather than employment. From another perspective, however, it resembles selection bias, just operating at the level of facts rather than topics.

In response to these challenges, our approach is guided by two general principles. First, we define bias as the presence of systematic patterns in the selection and framing of news that are observed across articles, topics, or time periods rather than isolated instances. Therefore, measuring bias

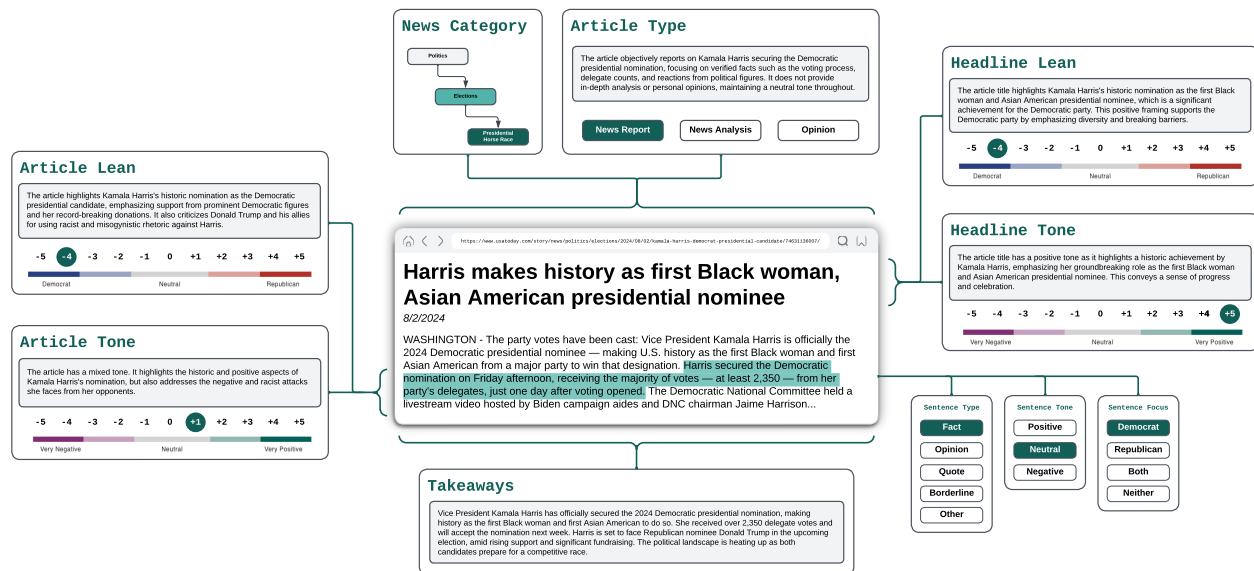


Figure 2.4: **Data points extracted from each article.** All labels and qualitative descriptions are obtained using LLMs, and can be aggregated across articles to investigate a broad range of research questions about media coverage.

requires some reference point against which these deviations can be identified to reveal consistent editorial choices. On the other hand, we define neutrality as the property of an individual article’s text which can be assessed more directly, such as whether its tone is positive or negative or whether it aligns with one political party’s position or another. An article that is neutral in tone and lean can still contribute to bias at a higher level through the publisher’s decision to cover that topic in the first place, or through a pattern of neutrality that masquerades as balance but actually functions as “bothsidesism”. By observing differences between publishers at one period in time, or by tracking shifts within a single publisher over time, we can detect and surface the signatures of potential bias, but we refrain from assigning explicit “bias scores” to individual stories or publishers.

Second, we also avoid making a categorical distinction between selection and framing bias, instead labeling our data at both the sentence and article level, thereby building in the flexibility to aggregate at either of these levels as well as higher levels such as topics, publishers, and even groups of publishers. Although our approach is in some respects less satisfying than definitive-seeming bias scores and cannot detect certain instances of bias (e.g., false equivalency at the article level), we believe that it is more robust and scalable than methods that assume a notion of objective truth.

Moreover, in instances where objective data are available (e.g., published economic indicators), our data can potentially be combined with such data to yield more definitive claims of bias for those specific cases.

Our framework therefore measures neutrality at the level of the elemental properties of a news article that are directly measurable, such as topic, tone, and lean. The flexibility of combining these building blocks in various ways then gives us the ability to indirectly measure sustained bias over a period of time using comparisons of coverage within and between publishers. Figure 2.4 shows the various data points we extract from each news article, a sample of which can be found in Text S1.

Topic Labeling

Each day we use the LLM to classify the content of every news story at three levels: Category (Politics, Culture and Lifestyle, Business and Economy, Disaster and Incidents, Sports, and Science and Technology); Topic (e.g., within politics: crime, elections, etc.); and Subtopic (within crime: violent crime, political corruption, etc.). Given the large number of articles per day and the many possible ways to classify their content, we created a top-down category-topic-subtopic hierarchy via an iterative human-in-the-loop process.

First, a selection of co-authors hand-coded the topic and subtopic of all of the front-page articles in The New York Times and The Washington Post from September 1, 2022, to the date of the midterm elections on November 8, 2022 (Rothschild et al., 2023). Second, we asked the LLM to freely generate these labels for one month of news. Then, with a mix of humans and LLMs, we checked which topics and subtopics clearly fell under our initial set from the 2022 study, and what new topics and subtopics needed to be generated. Third, we followed an iterative process of asking the LLM to pick its labels from this list or otherwise suggest others if the list did not work.

After several iterations, we landed on an initial list of topics and subtopics. Currently, the list contains 33 topics and 162 subtopics across the six categories, where we add new topics and subtopics when human editors see evidence of articles not cleanly falling within our current list. Each article is assigned to exactly one topic and subtopic, and each predefined topic is mapped to one of these

six categories, defaulting to Politics when an article may fall in several categories. Clearly, many articles could plausibly be classified into more than one topic, and many other topics and subtopics could have been included beyond the ones we chose. However, our preference for single assignment to a relatively short list of topics has the dual benefit of improving classification accuracy and simplifying aggregation. Figure A.1 shows the full list of topics and subtopics developed in this process, and Figures A.2 and A.3 show the topic and subtopic labeling prompts that ask the LLM to choose from within this hierarchy, respectively. The most popular news topics in 2024 by number of articles are shown in Figure 2.3B. As expected, most of the coverage was dominated by politics, in particular the U.S. Presidential Election.

Finally, for each article we also generate a short summary of its main takeaways using GPT-4o-mini (see Figure A.4 for the prompt). This summary distills the article into a few sentences that cover the key points a reader is likely to remember, and can be used as a compressed representation of the article content for downstream analysis. These summaries also facilitate manual exploration of the dataset by allowing users to quickly skim the takeaways instead of having to read the full article to understand its gist.

Article Lean, Tone and Type

Aside from topic categorization, the three article-level properties that we measure are political lean, tone, and type. Political lean, which we also refer to as lean, is defined as the extent to which the article either explicitly or implicitly aligns with the viewpoints, policies, or concerns of Republicans vs. Democrats. Lean can manifest itself as overt partisan bias, such as when an article directly criticizes a political party or its representatives, but can also be more subtle, such as when certain topics or perspectives are inherently more aligned with one party’s preferences or policies. For example, GPT-4o tends to code articles supporting social justice movements as Democratic-leaning, even when the article neither endorses nor critiques either party, on the grounds that such issues are more explicitly associated with Democrats than Republicans.

Lean, in other words, is itself a complex and multidimensional concept that is also somewhat subjective. To improve consistency and transparency, we first ask the LLM to generate a natural

language analysis of the article’s political lean and then provide an appropriate numerical score on an 11-point Likert scale in the range of $[-5, +5]$ (see Figure A.5 for the prompt). Following the chain-of-thought prompting approach (Wei et al., 2022), this encourages the model to articulate an analysis of the article before scoring, which both improves the consistency of its outputs and also provides an interpretable summary of the article’s lean. As we will describe in Section 2.2.4, human raters agree substantially with GPT-4o’s lean and tone ratings.

We caution that we cannot rule out that both GPT-4o and human ratings are themselves biased in the sense that a right-leaning rater might code lean systematically differently from a left-leaning rater. Nonetheless, we believe that lean ratings are more reliable than asking directly about bias in the sense that it is easier (both for LLMs and humans) to determine that a particular article is pro-Republican or pro-Democratic than what its “true” orientation ought to be, and hence how it differs from the truth.

In addition to lean, we also measure tone, defined as the extent to which the article overall has a positive vs. negative tone. As with lean, the tone of an article—or of a body of news coverage—also represents an editorial choice that can shape readers’ opinions. For example, talking about the economy in a pessimistic tone can be interpreted as a publisher taking a negative stance with respect to the incumbent government, and by extension, the ruling political party. Although tone tends to be simpler and more easily interpretable than lean, it too is open to a certain amount of subjectivity. As with lean, therefore, we also use chain-of-thought reasoning, asking the LLM both for a human-interpretable analysis of the article and a numerical rating in the range $[-5, +5]$ (see Figure A.6 for the prompt). Figure 2.4 shows an example of the output from both prompts illustrating the surprising sophistication and depth of the explanations. In Section 2.2.4, we provide results on the validation of these labels. We also tested for robustness of these ratings by re-running the same prompt five times on a set of 50 articles spanning diverse topics and the full range of lean and tone labels. The average Pearson’s r was 0.99 for lean and 0.97 for tone (Figure A.11). Of the 50 articles, the lean and tone ratings were identical across all runs for 35 and 37 articles, respectively. In cases with any variation, the confidence intervals were almost always within 1 scale

point of the mean.

We also repeat the above labeling process separately for the articles’ headlines in order to study how they may deviate from the actual article text in terms of lean and tone, which is itself a form of media bias. We use the same wording for both prompts with the addition of the article’s topic to give the model more context to interpret the headline. The prompts for headline lean and tone labeling can be found in Figures A.7 and A.8, respectively.

Finally, we also label the article’s type as a news report, news analysis, or an opinion piece. We consider an article to be a news report if it provides fact-based, objective reporting on an event. Meanwhile, a news analysis article is a more in-depth examination accompanied by context. An opinion piece is more subjective than either of these, and can reflect personal views or beliefs of the writer. Classifying the type of an article is important for studying media bias because different types carry different expectations of objectivity. We generally expect news reports to be impartial, whereas opinion pieces are understood to represent a particular viewpoint. These labels can thus be used in conjunction with lean and tone to help understand where in a publisher’s reporting bias shows up most strongly. As with lean and tone, we ask the LLM to provide a natural language analysis in addition to the categorical label (see Figure A.9 for the prompt). Since the boundaries between these categories can often overlap, as a news analysis, for instance, may blend factual reporting with interpretive framing in ways that bring it close to an opinion piece, the LLM’s analysis serves as a useful check on borderline cases.

Sentence Type, Tone, and Focus

In addition to article-level labels, we also label every sentence in an article with three features: type, tone, and focus. These sentence-level annotations complement the article-level labels by enabling more fine-grained analysis of how bias manifests within the structure of individual news stories.

The type of a sentence can be one of “fact”, “opinion”, “borderline”, “quote”, or “other.” We do not ask the model to evaluate the truth or falsity of any claims, and hence by “fact” we mean only that it has the form of a factual claim or statement, not that it is a true fact (Mitchell et al., 2018).

“Opinion” refers to sentences that express a viewpoint or belief. “Borderline” captures sentences that blend factual reporting with interpretive or evaluative language. “Quote” identifies direct quotations, which are handled separately in our quote extraction pipeline (Chapters 3 and 4). Finally, “other” is a residual category for sentences that do not neatly fit into any of the above types.

The tone of a sentence is defined in a similar way to tone at the article level but is assigned only one of three values: “positive”, “negative”, or “neutral.” We use this coarser rating scale to reflect the more limited information available at the sentence level. However, even these coarse labels can be useful when aggregated—for example, to study the proportion of positive or negative sentences in coverage of a particular topic or individual.

Finally, focus identifies whether the sentence refers to a particular political party, with possible values of “Democrat”, “Republican,” “neither,” or “both.” Focus differs from lean in that it only asks what is being referred to, not whether the reference is “pro” or “con,” and hence is simpler to answer reliably (we do not code individual sentences for lean, as these ratings do not appear to be reliable).

Although these labels are extracted at the sentence level, their coding still requires the surrounding context of the article to make accurate judgments. Therefore, we pass the entire article to the model in the form of an enumerated list of sentences, which we obtain by using the Stanza toolkit (Qi et al., 2020) for segmentation. The full sentence labeling prompt can be found in Figure A.10.

2.2.4. Data Validation

To evaluate the quality of labels generated by our system, we built a custom data annotation platform and conducted separate data labeling exercises for each facet of our data: article lean/-tone/type, article topic/subtopic, and sentence type/tone/focus. Each of the four data dimensions was independently labeled by three annotators. We used Pearson’s r to compute correlation for the numerical article lean and tone ratings and Cohen’s κ to compute agreement for the remaining categorical labels. We represent the inter-annotator agreement as the mean of all pairwise agreement scores between the three annotators, and model-annotator agreement as the mean of all pairwise agreement scores between the model and each of the three annotators. To evaluate whether the

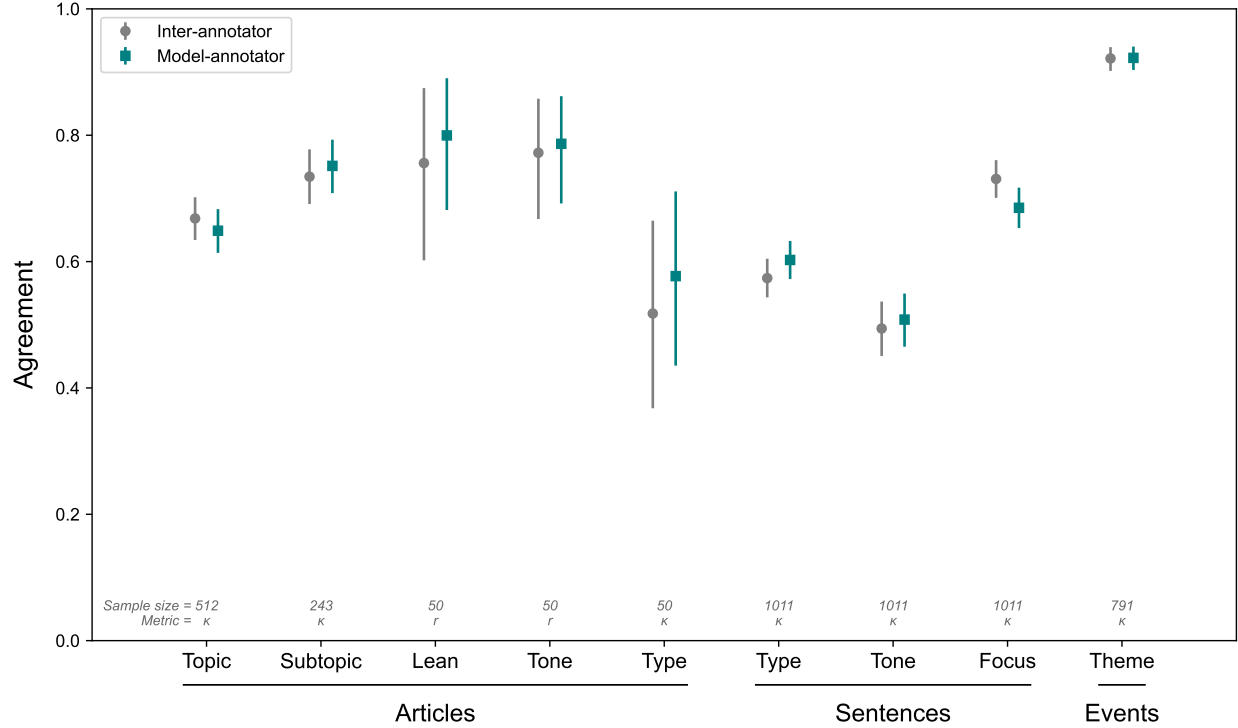


Figure 2.5: **Validation results across tasks.** Average pairwise inter-annotator and model-annotator agreement across article- and sentence-level validation tasks. The strong correspondence between human and model agreement, along with overlapping confidence intervals, indicates that the model’s labels are highly consistent with human labels.

level of model-human agreement is consistent with human-human agreement, we perform a bootstrap test with 10,000 resamples to show that their confidence intervals overlap significantly and that the agreement between the model and humans is statistically indistinguishable from that between the human annotators themselves. The following sections describe the results for each dimension of the data separately, with full results presented in Figure 2.5.

Article Topic and Subtopic

We sampled 512 news articles with equal representation from all topics and subtopics where possible and asked three annotators to independently assign appropriate labels to them at both levels. The full set of 512 articles was used for the topic-level analysis. For the subtopic-level analysis, however, we restricted the sample to 243 articles where all annotators and the model had selected the same topic, and compared the subtopic labels within that subset. We computed pairwise Cohen’s κ and

found strong inter-annotator agreement at both the topic (mean = 0.668, 95% CI [0.634, 0.702]) and the subtopic level (mean = 0.734, 95% CI [0.691, 0.778]). Comparing these labels with the model’s outputs, we find that it achieves comparable κ at both the topic (mean = 0.649, 95% CI [0.614, 0.683]) and subtopic level (mean = 0.751, 95% CI [0.708, 0.793]). These results show that the model closely tracks human-level performance as the model-annotator agreement was statistically indistinguishable from inter-annotator agreement for both topics and subtopics.

In most cases, we found that misclassifications were not outright wrong, but that the article existed at the intersection of multiple topics and could reasonably have been classified as either. Figures A.17 and A.18 show the confusion matrices for topic- and subtopic-level classification, respectively. For example, an article on the involvement of the U.S. in a foreign conflict might be classified as about “Foreign Policy” but also as about “War and International Conflict.” We continue to evolve our news topic and subtopic hierarchy in order to accommodate shifts in the news cycle as well as to update it retrospectively and make it more robust to future changes.

Article Lean, Tone, and Type

To evaluate our lean, tone, and type labels for news articles, we divided our 11-point Likert scale for lean and tone into five buckets each ([−5, −4], [−3, −2], [−1, 0, +1], [+2, +3], [+4, +5]) and sampled two articles from each of the 25 possible lean-tone bucket combinations while ensuring a balanced distribution over the underlying 11-point values, yielding a total of 50 articles. We recruited three PhD students in communication who had a deep understanding of the media and the current U.S. political landscape to read each of these articles and code their lean, tone, and type. The full interface for these tasks can be seen in Figures A.13 and A.14, and the codebook provided to the annotators can be found in Text S2.

To evaluate inter-annotator agreement for the numerical labels, we computed pairwise Pearson’s r and found high correlations between the annotators for both lean (mean = 0.756, 95% CI [0.602, 0.875]) and tone (mean = 0.772, 95% CI [0.667, 0.858]). The model’s average correlations with the human annotators closely tracked the inter-annotator values for both lean (mean = 0.800, 95% CI [0.682, 0.890]) and tone (mean = 0.786, 95% CI [0.692, 0.862]). For news type, the pairwise inter-

annotator κ indicated moderate agreement (mean = 0.518, 95% CI [0.368, 0.665]), reflecting the inherently subjective nature of the task, while the model’s labels achieved comparable agreement as well (mean = 0.577, 95% CI [0.435, 0.711]). In all three cases, the model’s agreement with human labels was indistinguishable from the inter-annotator agreement. Figure A.19 shows the scatter plots for lean and tone ratings and the confusion matrix for article type labels.

Sentence Type, Tone, and Focus

To validate sentence labels, we used a subset of 25 articles from the article lean and tone annotation task and asked a separate set of annotators to go sentence-by-sentence and label the type, tone, and focus of each. These 25 articles contained a total of 1,011 sentences, for an average of 40.44 sentences per article. The full task interface is shown in Figure A.16 and the codebook provided to the annotators can be found in Text S3.

Most sentences in the news articles were neutral facts that did not mention any particular political party, although this varied for different article types and topics. The inter-annotator κ was moderate for sentence type (mean = 0.574, 95% CI [0.543, 0.604]) and tone (mean = 0.494, 95% CI [0.451, 0.537]) and high for focus (mean = 0.731, 95% CI [0.701, 0.761]). This pattern likely reflects that focus, which involves identifying concrete references to political actors or parties, is a more unambiguous feature than type and tone, which require more subjective judgments that depend on contextual interpretation. The model’s agreement with the annotators showed a similar trend, with moderate values for type (mean = 0.603, 95% CI [0.572, 0.633]) and tone (mean = 0.508, 95% CI [0.465, 0.549]) and high κ for focus (mean = 0.685, 95% CI [0.653, 0.717]). Figure A.20 shows the confusion matrices for all three sentence labels.

2.3. Results

In this section, we apply our curated dataset from 2024 to a handful of questions about media practices and biases. It is important to emphasize that the analyses presented here are just illustrative samples of the full data. The comprehensive nature of our dataset allows researchers to answer much more focused questions by examining different subsets, enabling targeted investigations into specific aspects of media coverage, topical biases, or temporal trends.

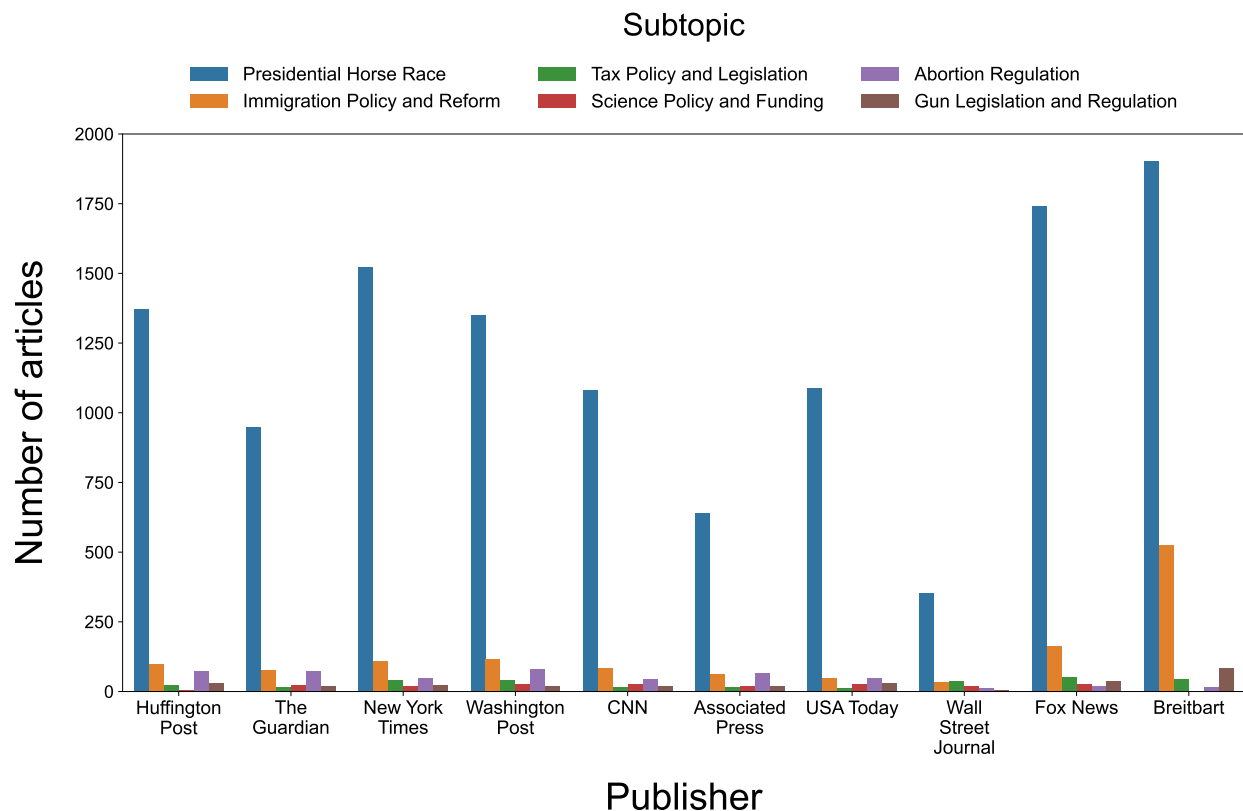


Figure 2.6: **The imbalance between policy and election horse-race coverage.** The media primarily covered the election as a horse race instead of focusing on more substantive policy discussions to inform their readers of each candidate’s and their party’s agenda. Across all publishers in 2024, more than four times as many articles were focused on the horse race as on all policy topics combined, with a notable share of policy coverage driven by Breitbart’s emphasis on immigration.

2.3.1. The imbalance between policy and election horse race coverage

News organizations operate under constraints of limited time and space, meaning they need to make careful choices about which stories to cover. This gatekeeping power has been extensively studied in the social sciences, particularly under the framework of agenda setting, which posits that while news media may not directly dictate what people think, they substantially influence what people think about (McCombs and Shaw, 1972). In making editorial decisions, news organizations assess the newsworthiness of events based on various factors such as prominence, timeliness, and proximity (Shoemaker and Reese, 1996). This assessment of newsworthiness is particularly crucial given that news organizations are often private entities driven by profit or political motives and aim to con-

form to their audience’s prior beliefs (Gentzkow and Shapiro, 2006). Furthermore, these editorial choices have a major impact on readers, whose understanding of the world around them is shaped by the news they consume. Ideally, news organizations should help voters make informed choices by informing them about policy discussions. However, existing literature has highlighted that media outlets tend to focus predominantly on the “horse race” elements of elections—emphasizing candidates’ polling numbers, strategic positioning, and projected wins—which can overshadow crucial policy issues (Iyengar et al., 2004; Vliegenthart et al., 2010).

Our data offer a more comprehensive quantitative analysis of the coverage of horse race dynamics compared with other critical policy issues in the lead-up to the 2024 U.S. presidential election than has previously been possible. Specifically, using our topic and subtopic classification system, we carried out a detailed comparison of the amount of coverage dedicated to the presidential horse race versus policy discussions. Figure 2.6 illustrates this disparity, showing that the coverage of the presidential race overwhelmingly dominated all news outlets, even receiving more attention than all policy issues combined. It also reveals how specific news values adapt by catering to different target audiences (Mullainathan and Shleifer, 2005; Welbers et al., 2016). For example, outlets like Breitbart and, to a lesser extent, Fox News allocate substantially more coverage to Immigration Policy and Reform in comparison to left-leaning media such as CNN or The Guardian.

2.3.2. Political lean varies across news topics

As discussed earlier, news producers can also influence public perception through their framing of issues (Chong and Druckman, 2007), defined broadly as choices in language and context as well as the inclusion or omission of certain details and perspectives (Gentzkow and Shapiro, 2006; Entman, 1993). Here we study lean—the extent to which a news article supports Democratic or Republican viewpoints and policies—which, as described above, can indicate framing bias. Current methods typically reduce news publishers to this single metric by assigning them a political lean rating that ranges from “Left” to “Right” (AllSides, 2025; Media Bias/Fact Check, 2025; Ground News, 2025). While these static classifications are useful, they often fail to capture the nuances of bias within a single publisher, across various topics, or even from one article to another. Understanding how parti-

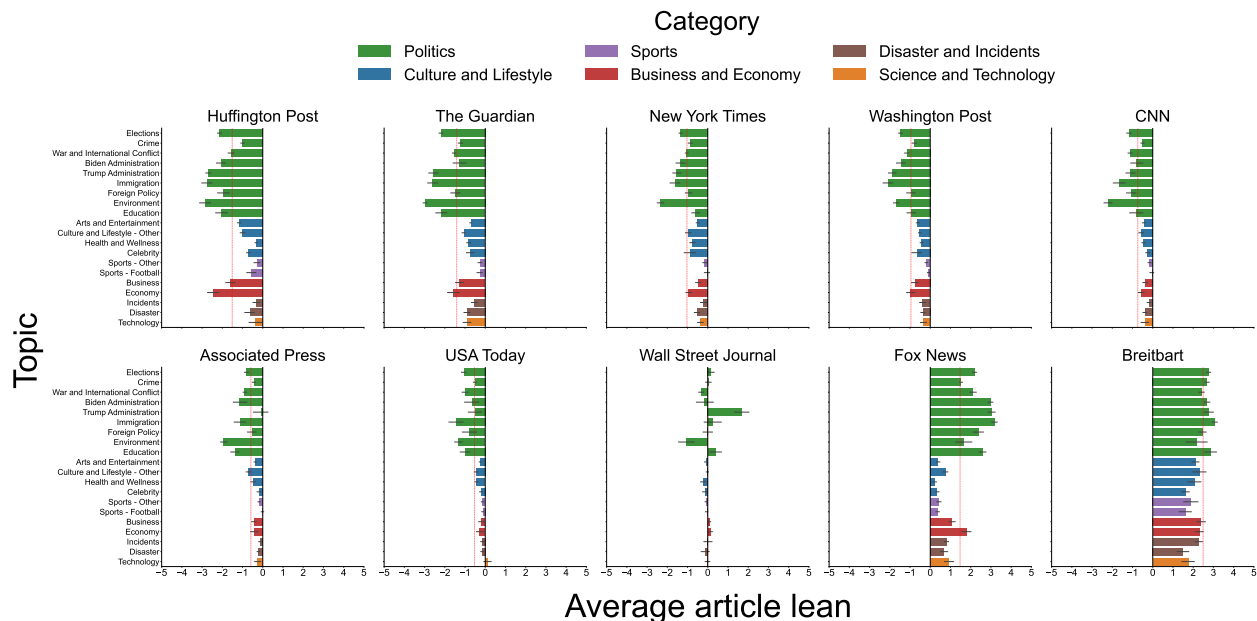


Figure 2.7: **Political lean varies across news topics.** This figure illustrates the average (indicated by the vertical red line) political lean of selected news outlets, categorized by topic. Negative values indicate a Democratic lean, whereas positive values suggest a Republican lean. The red line shows the average lean for a publisher across all topics; the variation highlights the nuanced and topic-dependent nature of media bias. Variations in the degree of political alignment provide insights into each outlet’s editorial stance and the complexity of bias representation in media coverage.

san alignment shifts depending on the subject matter can substantially enhance our comprehension of media dynamics and discourse.

As described in Section 2.2.3, the MEDIA BIAS DETECTOR provides lean labels at the article level (focusing on the substance of the articles, these labels ignore headlines, which we address in Section 2.3.4), allowing for a more nuanced analysis of media outlets’ political lean. Figure 2.7 presents the average political lean across publishers and different topics, with negative values indicating a Democratic lean and positive values indicating a Republican lean. Although the overall pattern aligns with commonly held perceptions of these publishers’ political lean, it also reveals intriguing insights into the subtleties of media partisanship. This granular approach uncovers variations in bias not only across different outlets but also within publishers on specific topics, offering a more detailed understanding of how partisan framing can shift.

As depicted in Figure 2.7, it is generally true that an outlet’s stance on one issue tends to correlate with its stance on others; however, the extent of this alignment can vary considerably depending on the topic. For instance, The New York Times exhibits a consistently left-leaning position on topics such as the environment, whereas it maintains a relatively neutral stance on technology. In contrast, Fox News presents a more right-leaning stance on both topics. Figure 2.7 also shows that The Wall Street Journal appears to be the most neutral outlet within this set of publishers, maintaining a relatively balanced stance on most topics. Although The Wall Street Journal is often perceived as having a conservative lean, its apparent neutrality here likely stems from its relatively heavy focus on business and the economy, which—along with sports, travel, technology, and health and wellness—tend to be less partisan than political topics. We also reemphasize that neutrality does not imply a lack of bias. Whereas the former can be evaluated directly from the text of an article, the latter requires knowledge of some (typically unobservable) ground truth, which could potentially favor one “side” over the other. In such a case, a neutral stance would in fact reflect bias, as is often asserted in accusations of “false balance” or “bothsidesism.” We also emphasize that even claims about neutrality are inherently relative and can only be made in the context of other publishers or, alternatively, the same publisher over time. The characterization of The Wall Street Journal as “the most neutral” publisher is therefore specific to this particular sample of publishers and likely will not generalize to other samples.

With these caveats in mind, we observe that the overall ordering of publishers based on our framework nonetheless aligns closely with past work that has focused on assigning ideological slant to news publishers based on their consumption patterns or crowd-sourced labels, as shown in Figure A.21. Here, we compare the average political lean score for each publisher across all articles with estimates from (Bakshy et al., 2015) and (Budak et al., 2016). Our estimates align closely with both studies, showing strong correlations of $R^2 = 0.946$ with (Bakshy et al., 2015) and $R^2 = 0.976$ with (Budak et al., 2016), demonstrating the consistency of our approach with established findings. We also show that this lean is not static but fluctuates over time (Figure A.22), highlighting changes in news outlets’ reporting in response to major events.

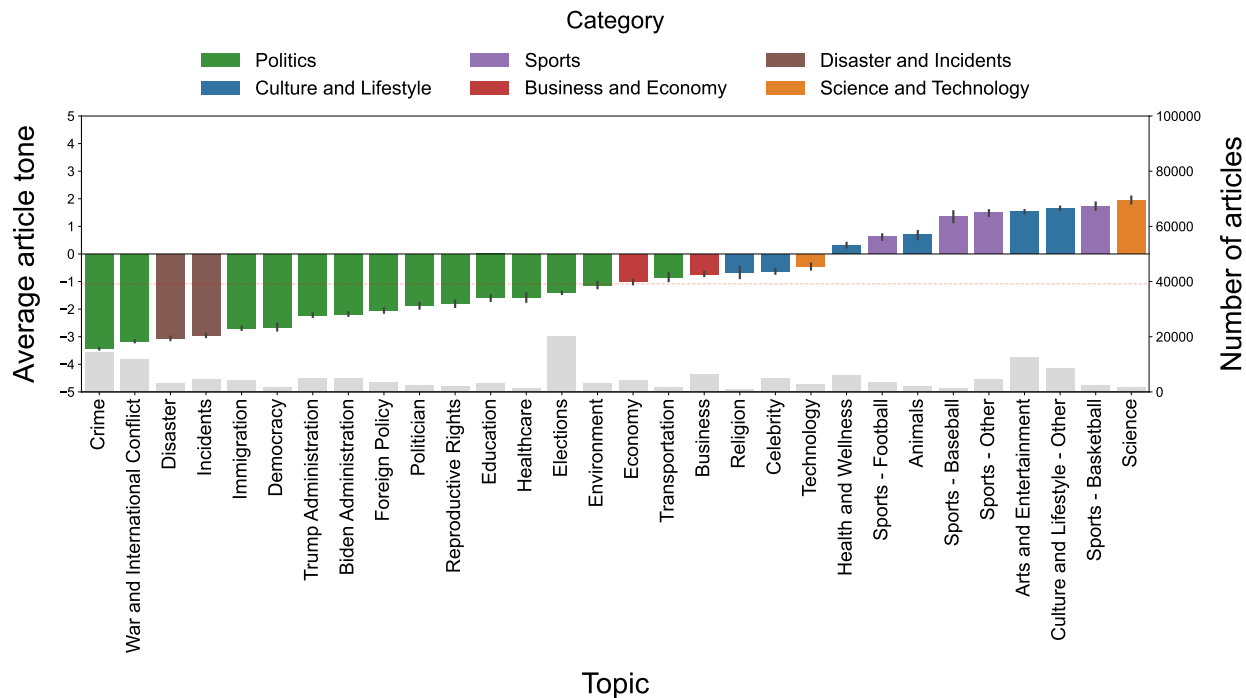


Figure 2.8: **Most news is negative, especially politics.** The colored bars represent the average tone of articles that fall in the top 20 topics by number of articles (shown in gray), sorted from most negative to most positive. Negative tone is prevalent, both by number of topics and by the total volume of articles. Except for news about weather and natural disasters, all of the remaining topics that are more negative than the average (red line) are related to politics, highlighting the pervasive nature of negative elements in media coverage.

2.3.3. Most news is negative, especially politics

As discussed earlier, the tone in which stories are presented can be another indicator of framing bias. Prior work has found that tone plays a critical role in agenda-setting by making a topic seem more important (Sheafer, 2007) and influences the salience of negative and positive elements within the public debate (Vreese and Boomgaarden, 2003). In politics, for example, voters have been found to be influenced by the tone of news coverage surrounding an election candidate (Druckman and Parkin, 2005), while in economic reporting, news sentiment predicts macroeconomic variables like consumption and output (Shapiro et al., 2022).

Here, we study negativity bias, an important case in which tone may function as a form of framing bias in the media, which has been extensively documented in both the production and consump-

tion of news (Lengauer et al., 2012; Robertson et al., 2023). This focus on the negative could arise from both the selection and framing of news. On the selection side, it often involves prioritizing stories inherently perceived as negative, such as those involving conflict, crime, and disasters (Van Der Meer and Hameleers, 2022). Additionally, the framing of certain stories can enhance negativity through the deliberate use of specific words and phrases (Trussler and Soroka, 2014). The tone classifications in our dataset capture both of these aspects, providing a comprehensive view of how negativity is embedded in news content across a wider range of topics and publishers than has previously been possible.

Figure 2.8 depicts the average tone (left y-axis) of articles across topics as well as the number of articles on that topic (right y-axis). We find that the majority of the news content analyzed contained negative elements, aligning with trends that are well documented in the aforementioned literature. However, we also extend this finding by showing how tone varies within political news, which is consistently negative yet also receives the highest volume of coverage. This combination of high negativity and high visibility in political news has important implications for democratic discourse, as it may contribute to political cynicism and disengagement (Trussler and Soroka, 2014). Moreover, our cross-topic analysis demonstrates how this negativity bias extends beyond politics (e.g., religion and economy), suggesting a systemic feature of contemporary news production that affects how the public perceives various aspects of social life.

2.3.4. Headlines are not always faithful to their articles

A headline is arguably the most important part of any news story and sets the stage for the rest of the article. Indeed, for many casual readers, the headline might be their only takeaway about an event, an effect that is exacerbated by the news feeds common on social media platforms that focus almost exclusively on headlines (Searles and Feezell, 2023). Consequently, any misalignment in lean or tone between news headlines and the accompanying articles (Marquez, 1980) raises concerns for at least two reasons. First, many individuals decide to share information based solely on the headline, without reading the full news article. A recent study on social media sharing demonstrated that more than 75% of the news articles shared on Facebook were not clicked on before being forwarded

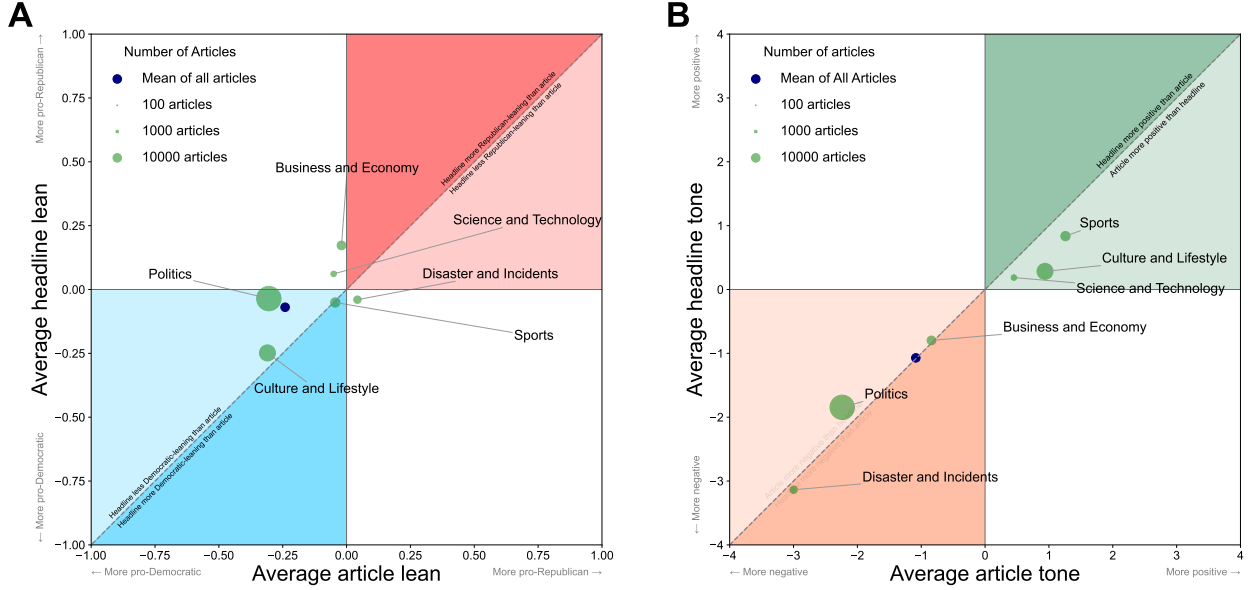


Figure 2.9: **Headlines are not always faithful to their articles.** Each point represents average headline (y-axis) versus article (x-axis) lean or tone for a topic category. Positive lean values indicate pro-Republican stances; negative values indicate pro-Democratic stances. All content is labeled on a $[-5, +5]$ scale (axes adjusted separately for clarity). **(A)** Headlines lean slightly more pro-Republican than their articles on average (blue point above diagonal), most notably for business and economy topics. **(B)** Headlines are more faithful to article tone, with the average (blue point) falling on the diagonal.

(Sundar et al., 2025). Second, even when people do read the entire article, the headline continues to exert a substantial influence on opinion formation. If the information presented in the headline is incongruent with the article’s content, individuals often struggle to adjust their initial impressions (Carcoppolo et al., 2022) and their ability to recall facts presented in the article is negatively impacted (Ecker et al., 2014).

The separate labeling of headlines and article texts with identical prompts in our framework enables us to explore this phenomenon across both dimensions of tone and political lean. Figure 2.9 presents the results broken down by news category. In Figure 2.9A, the average headline lean is represented on the y-axis, and the average article lean on the x-axis, with positive values indicating an increasingly pro-Republican lean (red) and negative values indicating an increasingly pro-Democratic lean (blue). The dark blue marker shows the mean across all articles, and the size of each scatter point reflects the number of articles within that category.

Our results reveal that news headlines systematically lean more pro-Republican than the articles themselves, as evidenced by the position of the dark blue marker—the overall average—being above the diagonal in Figure 2.9A. This pattern is most pronounced in the “Business and Economy” category, where articles are fairly neutral overall but are often accompanied by headlines that lean more pro-Republican. Similarly, for the “Politics” category, the articles lean slightly pro-Democratic but the headlines are fairly neutral. Figure 2.9B presents the same comparison for tone, with average headline tone depicted on the y-axis and average article tone on the x-axis. While the overall mean lies close to the diagonal, suggesting a close correspondence of tone between headlines and articles on average, this aggregate measure masks substantial heterogeneity across topics. That is, topics with inherently positive content (sports, culture and lifestyle, and health) tend to have headlines that are less positive than their corresponding articles. Conversely, for typically negative topics like politics and disasters, the headline tone either aligns with or is slightly more positive than the article text. Comparing headline and article labels on both of these dimensions also shows that the observed pro-Republican lean is not due to differences in length or style of headlines compared to the full articles (since the same shift is not seen in tone), but may instead reflect editorial choices about how to frame the news.

2.3.5. Media outlets’ partisan focus is inversely correlated with their lean

Partisan news outlets employ various tactics to advance their ideological agendas, including their choices about which political actors to cover. One intriguing example that has been studied previously is the tendency of partisan media to focus more on critical coverage of opposing political groups than on positive coverage of their ideological allies (Smith and Searles, 2014). This “attack over defense” pattern may be amplified by the fact that politicians’ denunciations of media can heighten perceived biases, irrespective of the politicians’ party affiliations (Smith, 2010). This behavior raises important questions about the role of media in shaping public discourse, potentially fueling partisan polarization and influencing how audiences perceive political legitimacy and the opposition. Our data can address this question by applying a combination of focus and lean measures at a larger scale than has previously been possible with human-annotated data.

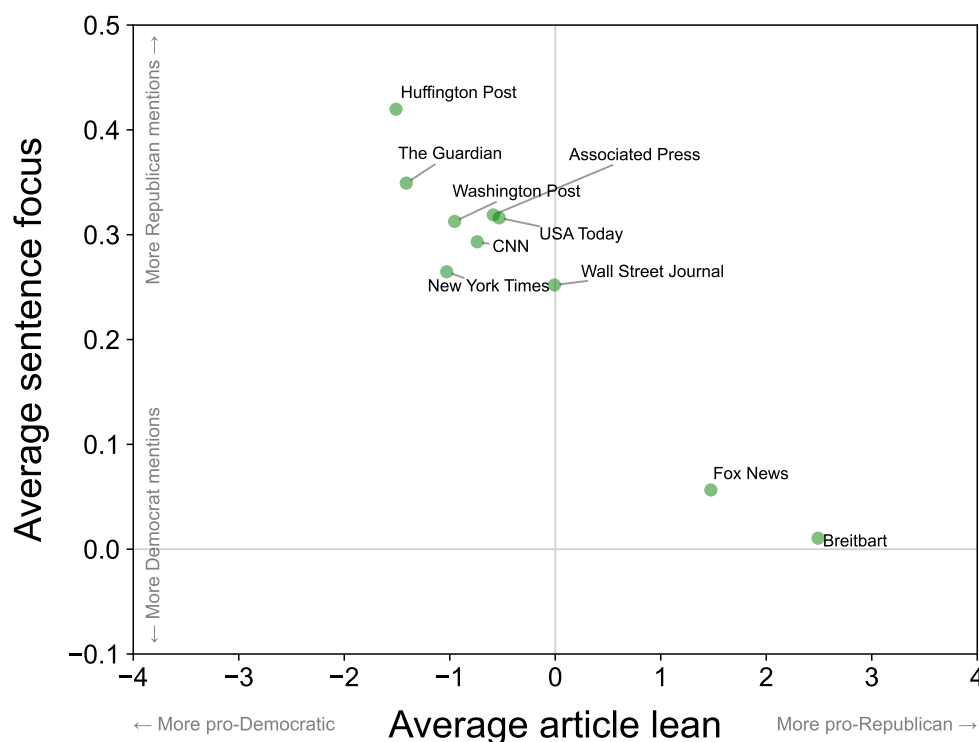


Figure 2.10: **Media outlets focus more on the opposing party than on their own side.** Association between publishers’ political lean and coverage of each political party. The x-axis shows the average political lean of all articles by a publisher (positive values represent more pro-Republican lean and negative values represent more pro-Democratic lean) and the y-axis shows the ratio of sentences that mention Republicans (positive) vs. Democrats (negative). The more a publisher talks about a particular party, the more they lean toward the opposition.

Specifically, we compare the political lean of articles—which measures the extent to which each article supports the viewpoints and policies of one party over the other—with the focus of the corresponding article. As noted earlier, focus is measured on a sentence-by-sentence basis, indicating whether each sentence references the Democratic Party, the Republican Party, both, or neither. From this, we compute an article-level focus score by averaging the individual sentence scores, with Republicans coded as +1, Democrats as -1, and neither or both as 0. Figure 2.10 illustrates the relationship between average lean and focus for the 10 publishers included in our dataset. The figure reveals a negative correlation between focus and lean, indicating that more left-leaning publishers tend to talk more about Republicans than right-leaning ones, and that the average focus across all outlets is skewed toward Republicans. We note that our focus measure conflates positive and

negative mentions; thus, although our findings are directionally consistent with the hypothesis that partisan outlets focus more on opposing parties, they do not rule out the possibility that outlets also engage in “cheerleading” for their own side.

2.4. Discussion

In this chapter, we have introduced the MEDIA BIAS DETECTOR, which leverages LLMs to analyze news media at the scale of hundreds of articles per day and in near-real-time to support and accelerate research on media bias. Our work builds on broader efforts to quantify the impact of news on democracy (Watts et al., 2021), in particular by constructing robust, scalable data infrastructure for studying how news moves from production to consumption and absorption. In addition, the MEDIA BIAS DETECTOR exemplifies the related goal of building web-based interactive visualizations to communicate insights to nonacademic stakeholders (Wang et al., 2025). By making our data and dashboards widely accessible, we hope to enable data-driven transparency in journalism, allowing stakeholders—including policymakers, educators, and media professionals—to engage with the findings in meaningful ways.

The MEDIA BIAS DETECTOR complements existing media datasets such as PeakMetrics, GDELT, and Media Cloud that already provide researchers with high-volume collections of online news articles, enabling large-scale studies of media coverage (Roberts et al., 2021), indeed at scales much larger than we have presented here. Although these datasets collectively constitute a valuable resource, like all such efforts they suffer from limitations. As noted earlier, for example, scraping thousands or tens of thousands of websites daily, or even more frequently, over long periods inevitably introduces numerous problems with data missingness and quality. Articles that are easier to extract—such as those from publishers without paywalls—tend to be overrepresented, while those from publishers with sophisticated paywalls are more sparse. Furthermore, these very large-scale scraping efforts generally treat content across the entire website equally, regardless of whether it is featured prominently on the homepage or buried deep within the category pages. In contrast, the MEDIA BIAS DETECTOR focuses on effectively capturing the *top* stories from each publisher’s homepage at regular intervals with high reliability. As such, our approach prioritizes capturing sto-

ries that receive substantial exposure, creating opportunities to study how publishers shape public focus through their editorial choices.

Symmetrically, our approach also reveals which topics are absent from the headlines, allowing researchers to study patterns of omission—cases where major political actions, policy decisions, or social issues receive substantial coverage from some outlets but are largely overlooked by others. By leveraging LLMs and measuring multiple quantities at the article and sentence level, our approach generates higher-quality and more granular labels than have been possible until very recently. Finally, the continuous human-in-the-loop monitoring ensures that our system is able to adapt to the news cycle and avoid concept drift. In this way, we ensure that our dataset is always up-to-date with the events and topics being covered in the media and classifies them accurately. While our current focus is on text-based news from mainstream media, our framework could also be adapted for social media by updating our prompts to more effectively label short-form text and by incorporating interaction data and network features such as likes and shares to measure the popularity and influence of various pieces of content.

Naturally, our approach also has important limitations. First, our data collection process does not provide exhaustive coverage of all online news. We limited ourselves to 21 major publishers (10 in the first stage of the project), but there are many more local and online news sources whose coverage may be of interest depending on the research question being studied. Further, by limiting ourselves to the top-ranking news articles from these publishers and only collecting them at regular intervals throughout the day, we may miss articles that either do not show up in the top 20 or appear only briefly and disappear before we are able to scrape them. The time frame of our collection is also limited to January 2024 onward, which limits the depth of temporal analysis that can be done with these data (although efforts are currently underway to backfill prior to 2024). In some cases, we also face issues with paywalls or broken links, which lead to gaps in our coverage.

Our data labeling and validation pipelines also have limitations. First, we use proprietary models that may become unavailable in the future, requiring us to switch to a different model, thus impacting the consistency of our data. While we used state-of-the-art models at the time of writing, future

models may be more capable on these tasks and may require relabeling past data. Beyond GPT-4o and GPT-4o-mini, we did not compare other models such as Claude, Gemini, or Grok, an exercise that could be helpful in understanding how models from different families and of different sizes analyze news articles’ political lean and tone. Our approach also used zero-shot prompts throughout the pipeline and did not attempt to fine-tune any models. We also note that the level of agreement varies considerably between labeling tasks. However, the model-annotator and inter-annotator agreement are statistically indistinguishable in all cases, which shows that some tasks—such as article type, sentence tone, and sentence focus—are inherently more ambiguous than others. This is therefore not a deficiency of the model, and, in fact, when aggregated across hundreds or thousands of articles, these random misclassifications are less likely to have an impact and the overall patterns remain reliable.

A final limitation is one that we have already mentioned: that in the absence of clear, objective ground truth, bias is inherently difficult to measure. In fact, some experts have concluded that the difficulty is so great as to render the detection of content that is misleading on account of bias rather than verifiable falsehoods unsuitable for scientific inquiry (Williams, 2024). Although we are sympathetic to this critique, we remain optimistic that something useful can be accomplished despite these challenges. Specifically, our approach of measuring “elemental” quantities such as topic, tone, lean, and focus is explicitly designed to avoid making judgments about the “true” or desired state of the world, and hence to result in more robust labels. For example, it is possible for raters with different biases to agree that an article is about the economy, has a positive tone, talks more about Democrats than Republicans, and hence is overall favorable to Democrats while maintaining very different views about its accuracy, informativeness, and even whether it should have been published at all.

As previously noted, a downside of this design principle is that we are unable to make direct assessments of article or even publisher bias—a limitation that media critics and political partisans alike may find frustrating. We maintain, however, that any attempt to impose a single notion of truth or objectivity against which bias can be measured will itself inevitably be biased, undermining

the integrity of the project. Moreover, by measuring many of these elemental quantities both at the sentence and article level, and by allowing analysts the flexibility to aggregate them to yet higher levels (e.g., publisher, groups of publishers, the entire collection, etc.), we believe our approach does generate valuable insights about media bias, albeit indirectly. For example, our demonstration that news publishers devote dramatically more coverage to horse race stories than to policy issues does not settle the question of how much coverage ought to have been devoted to each; but it does call into question public declarations by news organizations that their goal is to inform citizens about the most important issues in an upcoming election (Watts and Rothschild, 2017). Likewise, our demonstration that the political lean of headlines is systematically different from the text of the corresponding stories does not settle which one, if either, is correct; but the difference does show that writers of articles and headlines alike exploit their flexibility to cast the same story in different lights. Finally, our demonstration that article lean is negatively correlated with partisan focus does not determine which of our publishers is more or less biased than the others; but it does identify a possible mechanism (attacking the opposition rather than supporting one’s own side) by which all publishers express their bias.

In summary, by integrating systematic data collection, LLM-driven analysis, and human-in-the-loop validation, we have laid a scalable foundation for advancing research on media bias. Although no single study can fully encapsulate the complexities of media influence, and no single dataset can comprehensively define media bias, we believe this work represents a substantial step forward. By continually refining our methodology and expanding our dataset to encompass past and future time periods, we aim to enhance both academic understanding and public awareness of media bias dynamics through an empirical, data-driven approach.

CHAPTER 3

MEASURING POLITICAL REPRESENTATION IN NEWS QUOTATIONS

3.1. Introduction

In democratic societies, the media does not just report on politics but essentially constitutes the arena in which political competition takes place. Past research on agenda-setting has established that while media coverage may not directly determine what citizens think, it does significantly influence what they think about (McCombs and Shaw, 1972; Scheufele and Tewksbury, 2007). When the media devotes sustained attention to an issue, citizens perceive that issue as more important, evaluate political leaders through its lens, and demand governmental action accordingly (Iyengar and Kinder, 2010; Van Aelst et al., 2014). In doing so, the media also determine whose voice shapes public understanding of those issues. By selecting which politicians are given the platform to speak directly through quotation, the pattern of quotation selection operates downstream of both story selection and framing by selectively allocating attention and authority on those topics.

This allocation of voice has concrete consequences. Visibility in the media is a strong predictor of electoral viability, with evidence demonstrating that name recognition alone affects candidate support due to voters' inferences that recognized candidates are more viable (Kam and Zechmeister, 2013; Zajonc, 1968). These findings have been further reinforced by research showing that voters in areas with less press coverage of their congressional representative are less likely to recall the representative's name, evaluate their performance, or engage with their legislative activity (Snyder Jr and Strömberg, 2010). These effects are not just confined to election periods either, as media visibility shapes politician popularity during routine governing as well (Van Remoortere and Vliegthart, 2024). For political figures, being quoted represents a potentially important form of visibility. Unlike mere mentions, direct quotations position a politician as an authoritative voice, providing readers with direct access to a politician's perspective while simultaneously signaling to audiences that this individual's interpretation of events matters.

Who receives more voice on a given issue also interacts with well-established patterns of issue

ownership, which is defined as the tendency of voters to associate certain parties with competence on certain policy domains (Hill and Leighley, 1996; Petrocik et al., 2003). Republicans are traditionally perceived as more competent on crime, immigration, and defense, while Democrats hold advantages on education, the environment, and social welfare (Egan, 2013). Critically, research has shown that candidates receive more favorable coverage when news focuses on issues their party “owns” (Hayes, 2008), suggesting that journalistic source selection may serve as a mechanism through which issue ownership is perpetuated. If journalists disproportionately seek Republican voices on crime or Democratic voices on healthcare, the resulting coverage could play a role in reinforcing the very associations that give rise to issue ownership in the first place.

The political consequences of these dynamics were brought to attention during the 2016 U.S. presidential election, when Donald Trump received an estimated \$5 billion in earned media coverage, more than all other candidates combined (Confessore and Yourish, 2016). Trump’s dominance of the media landscape is considered one of the primary methods by which he was able to set the terms of political discourse by effectively using media attention as a substitute for campaign advertising (Stewart, 2016; Sultan, 2017). Despite the impact of these editorial decisions, the question of whether one party’s voice appears more frequently than another’s in news quotations has not been studied at scale in the U.S. context. Prior work on media bias has focused primarily on aggregate features such as coverage volume, sentiment, or ideological slant (Gentzkow and Shapiro, 2010; Sundar et al., 2025; Budak et al., 2016). Meanwhile, research on quotations specifically has concentrated on building large-scale datasets without connecting them to the political landscape (Vaucher et al., 2021; Čuljak et al., 2025), tracing how individual politicians alone are quoted (Niculae et al., 2015), investigating how the use of quotatives (grammatical devices used to introduce quoted text, e.g. “he said” or “declared X”) has evolved over time (Hu et al., 2023), or studying political quotes in non-U.S. contexts (Lazaridou et al., 2017).

In this chapter, we build on these efforts by conducting a systematic analysis of who, among U.S. politicians, gets to speak—and on which topics—across the full spectrum of mainstream media during a consequential political period in the U.S. To do so, we construct a large-scale dataset of

over 110,000 quotations attributed to 607 U.S. politicians, extracted from over 30,000 news articles published in 2024 by ten major news sources spanning the ideological spectrum. Analyzing this data, we document a consistent but uneven asymmetry, largely concentrated among the 2024 presidential and vice-presidential candidates. Donald Trump accounted for roughly 29% of all quotations in our corpus, and the five presidential and vice-presidential candidates together accounted for 56% of the total. Even after excluding these candidates, the Republican advantage did not completely disappear. Both right- and left-leaning publications were found to quote Republicans more than Democrats. Republican politicians were particularly overrepresented on policy topics central to domestic political discourse, such as crime, immigration, education, and the economy. The only topics on which Democrats received disproportionately more quotations were those tied to the responsibilities of the current administration in 2024, such as international conflict and foreign policy. We further used LLMs to analyze the emotional content of politicians’ quotes, showing that Republican politicians expressed higher levels of anger and fear, while Democrats expressed more unity and calm in their speech. These findings suggest that imbalances in media coverage extend beyond story selection and framing to the level of whose voice is included in news coverage, which has potentially significant implications for issue ownership, public perception, and democratic accountability.

3.2. Methodology

3.2.1. Corpus

We used the Media Bias Detector corpus introduced in Chapter 2, which contains the top-ranking news stories on the homepages of ten major news publishers (Associated Press, Breitbart News, CNN, Fox News, HuffPost, New York Times, The Guardian, USA Today, Wall Street Journal, and The Washington Post). These publishers are among the most widely-read and influential news organizations (Press Gazette, 2025) and represent a diverse range of ideological alignment, and serve as a useful testbed for evaluating political representation broadly. As mentioned earlier, the corpus was constructed by scraping the top 20 articles from each publisher’s homepage five times per day, yielding over 140,000 articles published between January 1 and December 31, 2024.

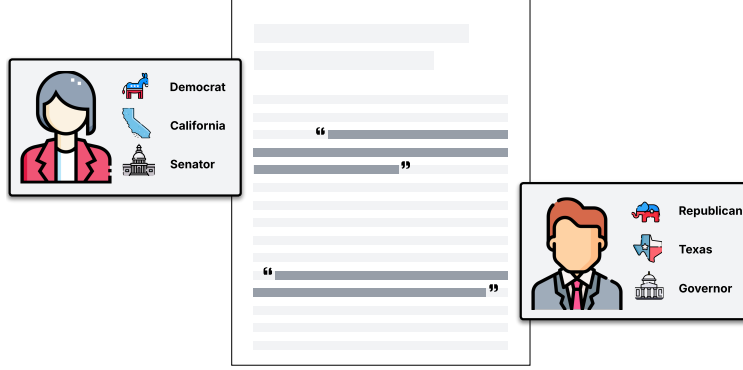


Figure 3.1: **Political quote extraction.** For each quotation in an article, we extract the full quote text, the quoted individual’s name, affiliation, and occupation, along with a classification of their professional domain and the capacity in which they are quoted. We then filter on U.S. politicians and link metadata on their party, office, state, district, age, and gender.

For each article, the Media Bias Detector corpus provides a variety of annotations produced by large language models. Out of the full set of labels discussed in Chapter 2, the ones most relevant to this study are topic and subtopic classifications, and article tone and lean ratings. The data validation results reported in Section 2.2.4 showed a high level of agreement with human annotators, with topic classification Cohen’s κ of 0.65 and tone and lean rating Pearson’s r of 0.77 and 0.8, respectively.

3.2.2. Quote Extraction

We used GPT-4o to extract all direct quotations from each article in the corpus. For each quote, the model was instructed to extract: (i) the full text of the quotation; (ii) the quoted individual’s full name, occupation, and organizational affiliation as reported in the article; (iii) the individual’s domain, classified into one of 13 categories (Politics, Corporate, Academia, Legal, Media and Arts, Science and Technology, Sports, Law Enforcement, Military, Health and Medicine, Religion, Community and Social Services, Other); and (iv) the capacity in which the individual was quoted (expert, observer, participant, subject, commentary, illustrative anecdote, spokesperson, other). The full quote extraction prompt is provided in Figure B.1.

3.2.3. Politician Identification and Matching

From the full set of extracted quotations, we retained only those where the quoted individual’s domain was classified as “Politics.” We then linked these names to a comprehensive roster of U.S.

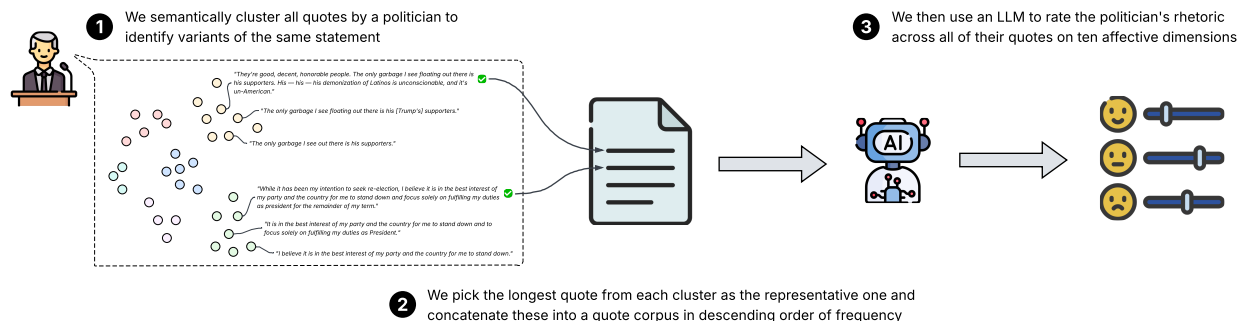


Figure 3.2: **Emotional analysis pipeline.** We obtain embeddings for each quotation using OpenAI’s `text-embedding-3-small` model and construct a similarity network with a cosine threshold of 0.8 to identify clusters of semantically similar quotes. The longest quote from each cluster is selected as the representative instance. Representative quotes are concatenated in order of cluster size to produce a per-politician quote corpus, which is then rated on ten emotional dimensions by an LLM.

politicians who held office at any point during 2024 or were elected on November 5, 2024. This roster was compiled from Wikipedia’s lists of members of the 118th and 119th United States Congress, state governors, and 2024 presidential election candidates, and includes members of the House of Representatives ($n = 442$), Senators ($n = 108$), Governors ($n = 52$), and the five presidential and vice-presidential candidates (Joe Biden, Kamala Harris, Tim Walz, Donald Trump, and J.D. Vance). The counts exceed the standard chamber sizes (435 House seats, 100 Senate seats, 50 governorships) because the roster includes politicians who served at any point during 2024, accounting for mid-term replacements, special elections, and members who resigned or were appointed during the year.

We used fuzzy matching to link individuals from the roster with those in the corpus to account for the variation in how names appear in news article text. Some common sources of variation are the presence or absence of middle names or initials, abbreviated first names, hyphenation, and diacritical marks. For example, the variants *Mario Díaz Balart*, *Mario Díaz-Balart*, *Mario Diaz-Balart*, and *Mario Diaz Balart* were all normalized to the same individual. We used the Levenshtein distance with a threshold of 0.95 and all fuzzy-matched names were manually reviewed to verify correctness and eliminate false positives.

3.2.4. Emotion Analysis

To characterize the affective content of politicians’ quoted speech, we developed a clustering approach that balances representativeness with computational feasibility. Analyzing each quote by each politician individually would omit the broader rhetorical context of their speech, while concatenating all of their quotes together would introduce extensive redundancy as many quotes are repeated multiple times across different articles and would exceed the context limits of current LLMs.

Our approach addresses this through a multi-stage pipeline. First, we obtain vector embeddings for each quotation using OpenAI’s `text-embedding-3-small` model. Then, we construct a similarity network in which quotations are connected if their cosine similarity exceeds 0.8, which yields connected components in the network representing clusters of semantically similar quotations. Third, within each cluster, we select the longest quotation as the representative instance, since it is typically the more complete version of the quote and provides more context for the model to evaluate it. Finally, we concatenate all representative quotations for each politician, ordered from the most frequently repeated (i.e., belonging to the largest cluster) to the least, producing a “quote corpus” that captures both the politician’s most prominent themes and their less frequent statements. For Donald Trump, Joe Biden, and Kamala Harris, whose quote corpus still exceeded the context limit for LLM classification, we truncated the input and ignored quotes that stretched beyond 120,000 tokens.

We gave each politician’s quote corpus to GPT-4o with instructions to rate the overall emotional tone across ten affective dimensions: hope, calm, unity, compassion, motivation, confidence, uncertainty, sadness, fear, and anger (see Figure B.2 for the full prompt). Each dimension was rated on an integer Likert scale from 0 (not present at all) to 10 (very strongly present). These dimensions were selected to span a broad range of emotions relevant to political communication, covering positive valence (hope, calm, unity, compassion), negative valence (anger, fear, sadness, uncertainty), and dimensions associated with mobilization and leadership (motivation, confidence). For stability, we ran each prompt ten times for each individual politician and used the average score to represent

their emotional profile.

3.2.5. Validation

To validate the accuracy of our GPT-4o quote extraction pipeline, we randomly sampled 50 articles and manually identified correctly and incorrectly identified or omitted quotations. GPT-4o extracted 142 quotes from this set of articles, and achieved a precision of 0.944 and recall of 0.947. Our definition of quotations included not only spoken remarks but also text reproduced from written statements or social media posts. Because we imposed no minimum length threshold, the extracted quotations also included directly quoted fragments consisting of only a few words. Attribution errors were rare and occurred primarily for joint statements, which were sometimes attributed to the politician who led or announced the statements rather than to all signatories. Because our two-stage pipeline first identifies individuals classified as politicians and then matches them against our roster of elected officials, we found no cases in which a quotation was incorrectly assigned to a person outside the roster, or vice versa. A sample of the extracted quotes for a subset of articles can be found in B.1.

Throughout our analysis, we treat the total volume of quotations attributed to each party as the primary quantity of interest instead of normalizing by the number of politicians in each party. We make this choice because, from the perspective of a reader encountering political news, the relevant question is how much Republican or Democratic voice they are exposed to in news coverage overall, and not the per-capita rate at which individual politicians are quoted. Additionally, the choice of a per-capita denominator is itself not a neutral one, as it treats all politicians as equivalent, even though senators, governors, House members, and party leaders differ substantially in their constituency size and their national relevance. Other denominators, such as the population represented by each politician, or measures of formal leadership status such as positions on House and Senate committees, would encode different assumptions about what equal representation in the news should mean. Because our focus is on the partisan voice encountered by audiences, we analyze the Republican and Democratic parties as the relevant units of comparison, since they function as collective political entities with comparable institutional standing and roughly balanced representa-

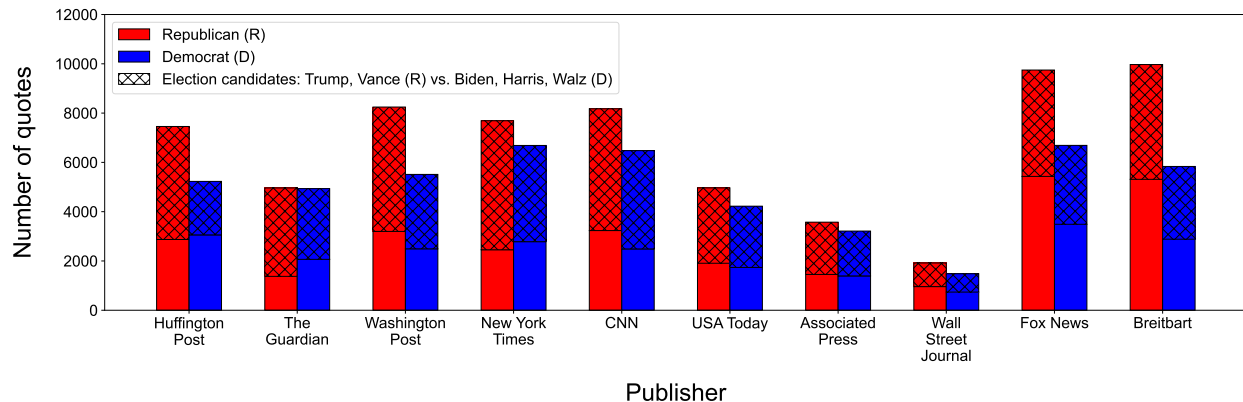


Figure 3.3: **Number of quotes by publisher.** Each set of bars represents the number of quotes attributed to Republican or Democrat politicians for a given publisher, with the shaded bars representing the amount that came from the five Presidential and Vice-presidential candidates in the 2024 Election: Donald Trump, J.D. Vance, Joe Biden, Kamala Harris, and Tim Walz. All publishers quoted Republicans more than Democrats by varying margins, with The Guardian being closest to parity.

tion across the offices included in our roster. The agenda-setting and issue-ownership mechanisms we focus on operate at the party level, since voters form impressions of what Republicans say about immigration or what Democrats say about healthcare from the cumulative voice they encounter in coverage, not from an implicit average across individual politicians. We therefore report raw quotation volumes as our primary measure, while noting in our discussion of individual-level patterns that the asymmetry is not driven solely by a small number of highly quoted figures.

3.3. Results

3.3.1. Quotation Volume Skews Republican Across Publishers

To investigate whether both parties received comparable quotation volume during the 2024 election cycle, we compared the number of quotations attributed to Republican and Democrat politicians across the ten publishers in our corpus (Figure 3.3). We find that Republicans received 66,704 quotations compared to 50,282 quotations for Democrats, a 32% advantage that exceeds what the actual ratio of politicians in our dataset would predict. A substantial fraction of the overall asymmetry was attributed to the five presidential and vice presidential candidates: Donald Trump, J.D. Vance, Joe Biden, Kamala Harris, and Tim Walz, whose quotes collectively accounted for

56% of the total dataset (65,666 quotes). However, even after excluding these candidates, the Republican advantage narrowed but did not disappear, and they still received 21% more quotes than Democrats across all publishers (28,196 vs. 23,124). This asymmetry was found across all publishers, although its magnitude varied considerably. Figure 3.3 shows the number of quotes attributed to individual candidates by each publisher. For some publishers like the Huffington Post and the Guardian, for example, the overall Republican advantage was explained entirely by the asymmetry due to the election candidates. At other publishers like the Washington Post and CNN, however, the asymmetry persisted even after excluding presidential candidates. It is interesting to note that, in all cases, the two Republicans (Trump and Vance) received more quotes than the three Democrats (Biden, Harris, Walz). As expected, Right-leaning outlets exhibited the largest gaps, with Breitbart and Fox News quoting Republicans 70% and 45% more often than Democrats, respectively, but this pattern also persisted at left-leaning and center-left outlets. For example, the Huffington Post quoted Republicans more frequently than Democrats by 43%, which suggests that even ideologically opposed news outlets give significant coverage to Republican voices.

At the individual politician level, we see that Donald Trump dominated news coverage with more quotes than both of his direct opponents, Joe Biden and Kamala Harris, combined (Figure 3.4A), which reflects his persistent centrality in the 2024 U.S. news cycle. Trump remained a focal point of political discourse throughout the year, not only as an election candidate but also in the context of his legal proceedings and his agenda-setting statements as the *de facto* leader of the Republican Party. We also see this effect extend to the rest of the party tickets, with J.D. Vance being quoted twice as much as Tim Walz, despite their comparable political profiles as a Senator and a Governor, respectively. They were both also announced as Vice-presidential candidates only two weeks apart, yet Vance still received twice the number of quotes on the day of the debate than Walz. Overall, we find that Republicans are quoted more than Democrats across multiple levels of office, comprising 29 of the top 50 most quoted politicians (Figure 3.4A inset), which indicates a consistent pattern rather than an artifact of individual prominence.

We also find that quoting both parties together is rare in the media. Across all publishers, only 20.7%

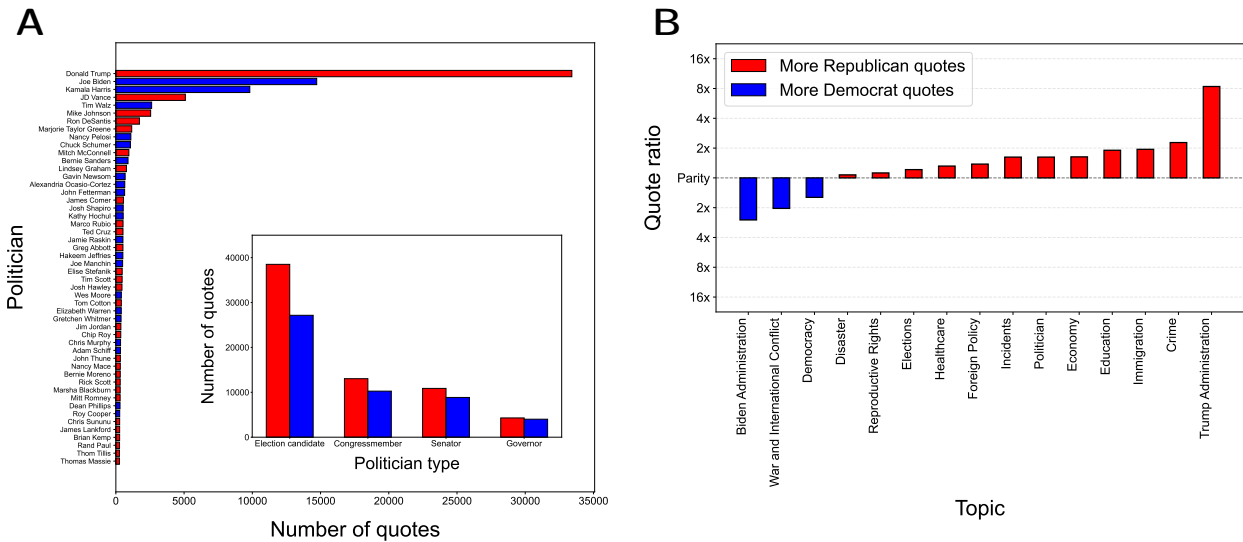


Figure 3.4: **Quotation patterns by politician, office, and news topic.** (A) The main panel shows the total number of quotations attributed to each politician. Among the 50 most quoted politicians, 29 are Republicans, with Donald Trump receiving the most quotes by far. The inset shows the breakdown by level of office. Trump’s advantage is reflected in the election-candidate category, but Republicans in the House of Representatives and the Senate also receive more quotes than Democrats on average. (B) Among the 15 topics with the largest total numbers of quotes, 12 have an overrepresentation of Republican quotes. Republicans are quoted disproportionately more on domestic policy topics such as crime, immigration, and the economy, whereas Democrats are quoted more only on international conflict and democracy.

of articles quote both Republicans and Democrats (Figure B.3), with the majority of articles only providing viewpoints from one side of the aisle. This proportion is lowest for Breitbart, which only quotes both parties in 10% of its articles. Articles that only quote one party are found to generally lean more towards it (Figure B.4), and when right-leaning publishers such as Fox and Breitbart exclusively quote Democrats, their coverage is more negatively framed than when they only quote Republicans (Figure B.5). The opposite is true for center-left and left-leaning publications, where Republican-only articles are more negative on average than bipartisan or Democrat-only ones.

3.3.2. Republican Quotations Are Overrepresented on Domestic Policy Topics

To investigate whether and how much each party dominated the discourse on different topics, we analyzed the ratio of quotes from Republicans and Democrats on the top 15 topics with the highest total number of quotations in our data set (Figure 3.4B). Out of these, Republican voices

were overrepresented in 12 topics (80%). As predicted by issue ownership theory, the Republican advantage was fairly pronounced on topics such as crime and immigration, where Republicans were quoted 2.27x and 1.94x more than Democrats, respectively. However, our results show that this Republican advantage also extended to topics such as education and health care, which voters have historically associated with Democrats (Egan, 2013). This suggests that the higher number of Republican quotes is not confined to topics traditionally associated with Republican competence but extends across policy domains, including those historically associated with Democrats.

The only topics where Democrats received more quotations were about the Biden administration, international conflict, and democracy, which are tied to the responsibility of the incumbent government more than issue ownership per se. Within these topics, the Israel-Gaza War and Biden’s Age subtopics were the two subtopics with the highest representation of Democratic quotes, reflecting the fact that Democratic officials were the primary individuals responsible for commenting on U.S. foreign policy and the president’s fitness. Comparing the Biden administration and Trump administration topics which are direct counterparts of each other, the Democrat advantage on the former was far smaller in magnitude than the Republican advantage on the latter (2.66x vs. 8.38x, respectively), which shows the asymmetry in how the media covers the two parties’ internal matters. In particular, we find that the topic of immigration was a sustained focus of Republicans throughout the year, while crime was primarily talked about in the lead-up to the election. This is in line with studies showing that right-leaning media such as Fox News give disproportionate coverage to crime before elections take place, and their coverage of it drops drastically afterward (Monroe and Savillo, 2022).

These topic-level results have significant implications when we look at them from the perspective of the election cycle. Research has shown that the issues that dominate public discourse during the lead-up to the election are the ones that shape the criteria by which voters evaluate candidates (Petrocik et al., 2003). If voters hear disproportionately more from Republicans on critical domestic issues like crime, immigration, and the economy, then they are likely to associate Republican candidates with competence on these issues, regardless of their actual policy records. Conversely, the lack

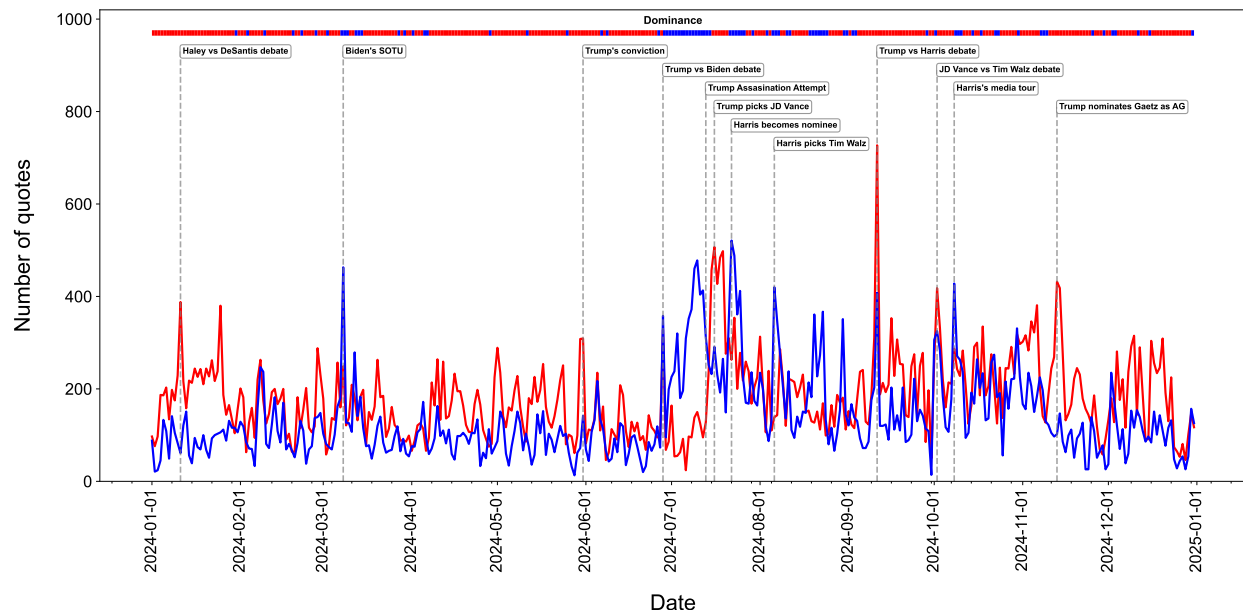


Figure 3.5: **Number of quotes per day by party.** This figure shows the daily number of quotations from Republican and Democratic politicians throughout 2024, with major political events highlighted. Quote volume closely tracked key moments in the election cycle, like Joe Biden’s State of the Union address and the presidential debates. Across the 366 days of the year, 280 saw higher numbers of Republican quotes. As shown in the dominance bar at the top, the first half of the year was characterized by sustained Republican dominance, with only brief intervals of Democratic surges. This pattern shifted after the first presidential debate, when Democrats were quoted more frequently during heightened coverage of intra-party reactions and post-debate commentary leading up to Joe Biden stepping down from the race.

of Democratic voices on these issues may lead to a decline in the party’s perceived competence on these issues that matter most to voters. One possible reason behind these observed patterns could be that Republican politicians have been more effective at engaging with traditionally Democrat-owned issues (Sigelman and Buell Jr, 2004; Sides, 2006), while another explanation could be that these results are a byproduct of the same dynamics driving overall quotation asymmetry where Republican voices are highlighted more regardless of the topic. In either case, the net result is that readers encounter Republican voices frequently on domestic policy topics, while Democratic voices are more prevalent primarily in foreign policy and international conflict news.

3.3.3. Quotation Volume Tracked Election Events

In addition to analyzing the overall asymmetry in quotation volume across publishers and topics, we also investigated how the attention given to each party fluctuated during the course of the 2024 election year.

Figure 3.5 shows the daily volume of quotes attributed to Republicans and Democrats across all publishers. Of the 366 days in 2024, Republicans were quoted more than Democrats on 280 of them (76%), with near-total dominance during the first six months. Democrats only sporadically received more quotes during this period, largely due to specific events, such as Joe Biden’s State of the Union speech in March. While the Republican party produced substantial amounts of political speech due to the Iowa caucuses, the Haley-DeSantis debate, and Trump’s legal proceedings, the Democratic party was less active due to Biden running as an uncontested incumbent.

However, this dynamic changed abruptly in late June, when Biden’s widely criticized performance in the first presidential debate triggered intense deliberation within the Democrat party regarding his future as an election candidate and led to a higher sustained quote volume than Republicans for the first time. This period lasted until the end of August and encompassed Biden stepping down and Kamala Harris becoming the nominee, and her subsequent selection of Tim Walz as the vice-presidential nominee. It’s important to note here that this period of Democratic prominence came only at a time when the party was in disarray and the internal turmoil became a dominant news story. Instead of leading in the news because of its messaging or agenda-setting, the party only came under the spotlight during a time of crisis. After the second presidential debate, the Republican advantage reappeared and persisted through the election to the end of the year.

These trends illustrate the close relationship between political events and media voice allocation, with a consistent Republican advantage that was only temporarily disturbed by major shifts in the election cycle. This pattern suggests that the Republican quotation advantage was persistent throughout the year, and the periods of Democratic prominence coincided with a moment of internal party crisis rather than proactive agenda-setting.

3.3.4. Attention Asymmetry and Concentration at the State level

To examine the geographic distribution of quotation ratios, we map the number of quotes from Republican and Democratic politicians across all 50 states (Figure 3.6). We find that 28 states have more quotes from Republicans, showing that the asymmetry extends to different media markets as well.

In many cases, we find that a small number of politicians dominate their state's quote share despite it being home to multiple members of the House, the Senate, and a Governor. For example, Bernie Sanders, as a senior figure in the Democratic party, accounts for 84% of all quotes from Vermont, while Mike Johnson, as Speaker of the House, accounts for 81% of quotes from Louisiana. These concentrations are highly correlated with the prominence of individual politicians either due to their leadership positions, membership on committees, or salience in national political discourse. We also see cases where the most quoted politicians from each state belong to the state's minority party, like Lauren Boebert in Colorado and Roy Cooper in North Carolina, both of which states lean towards the opposing party. A related pattern emerges in states where one party is represented by only a single politician, such as Joe Manchin in West Virginia or Susan Collins in Maine. Here, these politicians channel the voice of their entire party in that state and are the only sources for journalists seeking a cross-partisan perspective.

These patterns show that media attention at the state level is fairly concentrated on a small set of key politicians among many elected officials. These individuals play a disproportionate role in shaping political discourse in their state, and any analysis of that state's role in the national political discourse should pay special attention to how these figures influence agenda-setting and the framing of political issues, often shaping how their state is represented in the media.

3.3.5. Both Parties' Speech Exhibits Different Emotional Profiles

The previous sections analyzed the volume of quotations from either party on a variety of dimensions. We now turn to the content of those quotations and investigate whether the two parties differed in the emotional content of their speech.

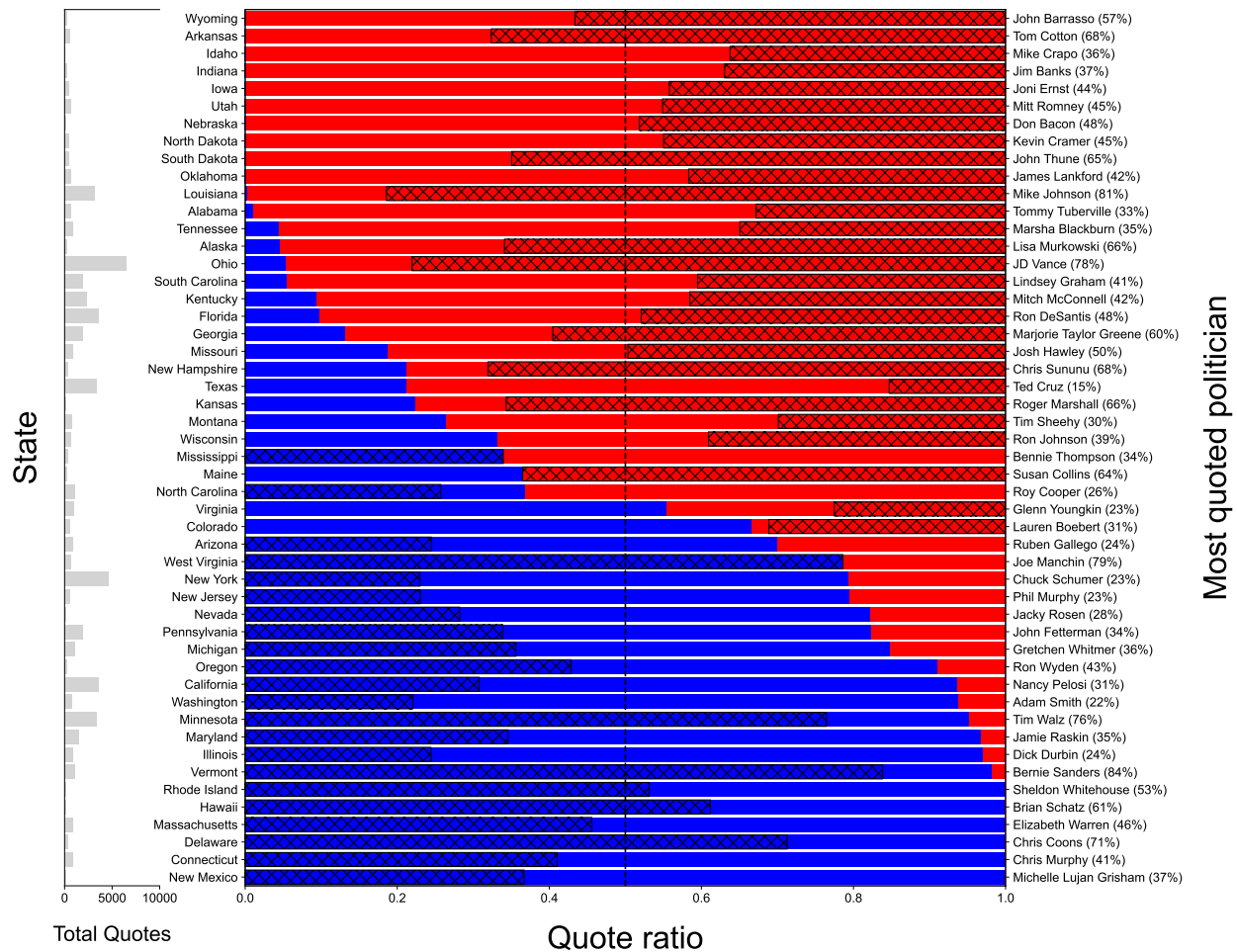


Figure 3.6: **Partisan quote ratio by state.** At the state level, 28 of 50 states show greater media representation of Republican voices. The total number of quotes originating from each state generally reflects the prominence of major politicians based there. For example, J.D. Vance from Ohio and Tim Walz from Minnesota contribute disproportionately to the quote counts of their respective states. The shaded regions indicate the share of all quotes in each state attributed to that state’s most quoted politician. In several cases, a single figure dominates state-level representation, e.g. Bernie Sanders for Democrats in Vermont and Mike Johnson for Republicans in Louisiana.

Figure 3.7 shows the resulting profiles for the two parties obtained by averaging over all politicians in our dataset with at least 50 quotes during 2024 (100 Republicans, 98 Democrats). We observe distinct profiles in affective register of the two parties. The clearest partisan differences emerged on the five dimensions where the 95% confidence intervals did not overlap between the two parties. The largest gap was on compassion, where Democrats scored substantially higher than Republicans ($\Delta = 1.93$; Democrats: $M = 5.39$, 95% CI [5.16, 5.63]; Republicans: $M = 3.46$, 95% CI [3.26, 3.66]).

Democrats also scored significantly higher on unity ($\Delta = 1.44$; $M_D = 5.08$ [4.82, 5.33] vs. $M_R = 3.64$ [3.40, 3.88]), hope ($\Delta = 1.13$; $M_D = 5.59$ [5.37, 5.82] vs. $M_R = 4.46$ [4.24, 4.68]), calm ($\Delta = 0.99$; $M_D = 3.36$ [3.17, 3.55] vs. $M_R = 2.37$ [2.19, 2.55]), and motivation ($M_D = 6.34$ [6.16, 6.51] vs. $M_R = 5.86$ [5.71, 6.00]). Taken together, these dimensions describe a rhetorical style oriented toward inclusion, reassurance, and collective purpose. On the other hand, Republicans scored markedly higher on anger ($\Delta = 0.74$; $M_R = 6.80$ [6.55, 7.05] vs. $M_D = 6.06$ [5.82, 6.30]) and fear ($\Delta = 0.42$; $M_R = 5.79$ [5.59, 5.98] vs. $M_D = 5.37$ [5.18, 5.55]). The Republican score of 6.80 was the highest score on any dimension between the two parties, indicating that anger was a defining feature of Republicans' speech.

While the overall party-level averages show clear partisan divides, breaking these down at the individual politician level also shows that there is substantial variation within each party. Figure B.7 shows the emotional profiles of the top 30 most quoted politicians, and we can see that the most extreme emotional profiles belong to politicians whose rhetorical style is well known, such as Marjorie Taylor Greene (R-GA), Josh Hawley (R-MO), and Tom Cotton (R-AR), all of whom score highly on the anger dimension.

These emotional profiles are consistent with a broader pattern documented in prior research on affective political communication. Anger and fear are often associated with heightened attention to political information, stronger in-group identification, and increased motivation for participation, while unity and compassion tend to be more effective for persuading undecided audiences and sustaining broad coalitions (Brader, 2006; Valentino et al., 2011; Marcus et al., 2000). The asymmetry we observe is therefore consistent with the two parties pursuing distinct rhetorical strategies in their quoted speech, with Republican politicians' language emphasizing anger and fear and Democratic politicians' language emphasizing unity and compassion. Whether these differences reflect deliberate messaging strategy, the selection effects of journalists choosing what to quote from whom, or some combination of the two cannot be determined from our data alone. However, the alignment between the emotional profile of Republican speech and the broader negativity bias documented in news selection (Soroka, 2014) suggests that these two factors may be connected, with emotionally

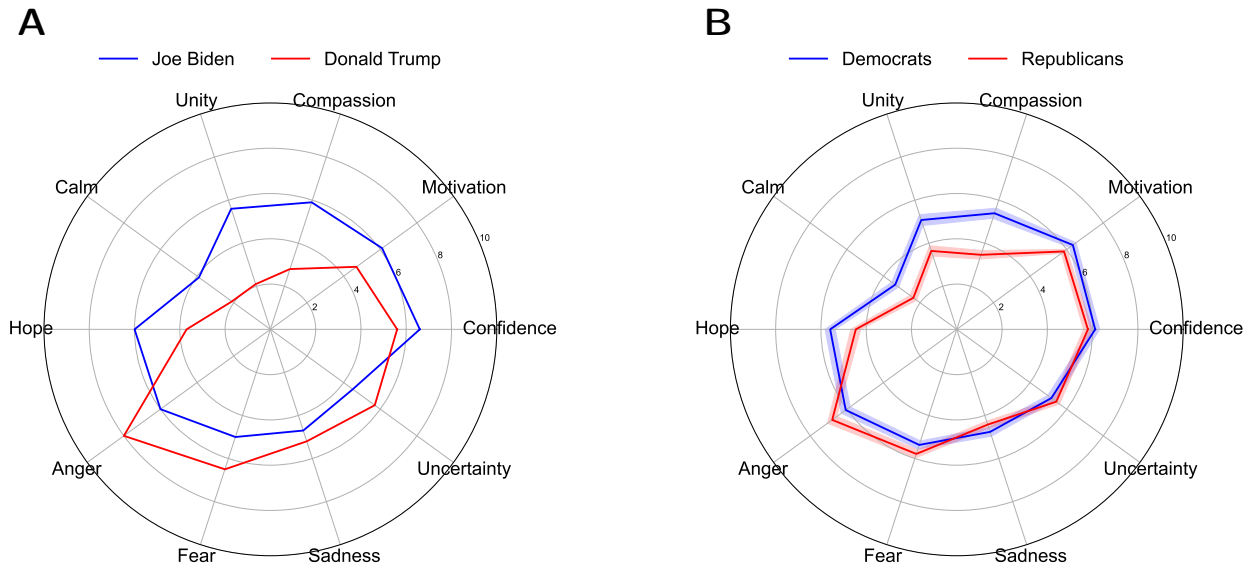


Figure 3.7: **Average emotional profiles.** (A) Joe Biden exhibited stronger expressions of hope, calm, unity, compassion, and motivation, while Donald Trump displayed higher levels of anger, fear, and uncertainty. (B) The same patterns extend to the rest of the parties. Averaging across all politicians, the non-overlapping confidence intervals indicate that these differences are statistically significant, highlighting the distinctiveness of each party’s communication style.

intense rhetoric attracting more attention, thereby shaping the tone and sentiment of political news that reaches readers.

3.4. Discussion

In this chapter, we found that Republican politicians received more quotation coverage than Democrats across mainstream U.S. news publishers in 2024. While part of this gap reflects Donald Trump’s exceptional dominance of the 2024 news cycle, a meaningful asymmetry persisted after accounting for the presidential and vice-presidential candidates, and across publishers and policy domains.

There could be several explanations behind this phenomenon. The first could be on the supply side, with Republican politicians simply producing more quotable speech. Donald Trump has been notably effective at generating statements that attract media attention, and his dominance of earned media in both 2016 and 2024 (Confessore and Yourish, 2016; Grynbaum and Koblin, 2024) illustrates how a politician who is willing to make provocative statements can command disproportionate coverage. Our emotional analysis is in line with this possibility, and the higher levels of anger and fear that characterized Republican politicians’ speech could have made them more quotable

for the media in an information environment where negativity drives engagement (Soroka, 2014; Soroka et al., 2019). Statements that are emotionally charged may be considered more newsworthy and thus more likely to be selected by outlets across the ideological spectrum (Harcup and O’neill, 2017).

A second reason could be on the demand side, where structural incentives to maximize audience engagement might influence which soundbites journalists choose to highlight. In this context, quotations serve a dual purpose, as they appear to let politicians speak for themselves while simultaneously enabling journalists to introduce a dramatic, conflictual element to their writing. If Republican politicians produce speech that is more emotionally charged, then the observed asymmetry in our findings may reflect editorial choices to attract audience attention. The party that more frequently generates attention-seeking speech will thus naturally occupy more of the media discourse (Mazzoleni and Schulz, 1999; Strömbäck, 2008).

Regardless of its origin, the asymmetry we observe has substantial implications for democratic representation. Research has shown that media visibility can be a critical factor in politics. Name recognition is linked to candidates’ election viability Kam and Zechmeister (2013) and media coverage has been found to influence voters’ engagement with their elected officials Snyder Jr and Strömberg (2010). Furthermore, how the media covers certain issues informs how citizens evaluate political leaders Iyengar and Simon (1993). Given all of these factors, when one party’s voices are consistently more present in news coverage, the resulting effect is that citizens hear more from that party and begin to evaluate critical issues through that party’s framing. This is how agenda-setting McCombs and Shaw (1972) operates at the level of voice rather than topic, and this dynamic is particularly consequential on critical domestic policy topics, where swing voters who encounter Republican voices more frequently may start to see them as more authoritative, while the absence of Democratic voices could lead to an erosion of the party’s perceived relevance on topics that are critical in every election.

Our work has several limitations. First, our dataset covers a single, highly-charged election year, and it remains an open question whether these results hold for prior elections and non-election periods.

The unusually large presence of Donald Trump in the media in 2024 for a variety of reasons may make this an outlier year. Extending this analysis to historical data, such as the 2016 and 2020 elections, as well as 2012 and 2008 for pre-Trump baselines, would clarify whether our findings are a structural feature of news production in the U.S. or a one-off. Additionally, while the ten publishers we study cover the ideological spectrum, this analysis could be extended to more outlets to get a broader sense of political representation in the media.

Another limitation of our work is that we cannot reliably determine whether the asymmetry we observe is due to deliberate media bias or is caused by other factors. If Republican politicians are more media-savvy and organize their messaging to make headlines, then the over-representation of their voices may not be a fault of the media. Understanding which of these factors is the cause behind the results we see would require measuring the volume and characteristics of all political speech in the U.S. and comparing this with how often they are quoted in the media.

To conclude, our analysis reveals that, on aggregate, Republicans consistently received more quotes during the 2024 presidential election than Democrats, with Donald Trump being the single largest contributor to this asymmetry. While it is hard to say for certain whether this reflects the greater “quotability” of Republican speech or editorial decisions on the part of the media, the result is a news landscape in which one party’s perspectives are more audible than the other’s. In a democracy that depends on citizens encountering a representative range of political voices, this asymmetry warrants sustained empirical attention and scrutiny.

CHAPTER 4

INEQUALITY IN EXPERT SOURCING

4.1. Introduction

In the previous chapter, I studied how the media platform politicians by selecting whom to quote in news coverage. Although politicians are often the central actors in political news, readers' understanding of their actions and current events is heavily shaped by third-party sources who are invited to interpret them. These sources are brought in by journalists to lend impartiality and objectivity to their reporting, and their role is to provide context, evaluate claims, and explain policy consequences. These individuals are termed "primary definers" (Hall et al., 1978), and research has shown that, in a push for credibility and authority in sourcing, journalists tend to over-index on individuals who occupy powerful and privileged institutional positions (Gans, 2004; Kleemans et al., 2017).

While these expert sources are helpful for providing technical rigor and supplying context that generalist news reporters may lack, the tendency of reporters to resort to a small set of trusted commentators leads to a monoculture where the same handful of individuals and institutions are repeatedly platformed as authoritative voices on a wide range of issues, while other perspectives are sidelined. This mechanism of news production results in reproducing the already problematic power structure in society which elevates elite institutions above the general public. This issue also extends to social media, where political elites and journalists have been found to propagate prestigious think tanks' framing of economic news (Anstead and Chadwick, 2018). Another concern in newsrooms is that of demographic representation. Women are consistently underrepresented in the media despite their increasing participation in other facets of public life (Shor et al., 2015), and this extends to citations in science journalism as well (Arabi et al., 2025). The Global Media Monitoring Project has documented a similar imbalance across a broad set of news organizations covering over 160 countries (Macharia, 2020).

However, most previous work has focused on a single outlet, a single topic, or a single demographic

dimension. The current literature lacks a cross-sectional picture that simultaneously characterizes which institutions are platformed, which individuals within these institutions are quoted most often, and how these patterns differ across the ideological spectrum. Furthermore, elitism in academia has been shown to drive systematic inequality and hierarchy in faculty hiring (Clauset et al., 2015; Wapman et al., 2022), and a natural question arises as to whether a similar disparity is also seen in media representation.

In this chapter, we tackle this question by analyzing over 56,000 quotes attributed to academics affiliated with over 1,000 institutions, extracted from over 17,000 news articles published by 21 news sources that span the political spectrum. Using this data, we study academic concentration and elitism in journalistic sourcing and analyze how these trends vary across news topics, experts’ domains, publishers, and universities.

4.2. Methodology

We extend the Media Bias Detector corpus presented in Chapter 3 to 2025 and expand the coverage to 21 publishers (Associated Press, Breitbart News, CNN, Fox News, HuffPost, The Guardian, The New York Times, The Wall Street Journal, The Washington Post, USA Today, Los Angeles Times, Chicago Tribune, Star Tribune, BBC, Financial Times, Reuters, Bloomberg, CNBC, New York Post, Newsmax, and The Daily Wire). As described before, all articles are labeled for their news topic and subtopic, and article lean, tone, and type. We use the quote extraction framework described in Chapter 2 to extract the following data for each quote: i) the full text of the quotation; (ii) the quoted individual’s full name, occupation, and organizational affiliation as reported in the article; (iii) the individual’s domain, classified into one of 13 categories (Politics, Corporate, Academia, Legal, Media and Arts, Science and Technology, Sports, Law Enforcement, Military, Health and Medicine, Religion, Community and Social Services, Other); and (iv) the capacity in which the individual was quoted (expert, observer, participant, subject, commentary, illustrative anecdote, spokesperson, other). See Figure B.1 for the prompt.

From this dataset, we keep all quotes where the speaker’s affiliation either contains one of the keywords “university”, “college”, “school”, “institute”, “center”, or “lab”, or the common abbreviation

for universities ranked by U.S. News (e.g. “MIT”, “UChicago”, etc.). Using GPT-4o-mini and the prompt in Figure C.1, we resolve variations in references to universities or any particular departments or schools and map them to the official university name with a consistent format, then use the Wikipedia API to search for these institutions and use the official name stated on their Wikipedia pages (e.g. “Harvard Medical School” → “Harvard University”, “UCLA” → “University of California, Los Angeles”, etc.). We also strip titles and post-nominals such as “Dr.” and “M.D.” from individuals’ names. For each quoted individual, we perform another round of classification using GPT-4-mini to determine whether they are being quoted for their expert commentary or because they or their institution is involved in the story for other reasons, such as controversies or lawsuits (see Figure C.2 for the prompt). To exclude coverage of varsity sports, we drop articles that have been placed under the “Sports” category as per the Media Bias Detector labels. To further remove any remaining quotes by students, we remove all instances where the individual’s occupation is listed as “student”, “undergrad”, “freshman”, “sophomore”, or “junior”. We also perform fuzzy matching between the names of the top 1000 most frequently quoted individuals and resolve variations to map them to the same person (e.g. “Stephen” vs. “Steve”). Our final dataset covering 18,223 articles contains 57,905 quotations attributed to 15,123 individuals associated with 1,611 institutions. In the following sections, we present results from these data.

4.3. Results

4.3.1. Expert sourcing across publishers

We first study how frequently different publishers rely on university scholars as expert sources. Figure 4.1 shows the percentage of quotations attributed to academics for each outlet. Across publishers, academic experts account for an average of 5.68% of all quotations, indicating that their voices are relatively uncommon in the news in comparison with those of politicians.

The prevalence of academic expert sourcing varies substantially by news topic and publisher. Left-leaning and centrist publications devote a larger share of their quotations to university experts (with the exception of two regional newspapers: the Chicago Tribune and Minnesota Star Tribune). Among financial media, Financial Times, Bloomberg, and CNBC quote fewer experts than the

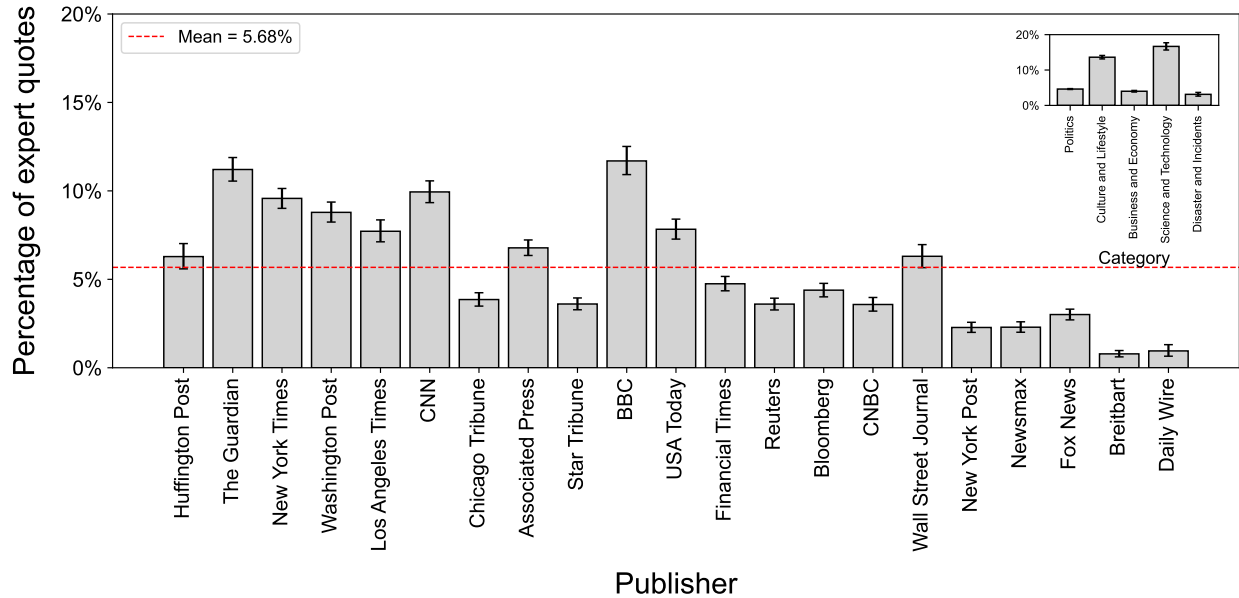


Figure 4.1: **Expert sourcing by publisher.** Percentage of all quotations attributed to university scholars quoted in an expert capacity by each publisher. Error bars indicate 95% confidence intervals estimated by bootstrapping articles within each publisher. On average, publishers quote academic scholars 5.68% of the time in their coverage, with right-leaning publishers consistently having the lowest rate of expert quotations.

global average while the Wall Street Journal quotes more. Figure 4.1 inset shows that the expert sourcing ratio for business and economics news is substantially lower than culture and lifestyle and science and technology across the entire corpus. Finally, we find that right-leaning media such as the New York Post, Newsmax, Fox News, Breitbart, and Daily Wire consistently quote the fewest academic scholars in their reporting. Breitbart has the lowest ratio across all publishers, quoting university professors only 0.79% of the time.

These patterns suggest that the share of academic voices in the news might be a function of each publisher’s editorial focus and ideological orientation. Outlets that quote scholars more frequently may contain more specialized knowledge and contextual analysis of current affairs provided by professors with expertise on these topics. On the other hand, lower rates of academic sourcing might indicate greater reliance on politicians, advocates, first-person accounts, or independent experts to help analyze and frame the events for readers. While we do not make normative judgments about whether greater reliance on academic experts constitutes a better form of journalism, these results

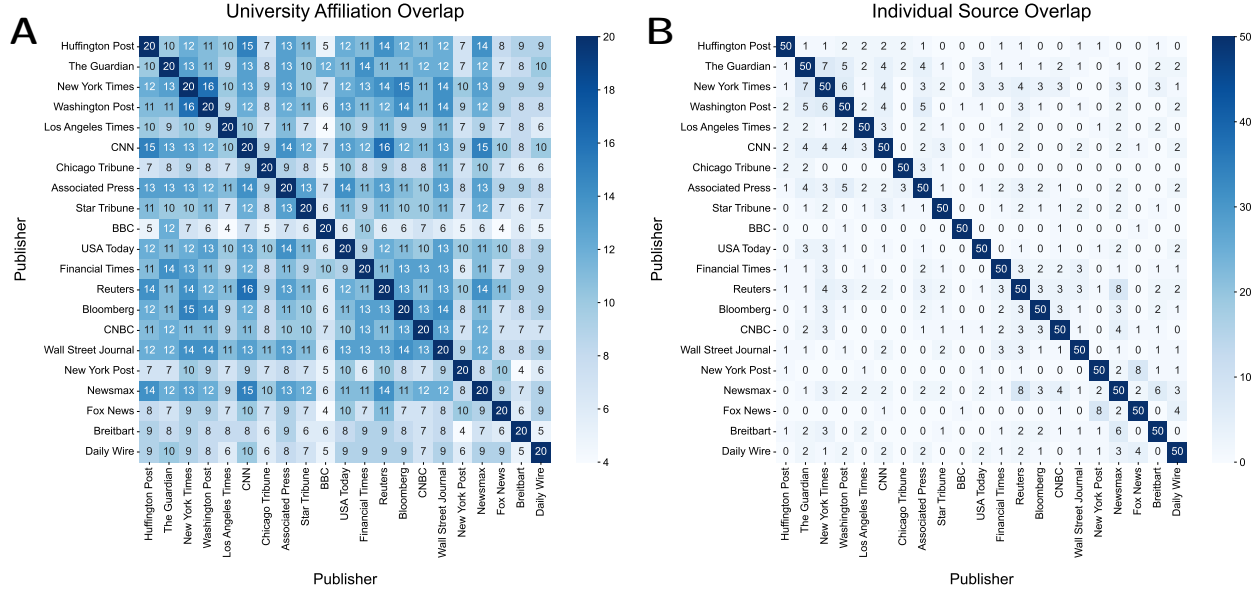


Figure 4.2: **Source overlap at the university and individual level.** Panel **A** shows how many of the top 20 most quoted institutions are common between the 21 publishers, while Panel **B** shows how many of the top 50 most quoted individuals are common between them. There is high overlap at the institution-level but a low overlap at the individual level. Although most publishers quote individuals affiliated with the same prestigious universities, the specific individuals whom they quote within those universities appears to vary widely.

reveal interesting variation in the sourcing practices of different publishers.

4.3.2. Source overlap at the institution and individual level

We next seek to understand whether ideologically dissimilar publishers draw on the same set of academic sources for their expert opinion. For each of the 21 publishers, we compute the set of 20 universities it quotes most often, and then compute the pairwise intersection of these sets between every pair of publishers. Figure 4.2A shows the resulting heatmap. The average overlap between publishers calculated using the off-diagonal values is $9.81/20$, or roughly 50%. This overlap is highest among outlets that are ideologically aligned or have a similar editorial focus. For example, Bloomberg, CNBC, the Wall Street Journal, and the Financial Times all have an overlap of 11-13 with each other since they primarily cover business and economy. BBC, being based in the United Kingdom, has a low average overlap of 6.25 with all publishers except the Guardian, the only other British publisher in our data, with which it has an overlap of 12.

A similar geographic trend is also seen within U.S.-based publishers, with Chicago Tribune and the Los Angeles Times having a lower overlap with the rest of the publishers due to a stronger local focus in their reporting. Finally, we observe that most right-leaning outlets, such as Fox News, the New York Post, Breitbart, and the Daily Wire consistently have lower overlaps with the other publishers in our set. The only exception to this is Newsmax, which, despite being a right-leaning outlet, has a higher overlap with other publishers due in part to its use of substantially more wire service content than the other publishers.

These patterns show, in general, a substantial fraction of universities that appear at the top of any one publisher's list also appear at the top of other mainstream ones. For example, Harvard is the most frequently quoted institution for 12 out of the 21 publishers we study. This is exactly the structure predicted by the theory of primary definers (Hall et al., 1978), where a stable set of institutional voices are platformed by the media across a wide variety of topics.

However, we find that this convergence of sources disappears at the individual level. For each pair of publishers, we also computed the number of overlapping individuals within their respective top 50 most frequently quoted experts (Figure 4.2B). The average overlap between publishers here is only $1.25/50$ or 2.5%, a substantial drop from the 50% average overlap at the institution level. No two publishers share more than 8 sources in common. The New York Times and The Washington post have the highest individual-level overlap of 14%, compared to the highest institution-level overlap of 80% between CNN and USA Today. The contrast between panels A and B in Figure 4.2 shows that, while all publishers frequently seek out expert commentary from the same elite universities, the specific individuals they choose to quote varies widely.

One explanation behind this is that each university houses hundreds of academics in every domain, so two journalists seeking an expert opinion on the same event may reach out to different scholars based on their preference. However, another reason for this divergence could be the routines and time constraints of news production. Due to deadline pressure, journalists tend to rely on familiar and accessible sources whom they have worked with in the past (Gans, 2004). This leads to a pattern where some journalists will have their go-to sources for certain topics and will reach out to

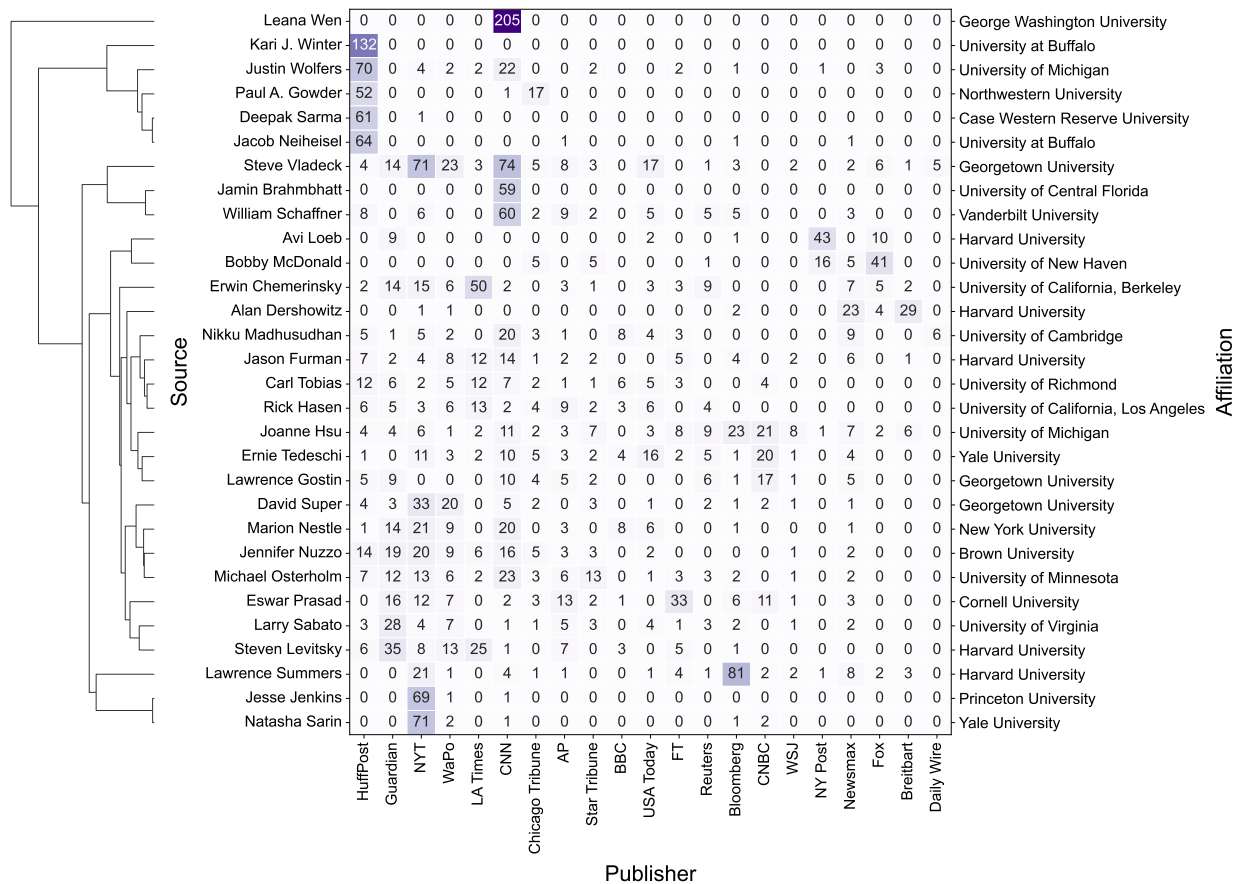


Figure 4.3: **Most frequently quoted individuals.** This heatmap shows the distribution of quotes attributed to each of the top 30 most prolific individuals by different publishers. The dendrogram on the left depicts agglomerative clusters based on individuals' quote distribution similarity across publishers. We see that some, like Joanne Hsu (U. of Michigan) are quoted broadly across the media, while others like Leana Wen (GWU) are quoted exclusively by a single outlet.

them repeatedly in their reporting.

We find some evidence of this when analyzing the quote distribution of the most prolific scholars in our data. Figure 4.3 shows the top 30 most frequently quoted experts, their institutional affiliation, and how many times they were quoted by each publication. We find that the most prolific experts are not necessarily quoted equally across the board, but can often be concentrated in a handful of publishers. For example, the most frequently quoted expert overall is Steve Vladeck, a law professor at Georgetown University, who received 242 quotes across 17 outlets. But the second-most frequently quoted expert was public health professor Leana Wen of George Washington University,

who was quoted 205 times, exclusively by CNN where she also serves as a medical analyst for the network. We further perform agglomerative clustering of these individuals based on their publisher distribution and visualize the hierarchical clusters in the dendrogram on the left of Figure 4.3. This allows us to find groups of experts who are frequently cited by the publishers at the same rates. For example, Justin Wolfers (U. of Michigan), Jacob Neiheisel (University at Buffalo), and Paul A. Gowder (Northwestern University) are all frequently quoted by CNN for their expertise on the economy, politics, and law, respectively.

4.3.3. Elitism in expert sourcing

Now that we know how often publishers quote academics and which individuals they platform most often, we examine the distribution of attention at the institution level. Across our corpus, we find that expert quotations are heavily concentrated among a small set of prestigious universities. Figure 4.4 shows the top 20 most frequently quoted institutions and the capacities in which their affiliates are quoted. Together, these universities account for 37.2% of all quotations in our corpus (21,541 out of 57,905), highlighting the outsized presence of elite, predominantly private institutions in media discourse (see Figure 4.5 for the cumulative distribution across all institutions). The long tail of the remaining 1,591 institutions contains many that are only quoted in a handful of articles during 2025. Attention is also skewed within this select group, as Harvard receives the most expert quotations in the media by far (2,924). Columbia, the second-most quoted institution, receives substantially fewer expert quotations (1,667) and is much closer in prominence to the other institutions in the top five (NYU, Stanford, and Georgetown). Out of these 20, only seven are public institutions (three of which are from the United Kingdom), with UCLA leading in that category with 1,079 quotations. Figure C.4 shows the breakdown of these quotes by news category of the articles.

Besides being quoted for their expertise, Harvard and Columbia also receive substantial media attention because they or their affiliates are themselves the subjects of news coverage. Approximately one-third of all quotations attributed to these institutions appear in articles focused on the university or the quoted individual (33.9% for Harvard and 37.4% for Columbia). The major themes in this coverage were these universities' legal and financial battles with the Trump administration, stu-

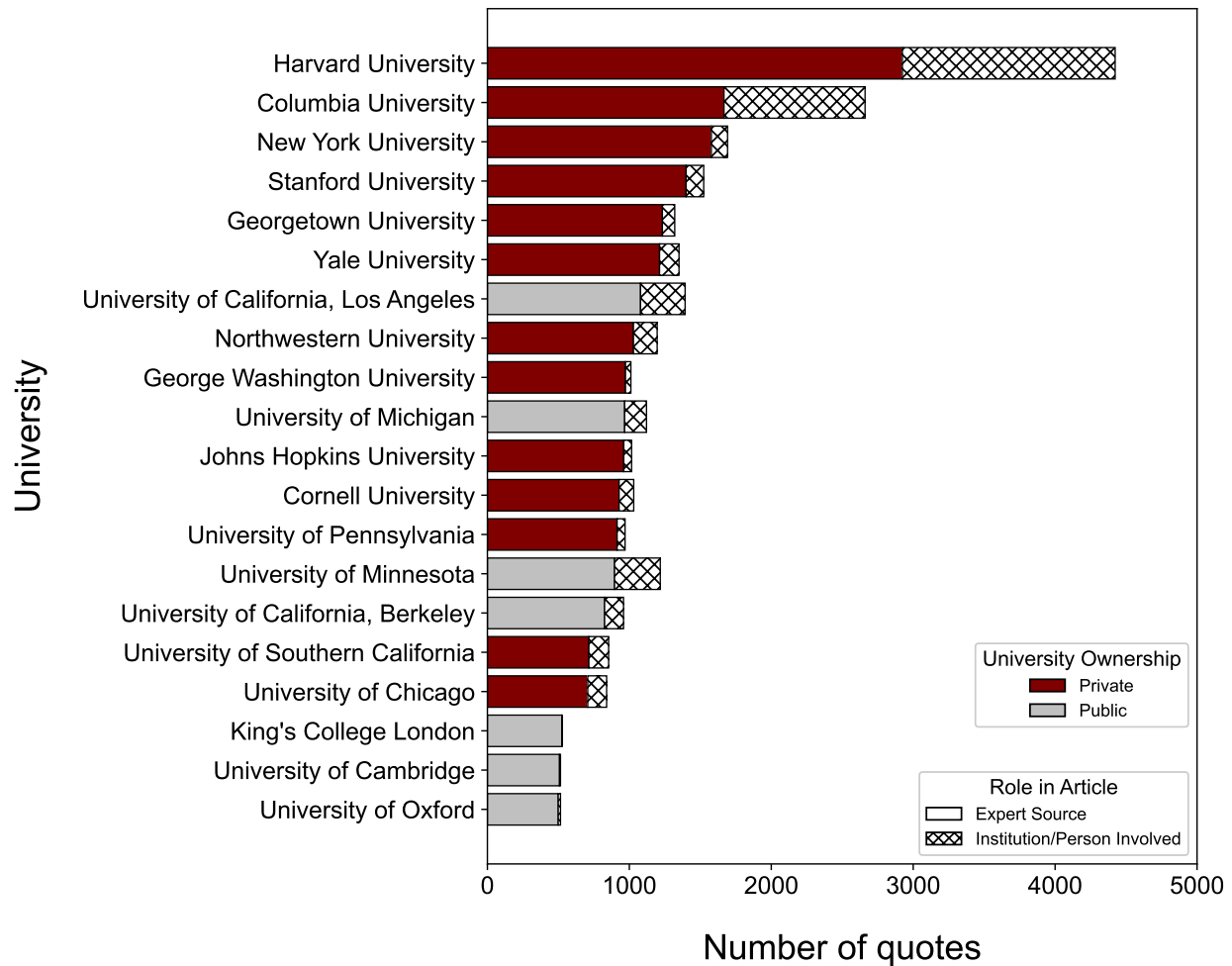


Figure 4.4: **Most frequently quoted universities.** Each bar shows the total number of quotes attributed to individuals associated with that particular institution, with colors denoting public and private universities. The solid bar segments represent the quotes where the individual is being quoted in their expert capacity as a scientist or scholar, while the cross-hatched segments represent instances where either they or their university are the subject of the story itself.

dent protests, and leadership changes. A similar, although less pronounced, pattern appears for two public universities, UCLA and the University of Minnesota (UMN), where subject-focused coverage accounts for roughly one-fourth of the total quotations (22.5% for UCLA and 26.4% for UMN). It's interesting to see that British universities, such as King's College London, Cambridge, and Oxford are rarely ever the subject of news articles in U.S. media, and their affiliates are primarily quoted in an expert capacity.

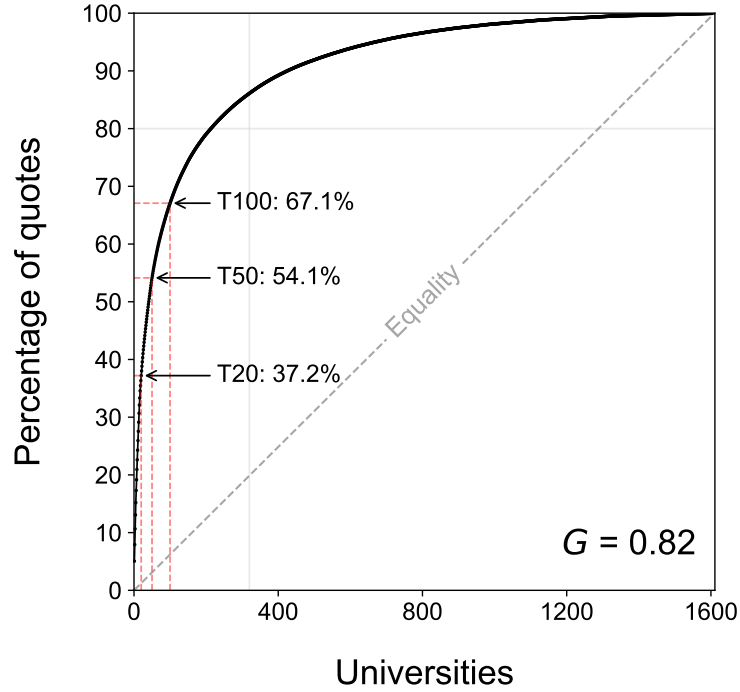


Figure 4.5: **Concentration of university expert quotations.** Out of more than 1,600 institutions, the top 20 universities account for more than a third (37%) of all quotes in our corpus, while the top 100 universities account for two-thirds of all quotes (67%). The gini coefficient of 0.82 shows that there is substantial inequality in quotation distribution.

This concentration in media attention has implications beyond the visibility of individual universities. By repeatedly turning to a small set of elite institutions, news organizations help reinforce existing hierarchies of prestige and authority, giving their readers the impression that the scholarship produced in these universities or the views of their faculty are especially credible and worthy of attention. These patterns may disadvantage researchers at less prominent institutions and narrow the range of perspectives that readers are exposed to.

4.3.4. Gender disparities in expert sourcing

Finally, we examine the gender disparities in expert quotations. Following Danús et al. (2026), we use the `genderize.io` API to determine an individual’s gender based on name. To distinguish between different stages of expert sourcing, we decompose the overall gender ratio into three components: the ratio of distinct male to female scholars quoted, the ratio of quotations per male scholar to the ratio of quotations per female scholar, and the ratio of words per male scholar quotation to

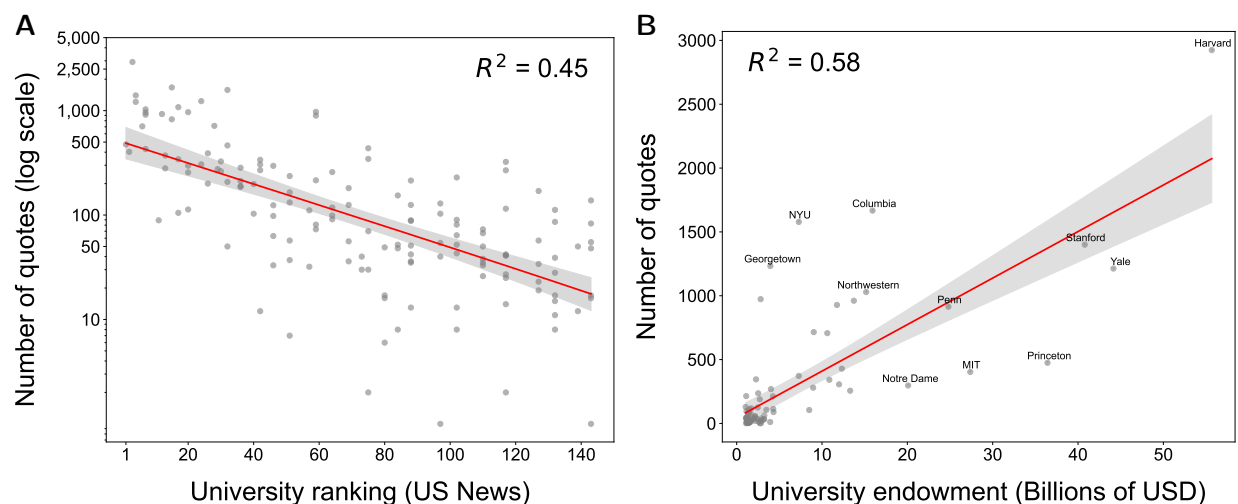


Figure 4.6: **Correlation of media attention with institutional prestige.** We observe moderate correlation between the number of quotes and the institution's ranking as per US News ($R^2 = 0.45$) and its endowment ($R^2 = 0.58$) for private institutions.

words per female scholar quotation. Through these three metrics, we aim to disentangle disparities in source selection at the initial stage of choosing whom to quote from disparities in the amount of attention allocated after that choice has been made. Figure 4.7 shows the ratio of male to female scholars for each publisher using these three metrics.

We find that the largest and most consistent gap occurs at the source-selection stage. The average gender ratio across all publishers is 2.51, meaning that outlets quote more than twice as many male scholars as female scholars. Every publisher has a ratio higher than 1, ranging from 1.61 at the BBC and 1.68 at the Guardian to 3.58 at Breitbart and 5.91 at the Daily Wire. All confidence intervals remain above one, showing that this underrepresentation of women is significant and consistent across all publishers.

However, once the source selection has been made, there is no further disparity in how much attention they are afforded. The average ratio of quotations per person, conditional on being selected as a source, is 1.08, although this is driven largely by higher ratios at Fox News (1.38) and the Daily Wire (1.89). At the Daily Wire, the wide confidence intervals [0.97, 3.55] are due to the small number of articles containing expert quotes. For all other publishers, the confidence intervals overlap with one, representing gender parity.

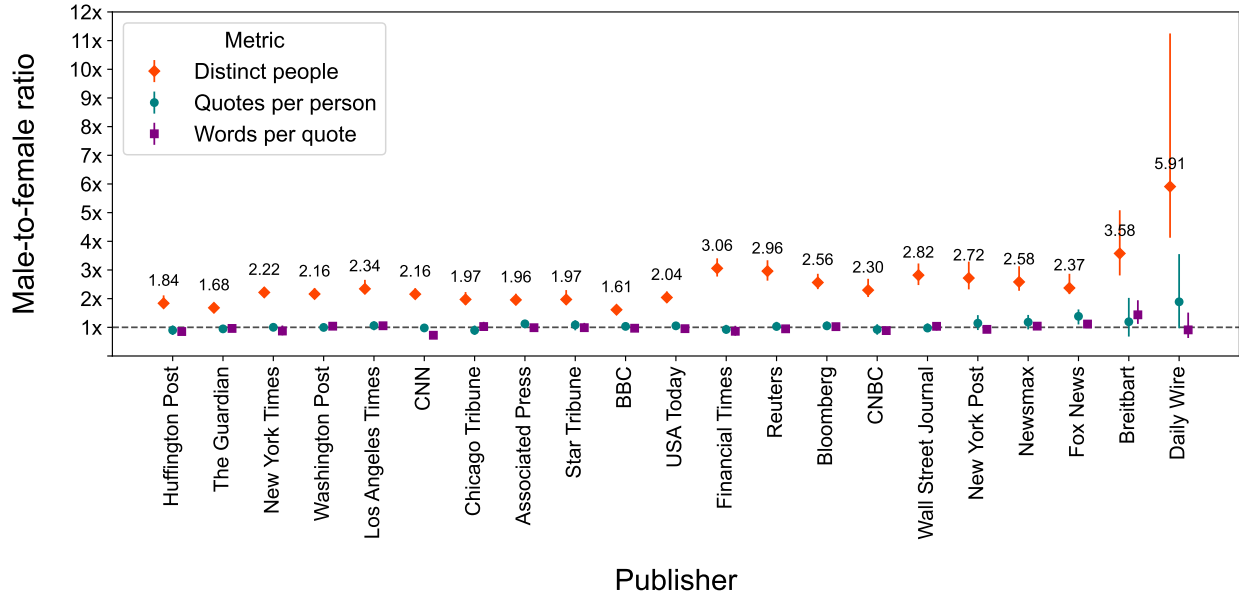


Figure 4.7: **Gender disparities in expert quotations by publisher.** Points show the male-to-female ratios for three distinct dimensions of expert quotation: the number of distinct individuals quoted, the average number of quotations per person, and the average number of words per quotation. Error bars indicate 95% confidence intervals obtained from bootstrapping at the article level. A ratio of 1 indicates gender parity while values above 1 indicate overrepresentation of men.

Similarly, we find no evidence that men receive longer quotations than women. The average male-to-female ratio of words per quotation is close to parity at 0.98, where the only significant confidence intervals are for HuffPost ($M = 0.85$, 95% CI [0.74, 0.97]), CNN ($M = 0.72$, 95% CI [0.64, 0.82]), Fox News ($M = 1.11$, 95% CI [1.03, 1.21]), and Breitbart ($M = 1.44$, 95% CI [1.12, 1.94]). These differences are much smaller in comparison to those in the number of people selected for quotations in the first place.

The ratio of distinct male to female scholars generally increases as we move from left- to right-leaning outlets, with Breitbart and the Daily Wire exhibiting the most imbalance as discussed before. Additionally, financial media, such as the Financial Times, Bloomberg, CNBC, and the Wall Street Journal, all have gender ratios above 2. Among wire services, Reuters ($M = 2.96$, 95% CI [2.63, 3.34]) has a markedly higher gender ratio than the Associated Press ($M = 1.96$, 95% CI [1.86, 2.17]).

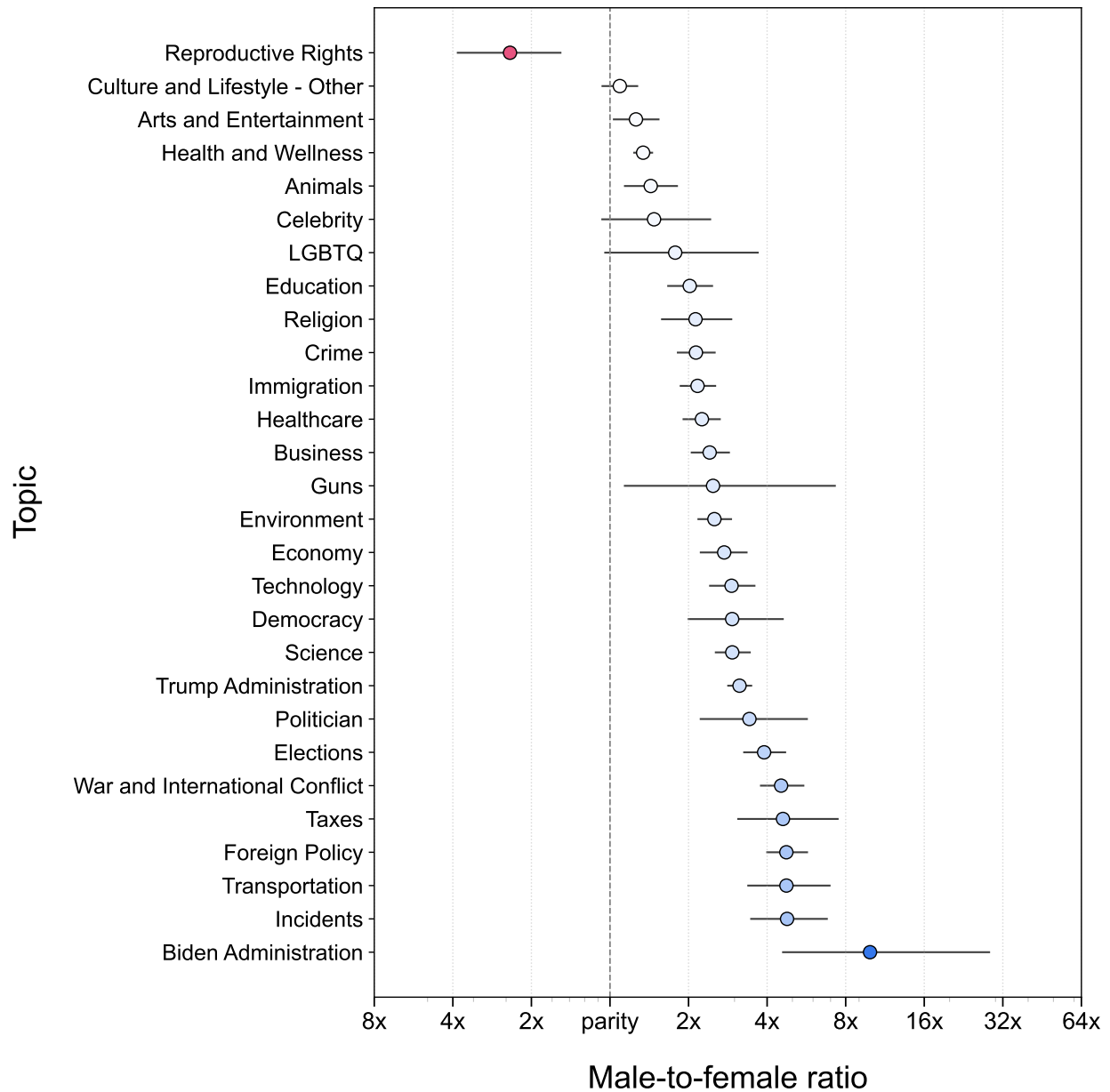


Figure 4.8: **Gender disparities in expert quotations across topics.** Across the full corpus comprising 28 topics, all but one contain more quotes from men than they do from women in their coverage. The only topic where women received more quotes than men was on “Reproductive Rights”, with the male-to-female ratio being highest for “hard news” topics and comparatively lower for “soft news” topics.

This gender disparity also varies substantially between news topics. Figure 4.8 shows the ratio of quotes attributed to male scholars vs. female scholars broken down by the topic of the news story.

Across all 28 topics covered in our corpus, only one, “Reproductive Rights”, contains more quotes from women than from men and had a male-to-female ratio below parity ($M = 0.41$, 95% CI [0.26, 0.65]). The only topics that come close to parity in gender representation are those related to culture and lifestyle, such as arts and entertainment ($M = 1.26$, 95% CI [1.03, 1.55]), health and wellness ($M = 1.34$, 95% CI [1.23, 1.46]), animals ($M = 1.43$, 95% CI [1.13, 1.82]), and celebrity news ($M = 1.48$, 95% CI [0.93, 2.44]). As we move towards “hard news” topics, the ratio rises to between 2-5x for most substantive policy areas, such as crime ($M = 2.13$, 95% CI [1.80, 2.54]), immigration ($M = 2.16$, 95% CI [1.85, 2.55]), healthcare ($M = 2.25$, 95% CI [1.90, 2.66]), the economy ($M = 2.74$, 95% CI [2.21, 3.37]), taxes ($M = 4.60$, 95% CI [3.07, 7.52]), and foreign policy ($M = 4.74$, 95% CI [3.97, 5.74]).

These results suggest that gender bias in expert quotations is influenced both by the publisher and the news topic. Women are comparatively less under-represented in domains conventionally associated with gender, wellness, culture, and entertainment, but their presence declines substantially in coverage of politics, economics, technology, and international affairs. This leads to gendered divisions in media coverage, where male scholars are disproportionately presented as authority figures on topics concerned with political and policy implications.

4.3.5. Representation of academic disciplines in the media

In this section, we turn to the question of which academic disciplines are most often cited as expert sources in the media. To identify scholars’ fields, we first searched for each individual using the Wikipedia API and matched them to candidate Wikipedia pages. We then used GPT-4o-mini to verify that the matched pages did indeed belong to the particular individual. For verified matches, we then used GPT-4o-mini to classify the scholar into a broad academic discipline based on the description of their research (see Figure C.3 for the prompt). This approach resulted in field classifications for 2,878 professors.

We find that academic disciplines differ substantially in how much they are platformed in the media. Law was by far the most frequently quoted academic field, with 3,315 quotations from 368 distinct scholars. It was followed by Economics with 1,969 quotes from 260 scholars and Political Science

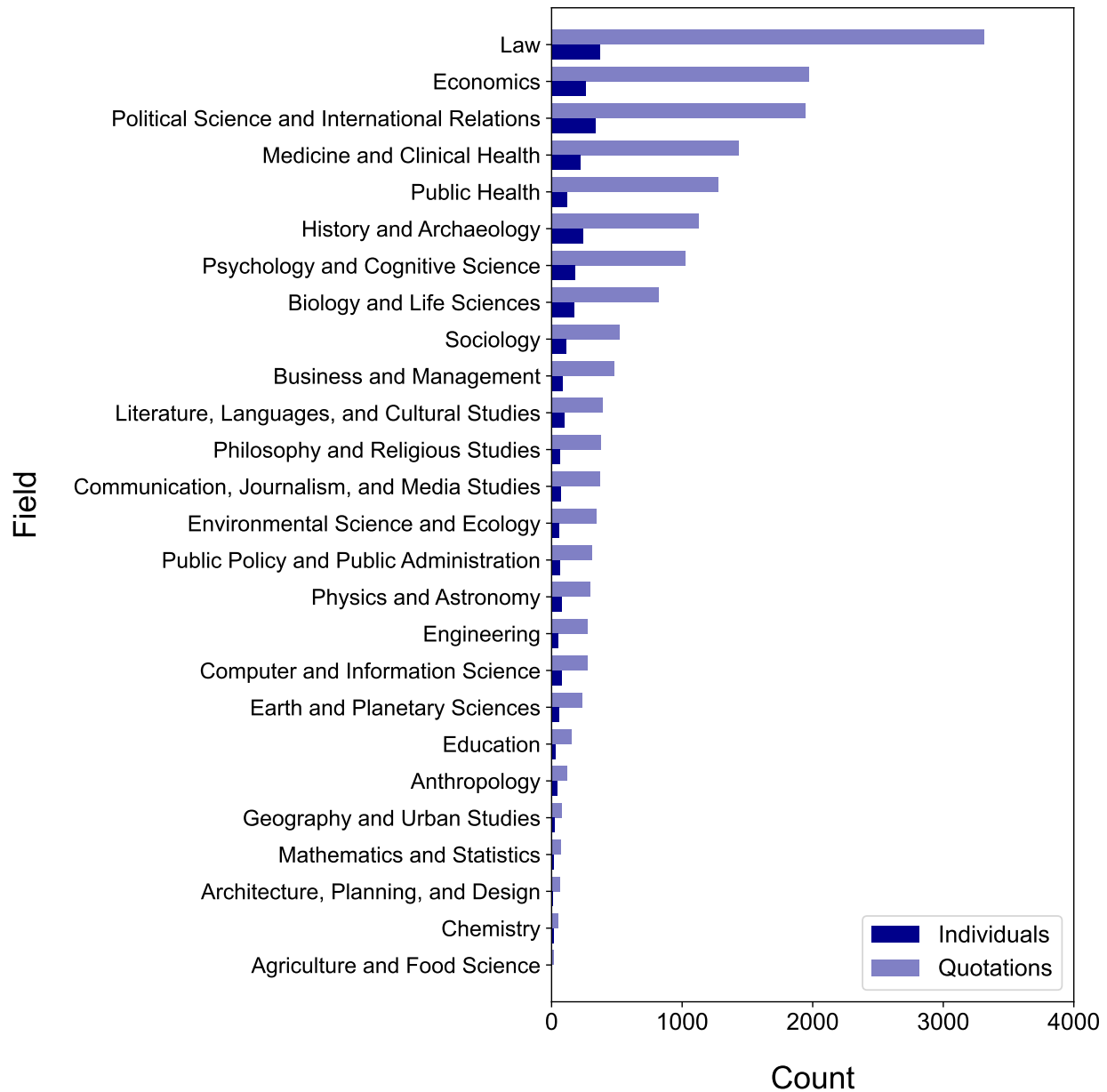


Figure 4.9: **Distribution of expert sources across academic fields.** The number of unique individuals quoted (blue) and total expert quotations (gray) are shown for each academic field, ordered by total number of quotes. Law, economics, and political science and international relations account for the largest shares of expert sources, followed by medicine and public health.

and International Relations with 1,947 quotes from 334 scholars. Although both of these domains differed by less than 20 quotes, economists received more quotes per capita than political scientists. These two fields were followed by Medicine and Clinical Health with 1,433 quotations from 224

individuals and Public Health with 1,276 quotations from 118 scholars. Together, these five fields accounted for roughly half of all quotations and scholars for which we could find a Wikipedia page and assign a domain. We see that fields related to politics and health receive significant attention in the media. This is consistent with the role academics most often play in the media, where journalists frequently seek expert voices to help contextualize legal matters, government actions, economic conditions, and healthcare stories for their readers.

In contrast to the social sciences and medicine, STEM fields received comparatively little media attention. Physics, Engineering, and Computer Science only received 298, 278, and 277 quotes, respectively. These differences show that academic expertise is not uniformly represented in the news, and university professors are most frequently invited to speak on issues that are directly related to public and political issues, such as law, politics, economics, and health, leading to higher representation of these fields.

4.4. Discussion

In this chapter, we studied how university scholars are represented as expert sources across a broad set of news publishers. Our results show three main patterns. First, we find that publishers differ substantially in how often they quote academic experts, with right-leaning outlets generally quoting fewer academics than left-leaning and centrist outlets. Second, we show that academic sourcing is heavily concentrated among a small set of prestigious universities, although the specific individuals quoted within these institutions varies considerably between publishers. Finally, we analyze the under-representation of women as expert sources and find that this issue exists largely at the source-selection stage, where journalists quote fewer women than men, although they receive similar amounts of attention once they have been selected.

Taken together, these findings show that expert sourcing is not simply a function of finding the individual with the most appropriate expertise for the topic at hand, but that it is shaped by the routines of news production and editorial priorities of news organizations. All publishers rely on well-known universities to source experts from, which shows that institutional prestige plays an important role in determining who is recognized as an expert. Although the publishers we

study vary greatly in their ideological slant, many of them repeatedly turn to the same set of elite institutions for expert opinion. Harvard, for example, is the most frequently quoted institution for more than half of the publishers in our corpus. At the same time, the low overlap between specific individuals quoted suggests that this is the stage in sourcing where editorial preferences have the most impact. Nevertheless, the consistent platforming of viewpoints from elite universities limits the range of perspectives presented in the media. This pattern can reinforce the same inequality in name recognition and resources that gives rise to this asymmetry in the first place, creating a feedback loop where institutional prestige leads to media exposure, which in turn further strengthens institutional prestige and individual scholars' public profiles. Researchers at less prominent institutions thus receive fewer opportunities to contribute to public debate and have their opinions heard on current events.

The particularly low rates of academic sourcing among financial outlets and right-leaning media are notable. This may reflect the nature of news topics covered, the format of their reporting, or these publishers' relationship with academic institutions. Financial publications, for example, may rely more on quotes from investors, executives, or market analysts than on university professors. In fact, we find that the business and economy news category has one of the lowest proportions of academic quotes. Similarly, the low number of university scholars in right-leaning outlets may reflect distrust of universities among conservative audiences or editorial preferences that prioritize pundits or political actors over academic interpretation in their reporting. Future work could study how academic sources quoted on the same event in right-leaning outlets differ from their counterparts quoted in left-leaning and centrist publications.

The gender disparities we find point to a different form of inequality in who gets platformed in the media. Across every publisher, we found that substantially more male scholars are quoted as expert sources than female scholars. However, we find no evidence that they also receive more or longer quotes per individual, conditional on their inclusion in the article. This suggests that the main source of disparity is not how much voice journalists give women after contacting them, but whether women are considered in the pool of prospective sources in the first place.

There could be several possible reasons for this selection gap. Men continue to be overrepresented in senior academic positions across many institutions and fields of study, leading to journalists' professional networks containing more men, particularly in politics, economics, law, and technology. The topic-level results show that male over-representation is highest in these "hard news" domains, while there is some degree of parity in culture, lifestyle, arts, and entertainment news. The only topic where women are quoted more than men is reproductive rights.

There are also some limitations in our work. We focus on one year of news coverage, where elite institutions such as Harvard and Columbia were consistently in the news due to leadership changes and events on campus. Although we differentiate expert quotes from these where the quoted individual or their institution are the subject of the story, it could be the case that these elite institutions' presence in the news cycle led to more expert quotes being sourced from them on a broader set of topics. Second, 2025 was a particularly charged year when it came to relations between elite institutions and the U.S. government, which may also have contributed to reinforcing their media presence across the board. Finally, we measure institutional prestige indirectly through the concentration of quotes from well-known universities. We do not model university research output, faculty size, geographic proximity, or the number of scholars working in each field. Larger, richer universities may be quoted more because they have more faculty across a wider range of disciplines. Future work could compare quotation rates with supply-side data to understand whether each university receives more or fewer quotes on a particular topic given its stature in that domain and faculty size.

Despite these limitations, the results provide a broad account of how academic authority is distributed in the news. Our findings point to practical ways in which sourcing could be broadened to address the disparities we observe. Journalists and newsrooms could consciously build and maintain more diverse expert databases and track the institutional and demographic distribution of their sources. By making a concerted effort to reach out to under-represented scholars, journalists can ensure that their reporting covers a wider range of viewpoints that their readers may be able to find similarities with. The selection of experts in the media determines whose knowledge and perspectives are used to frame current events for the general public and equip them with the

context to interpret them. By repeatedly platforming scholars from a small set of institutions and disproportionately selecting men, news organizations may be unintentionally reproducing existing hierarchies of prestige and gender in our society. Understanding these patterns is therefore an important step towards addressing these issues and ensuring that the media platforms as broad a slice of the community as the composition of its readership.

CHAPTER 5

SELECTION AND FRAMING IN EVENT COVERAGE

5.1. Introduction

In Chapter 2, we presented a framework for comparing the news coverage of publishers at the level of broad topics, partisan lean labels, or tone of reporting. These approaches are valuable for identifying persistent differences in coverage, such as whether one publisher pays more attention to certain topics than others. However, such topic-level comparisons flatten the variations in coverage given to certain events over a certain time period. For example, two publishers may appear to give the same level of coverage to the topic of immigration, but focus on different specific events within that topic. This would also mean that their framing of the same topic would be labeled differently, even though this might again be simply due to their focus on different events. Prior research has highlighted the diminishing state of “shared reality” in American media Hosseinmardi et al. (2025). Knowing whether publishers are covering the same underlying events is therefore necessary to separate differences in the composition of their coverage of a particular topic from differences in their presentation of common events. An example of this phenomenon is presented in Figure 5.1, which shows screenshots of the New York Times and Fox News homepages taken at the same time. While both publishers’ homepages are largely about political events, only one pair of articles is about the same event (Trump’s decoy Air Force 1 flight).

Event-level comparisons thus provide a more direct way to study these editorial decisions. By comparing publishers on a common set of news events, we can examine which events each publisher covered or did not cover, which facts they included or omitted, and how they framed the story. Research has shown that comparing how different articles report the same story can help identify partisan bias that is hard to detect from individual articles alone (Liu et al., 2023). Such event-centric approaches can also capture framing differences that are lost in topic-level comparisons (Das et al., 2024; Zhao et al., 2025). However, the core challenge for this approach is to accurately identify the event referred to in an article and group it together with others that refer to the same

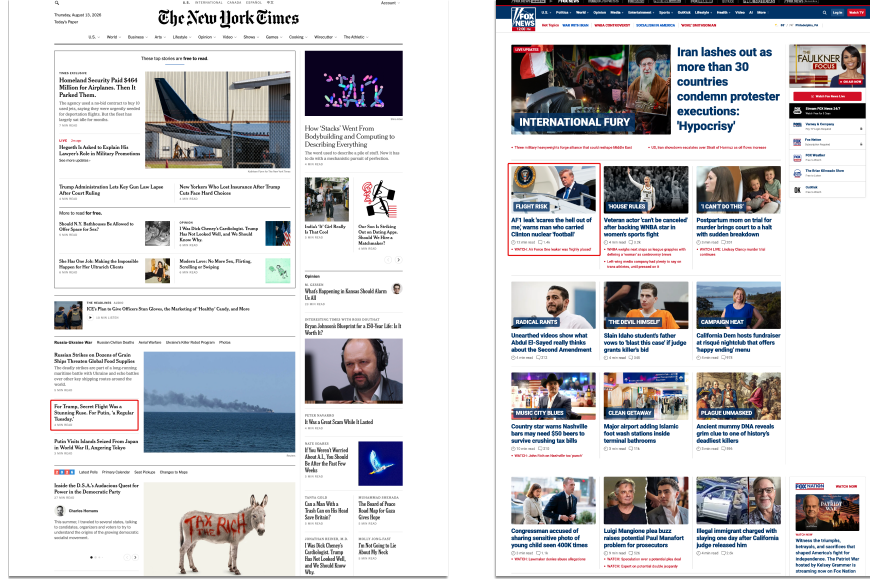


Figure 5.1: **A comparison of the New York Times and Fox News homepages at the same time.** These screenshots were taken seconds apart at 12:00 PM ET on August 13, 2026. Only one event (Trump’s decoy Air Force 1 flight from the day before) was tangentially covered by both.

event.

5.2. Methodology

Our event clustering pipeline operates in multiple stages. Following prior work on semantic clustering for event detection (Laban and Hearst, 2017; Liu et al., 2020), we use long-context document embeddings (in this case, OpenAI’s text-embedding-3-large model to obtain vector embeddings for all news articles published on a given day. We then compute cosine similarity between all pairs of articles and represent them as nodes in a network where the edge weight between any two articles is equal to their similarity. We drop edges below an empirically determined threshold of 0.85, chosen through manual inspection of cluster quality during development, and keep only connected components containing at least three articles across all publishers. These connected components form cohesive article clusters around major news events. Importantly, this approach does not constrain the number or size of clusters: on days where the news cycle is dominated by one major story, we might observe a skewed distribution with one large cluster, while on other days with dispersed coverage, a larger number of smaller clusters may emerge. Finally, we generate a natural language theme for each event based on the headlines of articles assigned to it (see Figure D.1 for the prompt).

In addition to clustering articles into events, we extract the key facts associated with each event to study which facts certain publishers choose to include (or omit) and how they frame them. We begin by retaining only those sentences within an article that were labeled as facts in the sentence labeling pipeline (Section 2.2.3). Then, we apply the same embedding-based clustering process described above to all factual statements across all articles in that event, this time at the sentence level using the smaller text-embedding-3-small model and a higher cosine similarity threshold of 0.85. This yields clusters of highly similar statements coming from different publishers, where larger clusters represent the most widely repeated and crucial pieces of information surrounding the event. We then summarize each fact cluster, which contains variations of the same core information, into a single, representative, synthetic statement using GPT-4o-mini (see Figure D.2 for the prompt). These summary facts capture the most widely reported and important pieces of information about a particular event, and can be used to study bias in fact selection between publishers at the event level. Figure D.3 shows a sample of the output from our event clustering pipeline.

To validate the quality of clusters obtained through this pipeline, we manually annotate one day of data (June 1, 2025), which contains 424 articles in total, out of which 107 articles were clustered into 19 distinct events by our framework, with event sizes ranging from 3 to 12 articles. We achieve an accuracy of 0.943 and macro-averaged precision of 0.996, recall of 0.883, and F1-score of 0.917.

5.3. Results

5.3.1. The event structure of the 2025 news cycle

Our dataset contains 261,126 articles published in 2025 by the 21 publishers we study. Out of these, 59,471 articles (22.8%) belong to 9,542 event clusters containing at least three articles, which is our threshold for identifying a cluster as a notable news event. We find that event sizes are strongly right-skewed, with the median event containing 4 articles compared with a mean of 6.23, and 75% of events contain 7 or fewer articles. However, there are certain major events that receive wide coverage, as the largest event in our data contains 133 articles across all 21 publishers.

Figure 5.2 shows the size of the largest event on each day of 2025. This timeline captures many

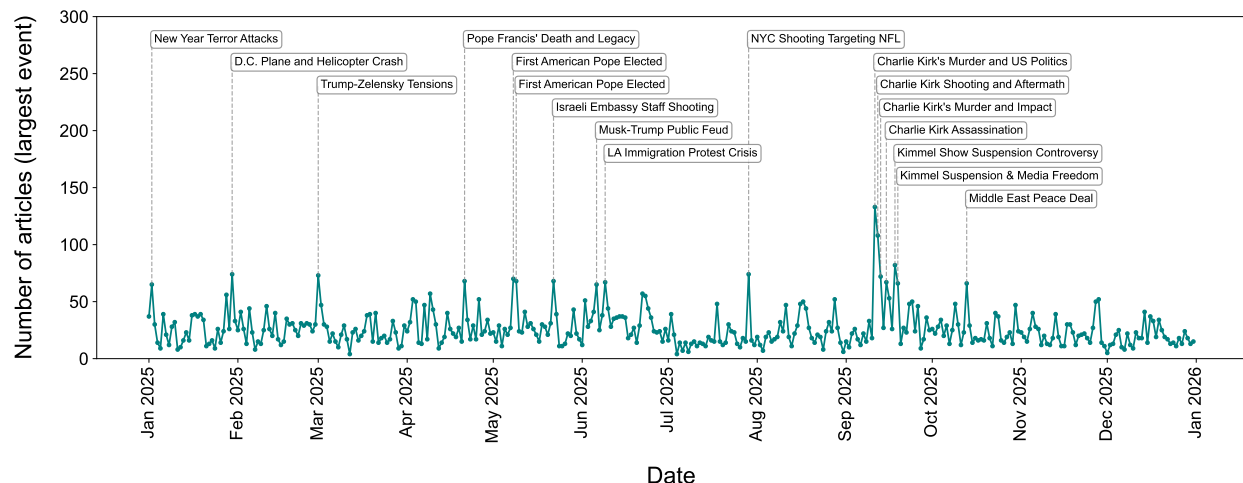


Figure 5.2: **Largest news events in 2025.** This figure shows the size of the largest event cluster on each day based on number of articles. Selected high volume events are labeled, and we see that major events often lead to sustained periods of follow-up coverage in the days after.

of the major news stories that dominated the news cycle during the year. These include the plane and helicopter collision near Washington, D.C. in January, tensions between Donald Trump and Volodymyr Zelensky in March, and the death of Pope Francis and the election of the first American pope in April and May, among others. We also see that some events persisted across multiple days. The largest event of the year as measured by our approach was the killing of Charlie Kirk on September 10, 2025, which resulted in 133 articles being published about it the following day. The next few days continued to cover the aftermath of this story, eventually segueing into coverage of Jimmy Kimmel’s suspension due to his remarks about Charlie Kirk on his talk show.

We find that the number of publishers covering an event varies considerably. Figure 5.3 shows the distribution of events according to the number of publishers that covered them. Out of the 9,542 events identified during the year, 5,789 (60%) were covered by 4 or fewer publishers. Additionally, 8,805 (92.3%) of events were covered by 10 or fewer publishers out of the full 21, which is less than half. These results show that there is substantial selection at the event level. However, we note here that our definition of events focuses strongly on precision, in that all articles assigned to a cluster should clearly be about the same real-world event, and not tangential commentary. Therefore, even though the event-level overlap between publishers is low, they may still devote similar amounts of

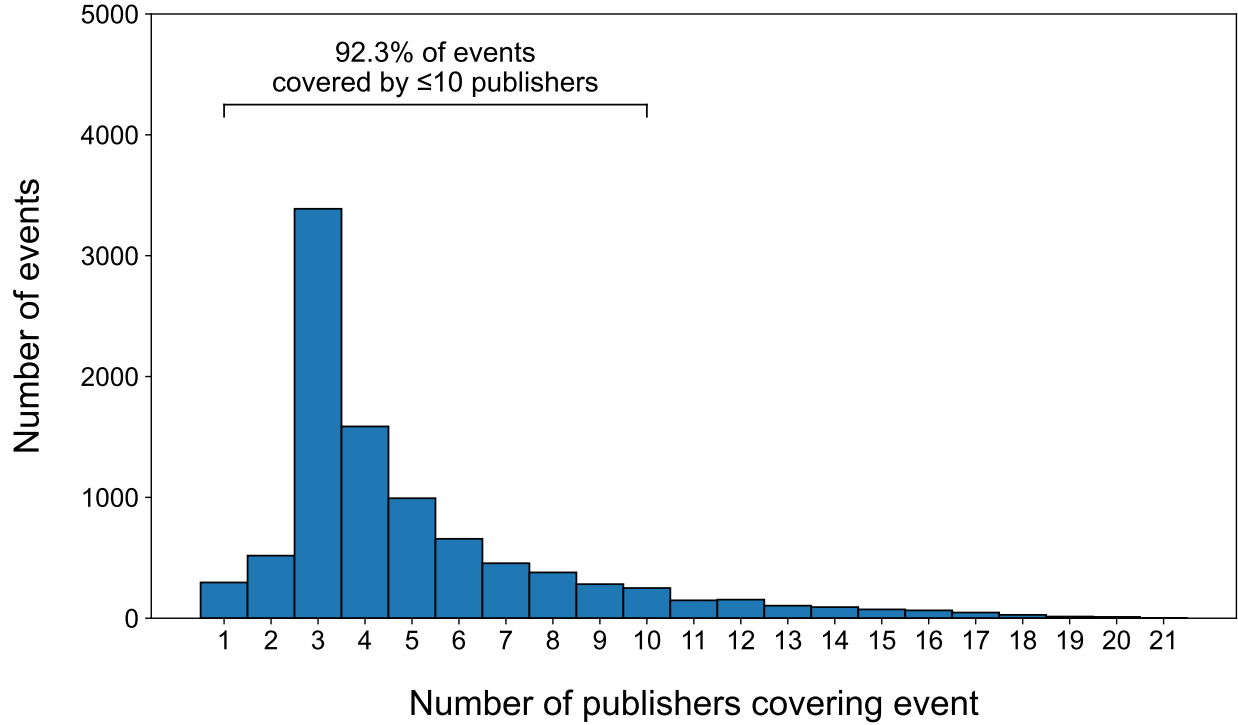


Figure 5.3: **Number of publishers covering each event.** This histogram shows the distribution of 9,542 event clusters according to the number of publishers that covered each event prominently (i.e. among the top 20 articles on their homepage). Coverage is highly skewed, with 92.3% of events being covered by fewer than half of the 21 publishers. These results hold even for more relaxed clustering thresholds (see Figure D.6 for robustness tests).

coverage to the overarching news topic.

5.3.2. Publishers differ systematically in the events they select

We next compare publishers based on which events they covered. For each pair of publishers, we computed the Jaccard similarity between their event sets, which we define as the number of events covered by both (intersection) divided by the total number of events covered by them (union). A value of one here would indicate that both publishers covered exactly the same set of events, while a value of zero would indicate no overlap in their coverage. Since the Jaccard similarity is dependent on the size of the union due to its symmetry, a value of 0.33 can be interpreted as both publishers sharing half of their events in common with each other (if they each covered the same number of events). It must therefore be read as a lower bound on the respective fraction of shared events in each publisher’s set.

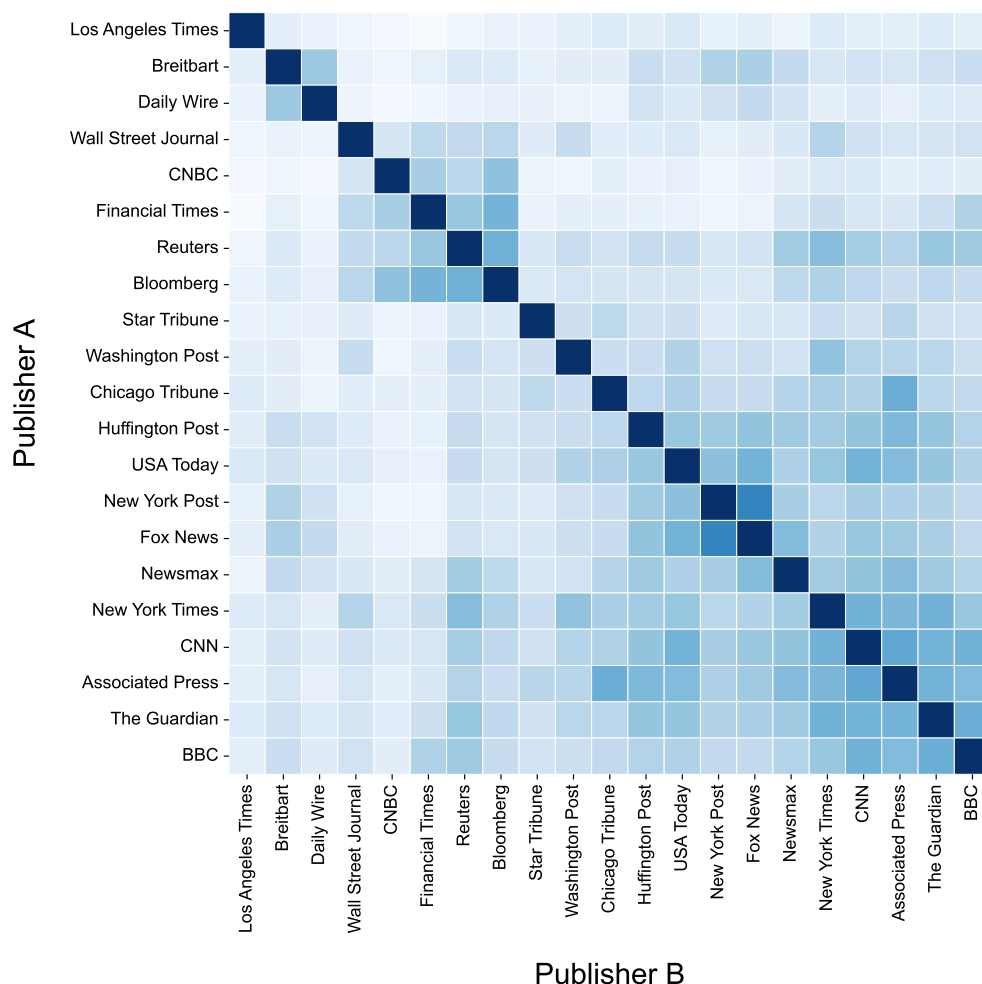


Figure 5.4: **Similarity in event selection between publishers.** This heatmap shows the pairwise Jaccard similarity based on the set of events covered by each publisher. Larger values indicate that the two publishers covered more similar sets of events. We see that publishers with similar editorial focus cluster together, indicating that event selection is related to the type of news an outlet focuses on.

Figure 5.4 shows the pairwise similarities across all 21 publishers. Although the differences are subtle, we find that the structure is not random, and publishers with similar editorial focus tend to cover similar sets of events. For example, we see a relatively high overlap between publishers focused on business and financial news, such as the Wall Street Journal, CNBC, Financial Times, Bloomberg, and Reuters. Another group of publishers that have a higher similarity are those that produce broad national and international news coverage, such as CNN, BBC, the New York Times, the Guardian, and the Associated Press.

The largest pairwise overlap was between the New York Post and Fox News, with a Jaccard similarity of 0.36 (Figure D.4), followed by CNN and Associated Press with a similarity of 0.29, The Guardian and BBC with 0.28. On the other end of the spectrum, several pairs, often including one business publication such as CNBC or the Financial Times, have similarities around 0.1. These differences are consistent with the distinct editorial domains of these outlets.

These results illustrate an important distinction between topic- and event-level selection in news production. While these publishers may cover broadly the same topics with similar levels of frequency, it does not automatically mean that they report on the same underlying set of events. Our event clustering approach makes this variation directly observable and gives us a useful unit of analysis on which publishers can be compared.

5.3.3. Separating differences in selection from difference in framing

Event clustering allows us to separate the two principal dimensions along which publishers can differ. We define *selection distance* between two publishers A and B to be equal to $1 - \text{Jaccard}(A, B)$, so that larger values indicate less overlap in shared event coverage. We separately define *framing distance* using only events covered by both publishers, and calculate the mean article lean for each publisher, then take the absolute difference between their values. The framing distance between two publishers is thus the mean absolute difference of the political lean of their coverage on the same set of events.

Figure 5.5 plots these two quantities for all 210 pairs of publishers. Selection distance ranges from 0.64 to 0.91, while framing distance ranges from approximately 0.5 to 3.5 points on the $[-5, +5]$ article lean scale. We find that these two dimensions are positively correlated ($r = 0.41$), showing that publishers that cover different sets of events also tend to frame these events differently on average.

Several publisher pairs illustrate this distinction. For example, the New York Post and Fox News have the smallest selection distance of 0.64 among all pairs, indicating that they focus on unusually similar sets of events. Their framing distance is also low (0.92), suggesting that they are also fairly

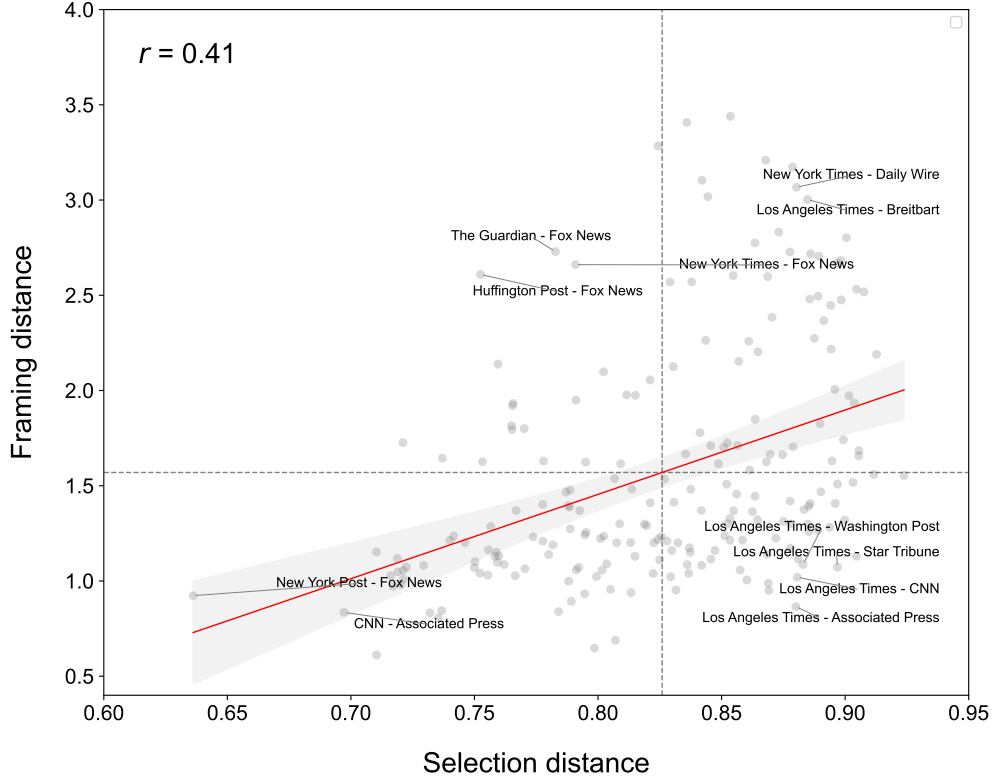


Figure 5.5: **Selection and framing distance between publisher pairs.** Each point in this plot represents one of the $\binom{21}{2} = 210$ publisher pairs. Selection distance between publishers with event sets A and B is defined as $1 - Jaccard(A, B)$. Framing distance is calculated as the mean absolute difference in publisher-level article lean for common events. The regression line shows the best fit with 95% confidence intervals shaded. Selection and framing are mildly positively correlated with Pearson’s $r = 0.41$, indicating that these two dimensions of publisher behavior can be distinguished.

united in their framing of the events that they share in common. Similarly, we see that CNN and the Associated Press have a low selection (0.70) and framing distance (0.84).

We also observe instances where publishers differ substantially on one dimension and less so on the other. For example, Fox News and the New York Times have a selection distance of 0.79 but a framing distance of 2.66. Fox News also has similarly large framing distances from the Guardian (2.73) and Huffington Post (2.61), showing that, while these publishers may frequently report on overlapping parts of the news cycle, they frame their coverage quite differently as determined by our article lean measure.

There are also cases where the reverse is true. The Los Angeles Times, for example, has large

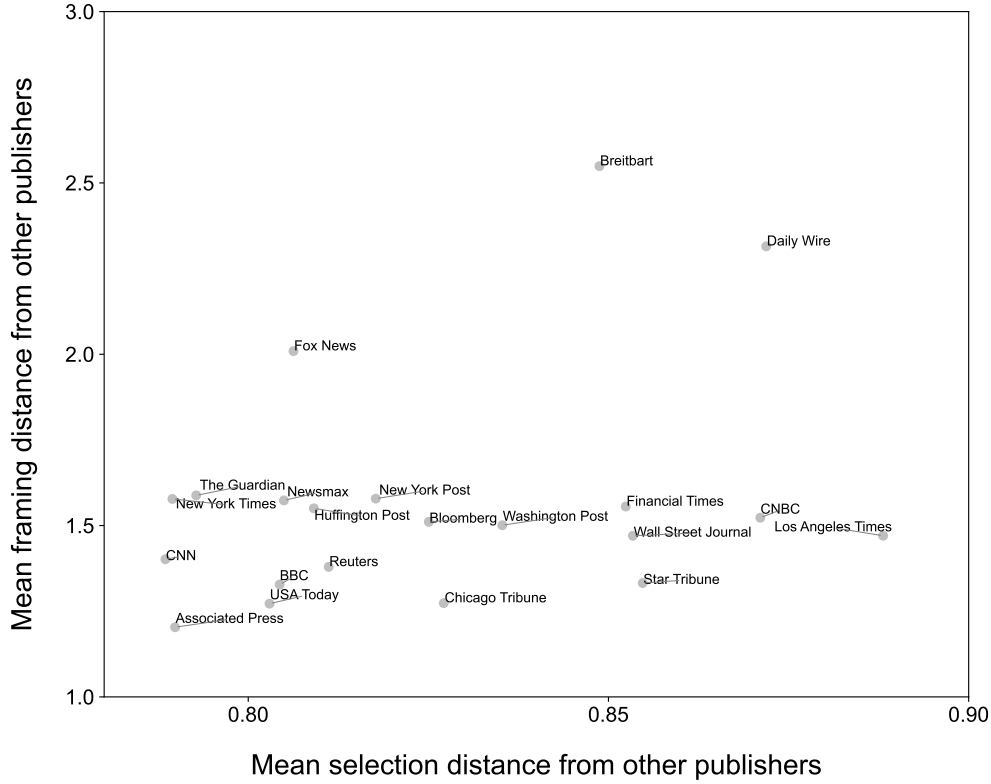


Figure 5.6: **Publisher-level differences in selection and framing.** Each point represents one publisher, and its position is determined by its mean selection difference from all other publishers (x-axis) and mean framing difference from all other publishers (y-axis). Publishers can therefore differ from the rest of the media landscape because they cover different events, because they frame common events differently, or both.

selection distances from several national publishers such as CNN, the Washington Post, and the Associated Press, but comparatively small framing distances. This shows that these publishers might focus on different events, but their framing of the ones they do cover together is fairly similar.

Finally, we have cases where the publishers differ substantially in both selection and framing, such as the New York Times and the Daily Wire, and the Los Angeles Times and Breitbart, both of which pairs have selection distances around 0.89 and framing distances around 3.0. These distinctions we are able to uncover through this approach would be difficult to identify directly from overall publisher-level averages.

5.3.4. Situating publishers in the wider selection-framing landscape

To characterize individual publishers rather than comparing all publisher pairs, we compute the average of each publisher’s selection and framing distances from all other outlets. Figure 5.6 places each publisher in this two-dimensional space, where the x-axis measures how different a publisher’s event selection is from the rest of the media, while the y-axis measures how different its framing of shared events is.

We see that CNN, the Associated Press, and USA Today lie relatively close to the center of the publisher landscape on both dimensions. Their mean selection distance ranges from 0.79 to 0.80, while their framing distances range from 1.20 to 1.40. These outlets therefore tend to cover events that are also covered by other publishers and frame shared events relatively similarly to other publishers as well.

Other outlets are distinctive primarily along one dimension, such as the Los Angeles Times, which has the largest mean selection difference from other publishers in the dataset (0.89) but a more standard framing distance (1.47). This suggests that what sets it apart is primarily its editorial focus on events that are not covered as much by other publications. One plausible reason behind this might be its focus on local California news and on arts, entertainment, and culture due to its proximity to Hollywood. However, we note that our measure does not directly identify the cause of this difference.

Fox News has the opposite pattern, where its mean selection distance of 0.81 is close to that of other broad-coverage publishers like BBC and USA Today, but its mean framing distance of 2.01 sets it apart from the rest. We can therefore interpret this as meaning that Fox News covers a relatively typical set of news events but differs more strongly in how it presents those stories to its readers. Other right-leaning outlets like Breitbart and the Daily Wire are distinctive on both dimensions, with selection differences of 0.85 and 0.87, respectively, and framing distances of 2.55 and 2.31, which are the largest among all publishers in our dataset.

Together, these results show why overall publisher similarity cannot be accurately described along

a single ideological or editorial axis. Two publishers could appear different for two intertwined reasons. It could be because they select different events to focus their coverage on, or because they frame shared events differently, or some mixture of both of these factors. Our approach makes it possible to distinguish these two mechanisms to some degree.

5.3.5. Polarization across news topics

We now shift from comparing publishers to comparing events, and, by extension, the topics they fall under. An event is considered more polarizing in our framework when publishers covering it do so through articles that are identified as having substantially different political lean. To quantify this phenomenon, we first compute the standard deviation in article lean for all articles about a certain event. We then calculate the mean event-level standard deviation across all events within each topic, bootstrapping within each topic 10,000 times to estimate 95% confidence intervals. Figure 5.7 shows the framing polarization results for all 32 topics in our news categorization, along with the number of events that fall under each topic.

We observe fairly large differences across topics. Events regarding the immigration had the highest average polarization in 2025 ($M = 1.7$, 95% CI [1.59, 1.8]), followed by those about the Trump administration ($M = 1.63$, 95% CI [1.56, 1.7]), taxes ($M = 1.61$, 95% CI [1.31, 1.91]), education ($M = 1.49$, 95% CI [1.26, 1.73]), and foreign policy ($M = 1.47$, 95% CI [1.38, 1.57]).

We also observe substantial disagreement in news about elections ($M = 1.41$, 95% CI [1.32, 1.52]), reproductive rights ($M = 1.35$, 95% CI [0.99, 1.73]), environment ($M = 1.33$, 95% CI [1.01, 1.67]), and war and international conflict ($M = 1.31$, 95% CI [1.24, 1.38]).

In contrast to this, events that fall under less politically charged topics show much smaller differences between publishers. health and wellness ($M = 0.73$, 95% CI [0.6, 0.85]), technology ($M = 0.72$, 95% CI [0.6, 0.85]), business ($M = 0.71$, 95% CI [0.64, 0.77]), science ($M = 0.45$, 95% CI [0.26, 0.66]), and celebrity ($M = 0.38$, 95% CI [0.3, 0.47]) had fairly low polarization. The least polarizing topics in the news during 2025 were sports: football ($M = 0.21$, 95% CI [0.14, 0.29]), baseball ($M = 0.2$, 95% CI [0.11, 0.31]), and basketball ($M = 0.11$, 95% CI [0.05, 0.17]) had barely any difference in

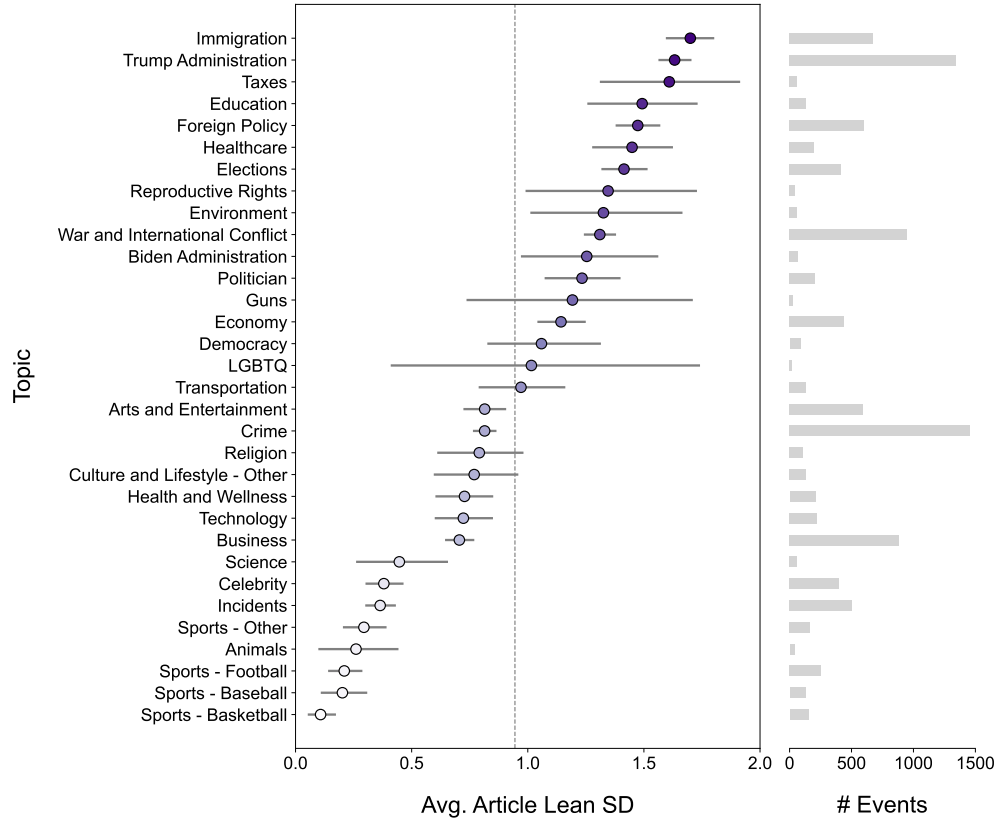


Figure 5.7: **Event-level framing divergence across topics.** Framing divergence is measured as the standard deviation of article lean among publishers covering a particular event. Each point shows the mean event-level framing divergence within a topic, with error bars representing 95% confidence intervals obtained via bootstrap sampling.

framing between publishers.

This ordering is notable because we calculate polarization only after conditioning on publishers covering the same event. Therefore, these differences in framing cannot be attributed to outlets simply choosing to cover different events within a topic. This approach thus measures variation in framing of the *same* underlying event. As expected, we find that politically salient topics tend to generate the most polarizing news coverage. However, the specific ordering with political news at large is interesting to observe. For sporting news, the media is fairly uniform in its reporting. Additionally, events with fewer articles, such as LGBTQ issues, guns, taxes, and reproductive rights have wider confidence intervals reflecting greater uncertainty in their estimated polarization.

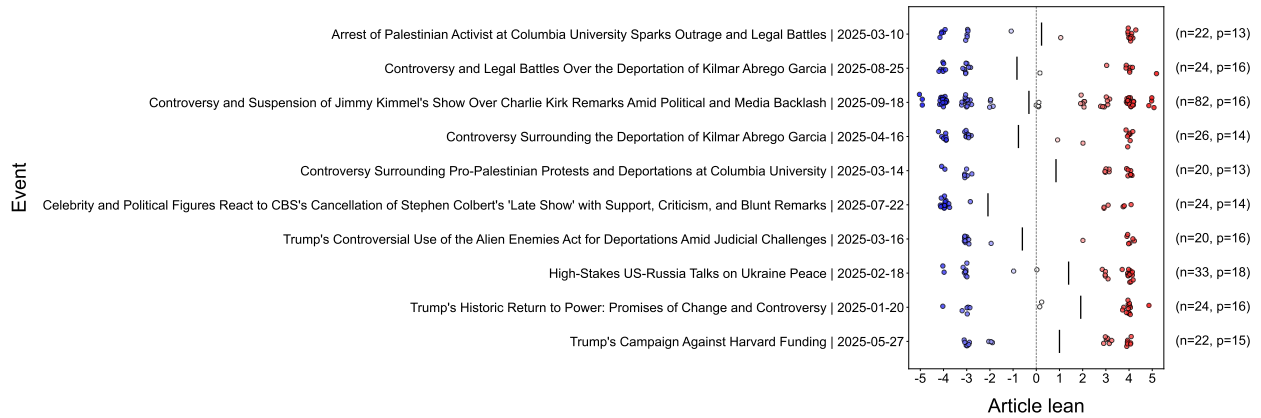


Figure 5.8: **Most polarizing major events of 2025.** This figure shows the 10 events with the largest standard deviation in article lean among events covered in at least 20 articles from at least 10 publishers. Each row represents an event, with each point representing an article about it and its political lean rating.

5.3.6. The most and least polarizing events of 2025

2025 was the first year of Donald Trump's second presidency and was marked by significant acrimony in the media and in public discourse regarding politics in America. In this section, we examine individual events that were especially polarizing as seen through the lens of news coverage. We restrict our attention to events containing at least 20 articles and coverage from at least 10 publishers, and then rank the events according to standard deviation of mean article lean at the publisher level.

The most polarizing events during 2025 were overwhelmingly connected to actions of the Trump administration. The event that received the most polarizing coverage was the arrest of a Palestinian activists at Columbia University and the resulting legal battle. Reporting on this event nearly spanned the full spectrum of our political lean rating scale, with articles' lean ranging from -4 to +4 with a standard deviation of 3.69. Other highly polarizing events were the deportation of Salvadoran citizen Kilmar Abrego Garcia ($SD = 3.60$) and Jimmy Kimmel's suspension ($SD = 3.55$).

We observe that the themes of immigration and the use of executive power to influence government operations and economic policy reoccur across most of the highly polarizing events. These cases show that it is not simply that all political news is equally polarizing, but the largest differences tend to occur around events concerning immigration, civil liberties, and governmental actions.

On the opposite end of the spectrum, we find that the least polarizing events of 2025 were essentially stories with no partisan angle (Figure D.5), such as accidents and disasters, celebrity news, and deaths of public figures. Some examples of such events are an NBA gambling scandal, the death of fashion designer Giorgio Armani, and Warner Brothers' and Paramount's bids to take over Netflix.

The contrast between these two sets of events illustrates the nature of event-level framing. Despite both types of events receiving wide coverage, some result in highly polarizing coverage while others do not. This emergence of publisher-level differences is thus strongly dependent on the content and context of the event in question.

5.4. Discussion

Our event clustering pipeline and the resulting analysis quantitatively show that differences between publishers are driven by two separate but intertwined factors. At the first level, publishers differ in which events they choose to cover. At the second level, they differ in how they frame those events in their coverage. These two processes are hard to tell apart at the topic level because differences between coverage might simply be because two outlets chose to focus on different events. By identifying a shared set of underlying events, we can determine whether these differences are actually due to that cause or because they frame the same events differently.

Our findings show that there are substantial selection differences across the news landscape. More than 92% of news events were covered by less than half of the publishers in our dataset, meaning that the readers of one outlet may be receiving a different picture of world events compared to those who follow another one. We find that publishers who cover similar events tend to have a shared editorial focus, such as financial news outlets such as FT, CNBC, and Bloomberg, and those that broadly cover national and international news, such as CNN, BBC, and AP.

By comparing how much each pair of publishers differs in both selection and framing dimensions, we find a small positive correlation of 0.405, which indicates that publishers that tend to focus on similar events are also more likely to frame them in similar ways. Our analysis also highlights publishers that differ substantially from the rest of the news landscape in one dimension or another.

For example, the Los Angeles Times has similar framing to the rest of the publishers but selects very different events to cover. On the other hand, Fox News covers largely the same events as the rest of the major news publishers but frames them very differently. Breitbart and Daily Wire differ on both dimensions.

These distinctions between publishers highlight the importance of decomposing news coverage into these two dimensions. A publisher can produce a substantially different view of the world for its readers without overtly framing its events in a certain way, but by simply selecting a different set of events to cover. Conversely, two publishers might cover the same events, but frame their stories in ways that portray them very differently to their readers. Aggregate topic-level measures conflate these two differences and are not as informative for identifying precisely how two publishers differ.

The event-level analysis further shows that framing differences are concentrated within certain news topics. During 2025, publishers disagreed the most when they covered events related to the presidency, immigration, elections, and policy issues. Because we measured variation in framing conditional on the same underlying event, these differences cannot be considered an effect of publishers simply covering different events. Our analysis of the most polarizing single events overall during 2025 reinforces this finding, as the most contentious events of the year were largely related to the Trump administration’s actions on immigration, foreign policy, and higher education. For the events we highlight, publisher-level article lean often spanned the full range of our scale from -5 to +5. By comparison, events related to non-political topics like sports and entertainment, which we found to be the least polarizing, often had zero political lean and disagreement in their coverage.

There are several limitations to this work. First, our dataset is not a comprehensive collection of all news events covered by American news publishers in 2025. We only focus on the 21 publishers tracked in the Media Bias Detector, and only scrape the top 20 articles every four hours for each of them. Second, our use of a high threshold for clustering articles into events ensures high precision in the resulting clusters, but may miss articles that are tangentially related to the core event. Third, although the Jaccard similarity metric makes our analysis robust to the inadvertent exclusion of articles from clusters, it does not capture how much attention a publisher gave a certain event. A

combination of number of articles published and their rank on the webpage could be used to address this. Finally, our definition of framing focuses on article lean, but this is only one dimension of framing and there could be other ways in which publishers frame shared events differently.

Despite these limitations, however, our framework shows the importance of going beyond topic- and keyword-level analyses of news coverage. Matching articles together based on specific events produces a richer dataset for analyzing the news that can be utilized by other researchers in the future. Additionally, our generation of natural language themes for each event makes the data easier to use than keyword-based event cluster labels which omit important context. Together, these results provide an approach for comparing news publishers at a more granular level than overall topic-level coverage differences to understand where and how polarization emerges.

CHAPTER 6

CONCLUSION

News organizations make a large number of choices in the process of relaying information about the world to their readers. They decide which stories deserve attention, which facts to emphasize and whose voices to include, and how these stories should be framed. The final output of this process, the news article, need not include any factual inaccuracies to shape public understanding of an issue. Instead, it is the selection of topics and framing of facts which influences which events and people readers should pay attention to and how they should understand various issues.

In this dissertation, I develop a computational framework for studying how these choices manifest in published news at scale. The foundation of this work is the Media Bias Detector, a system which continuously collects news articles from prominent news publishers and uses LLMs to extract a variety of data points from them. Across more than 200,000 articles collected in 2024 and 2025, I use these annotations to study news production at multiple levels, showing how media bias is a complex phenomenon that is hard to summarize with single, static publisher-level labels. Differences between news organizations actually emerge through a sequence of choices regarding what to cover, whom to platform, and how to present events.

I introduce the Media Bias Detector in Chapter 2, demonstrating how LLMs can be effectively used to analyze and annotate news articles at scale. The resulting data and the modularity and reusability of the accompanying annotations provide a valuable testbed for studying communication. Through a series of illustrative examples, I show the versatility of this dataset and how it can be used to not only discover patterns documented in prior research but also to answer new research questions.

In Chapter 3, I build on this by extracting more complex, structured information from news articles such as quotations and metadata around them. I use this data to study the allocation of voice to politicians during the 2024 presidential election year. I find that Republicans consistently received

more quotes than Democrats across publishers, news topics, states, levels of office, and time. Additionally, by using LLMs to measure the affective profile of each politician’s aggregated speech across the full year, I showed how both parties differ significantly on key dimensions like anger, fear, hope, calm, and unity.

Chapter 4 extended this study of voice allocation beyond politicians to academic experts. As “primary definers” in the media, scholars are regularly quoted to contextualize current events and help readers understand their implications. By analyzing university-affiliated scholars quoted in 2025, I showed that expert quotations in the media are highly concentrated. A small set of elite universities produce a disproportionate amount of quotes, and men receive significantly more quotes than women. The media thus appear to reproduce existing hierarchies of academic prestige in the process of selecting whom to quote on which topics.

Finally, Chapter 5 moves from topic- and quote-level analysis to specific news events. By developing a clustering framework to group articles together based on their semantic similarity, I create a dataset for analyzing selection and framing in the media. By decomposing differences in publishers’ news coverage along these two dimensions, I show how some publishers cover the same events but frame them differently, while others produce a similar framing but cover very different sets of events. By analyzing the variation of coverage within specific events, I showed which politically-charged events produced the most polarizing news coverage in 2025.

Together, these studies suggest that media bias is a multidimensional construct rather than a fixed characteristic of a news publisher. Furthermore, a reader’s information diet and understanding of the world is shaped not only by the cumulative effect of various decisions in the news production process, but also by the mix of sources they read.

In the future, the framework developed in this dissertation creates several opportunities for extending the study of news production. The most immediate is the expansion of this dataset across sources, media formats, and time. The current dataset is centered on major U.S. news publishers, and only contains the top-ranking articles on their homepages since 2024. In the future, this list of publishers

could be expanded to include prominent local outlets, as well as major international media sources. Additionally, extending this dataset backwards in time would allow us to study temporal trends in news coverage and validate whether the results found here generalize to prior years. Finally, expanding this framework to other modalities, such as television and social media, would allow us to analyze two crucial sources of news in the present age, and help us understand how they differ from online news.

A second direction which I find interesting is to attempt to answer *why* certain patterns in the news exist. In this dissertation, I primarily focused on the output of news production process, that is the content of the articles themselves, but it is equally important to identify the mechanisms which are responsible for the patterns we discover. For example, is the asymmetry we observe in political quotations simply because Republicans produce more quotable speech than Democrats, or because of deliberate editorial choices. Similarly, the over-representation of elite universities and male scholars could be driven by the unbalanced composition of academia or by existing journalist-source relationships.

Third, future work could connect the patterns in news productions with data on their consumption. Simply because a publisher chooses to focus on a particular topic or event more than others does not necessarily mean it meaningfully influenced its readers if they didn't read those stories. While home-page ranking provides us one measure of an article's popularity, linking our article-level measures with traffic or panel data would allow us to understand the information environment experienced by audiences of different publishers.

Finally, the computational methods themselves continue to improve. Although LLMs have become remarkably effective at a variety of social science measurement tasks, their outputs are still susceptible to variations in prompting strategies, and their estimates of ideological lean vary substantially between model families. Furthermore, using frontier models for labeling news articles can become prohibitively expensive at scale, especially if we aim to study decades of news across hundreds of publishers. Therefore, we need to more carefully explore the capabilities of different LLMs, especially open-source and open-weight models, to identify how high quality computational social

science research can be done in a reproducible manner.

In summary, the broader goal of this work is not to determine which news organizations are “biased” or not, nor is it even to define what it means to be biased. News organizations’ job is to distill everything that happens around us in the world into pieces of information that we can digest and which are most relevant to us. This process will, by definition, omit some events and voices and highlight others. My focus instead has been on how these choices differ *between* publishers at the same time, and *within* publishers across time. By using computational techniques to make these subtle differences measurable at scale, my aim is to help answer debates about media bias using empirical, verifiable data. As the media environment continues to expand and fragment, understanding this news production process will become increasingly important for understanding the information through which people make sense of the world around them.

APPENDIX A

APPENDIX FOR CHAPTER 2

Category	Topic	Subtopic													
Politics	War and International Conflict	Israel-Gaza War	Israel-Lebanon Conflict	Israel-Iran Conflict	Russia-Ukraine War	Syrian Conflict	Other								
	Biden Administration	Cabinet Appointments	Legal Challenges	Presidential Pardons	Biden's Age										
	Democracy	Voting Access/Rights	Fate of US Democracy	Election Security	Constitutional Crisis										
	Crime	Violent Crime	Financial and Corporate Crime	Police Misconduct	Cybercrime and Security Threats	Political Corruption	Terrorism	Sexual Misconduct	Drug-Related Crime						
	Guns	Gun Legislation and Regulation	Gun Safety and Violence	Legal and Societal Implications of Guns	National Rifle Association (NRA)										
	Immigration	Border Security	Immigration Policy and Reform	Government Action	Asylum and Refugees	Migration Trends									
	Environment	Climate Change	Wildlife Conservation	Agriculture	Renewable Energy	Government Regulation									
	Taxes	Tax Policy and Legislation	IRS and Tax Compliance	Personal Taxes	Corporate Taxes										
	Education	Diversity and Inclusion	Student Loans	Higher Education	Technology and Modern Learning	Academic Integrity									
	Trump Administration	Cabinet	Legal Challenges	Musk and Associates	Trump Legal Matters										
	Elections	Presidential Horse Race	Senate Horse Race	House Horse Race	Governor Horse Race	Voter Concerns	Local Elections	Foreign Elections							
	Foreign Policy	Russia	Canada and Mexico	Middle East	Africa	South America	Europe	Central America	Oceania	Asia	China				
	Politician	US House Representative	US Senator	US State Governor	US State Official	Non-US Political Official									
	Reproductive Rights	Abortion Regulation	Abortion and Politics	Abortion Advocacy and Activism	Birth Control	Fertility Treatment									
	Healthcare	Health Insurance	Medical Malpractice	Government Regulation											
	LGBTQ	Transgender Rights	Christian Church Divide	Rights and Advocacy	Hate Crimes Against LGBTQ Community	Same-Sex Marriage Rights	LGBTQ Activism	LGBTQ Representation in Media							
	Transportation	Public Transit	Transportation Safety												
Culture and Lifestyle	Animals	Animal Behavior	Animal Welfare												
	Arts and Entertainment	Art Industry	Award Show	TV Industry	Literature	Film/Movie Industry	Fashion Industry	Music Industry	Theatre						
	Celebrity	Celebrity Scandal	Royal Family	Celebrity's Health and Death	Celebrity Tribute										
	Religion	Religious Leaders and Influence	Religion and Social Issues												
	Health and Wellness	COVID-19	Nutrition and Diet	Exercise	Skincare	Sleep	Drugs and Alcohol	Mental Health	Vaccination						
Business and Economy	Culture and Lifestyle - Other	Home and Lifestyle	Travel and Tourism	Holidays											
	Business	Tech	Finance	Retail	Energy	Healthcare	Media	Transportation	Real Estate	Manufacturing	Food and Agriculture	Other			
Disaster and Incidents	Economy	Interest Rates	Government Spending	Job Market	Market and Trade	Banking and Investment	Inflation								
	Incidents	Traffic Incidents	Fire Incidents	Infrastructure Failures	Public Space Incidents										
Sports	Disasters	Natural Disaster	Man-Made Disaster												
	Sports - Baseball	MLB	College Baseball												
	Sports - Basketball	NBA	WNBA	Men's College Basketball	Women's College Basketball										
	Sports - Football	NFL	College Football												
	Sports - Other	NHL	Soccer	Olympics	Golf	Tennis	Boxing	Auto Racing and Motorsport	Other						
Science and Technology	Science	Astronomy	Paleoanthropology	Biology	Psychology	Physics	Neuroscience	Social Science	Science Policy and Funding						
	Technology	Artificial Intelligence (AI)	Cybersecurity and Privacy	Smartphone Technology	Space and Aerospace Technologies	Electric Vehicles	Digital Media and Advertising	Consumer Technology Trends	Scientific Exploration and Innovation	Social Media					

Figure A.1: **The MEDIA BIAS DETECTOR news classification system.** Our full topic hierarchy consists of 6 broad news categories composed of 33 topics and 162 subtopics.

Article Topic

The following is a news article. Read it and perform the task that follows.
Respond with a JSON object of key-value pairs.

#####

{article}

#####

Task: Classify the article into one of the listed topics.

Instruction: Try your best to bucket the article into one of these topics. DO NOT write anything that is not listed.

Key: "topic"

Value: One of: {topic_list}.

Do not return anything except the JSON object of key-value pairs as output.

Figure A.2: **Article topic labeling prompt.** The LLM is provided with a list of the 33 topics in our list and asked to choose the most appropriate one from them. Model: GPT-4o-mini.

Article Subtopic

The following is a news article on the topic of {predicted_topic}. Read it and perform the task that follows. Respond with a JSON object of key-value pairs.

#####

{article}

#####

Task: Classify the article into one of the listed subtopics under the predicted topic.

Instruction: Try your best to bucket the article into one of these subtopics. Label it as 'Other' if the article does not fit any possible subtopics.

Key: "subtopic"
Value: One of: {subtopic_list_under_the_topic}.

Do not return anything except the JSON object of key-value pairs as output.

Figure A.3: **Article subtopic labeling prompt.** Once the article topic has been labeled, we ask the LLM to pick the most appropriate subtopic from those that fall under that topic in our hierarchy. Model: GPT-4o-mini.

Article Takeaways

The following is a news article. Read it and perform the task that follows.
Respond with a JSON object of key-value pairs.

#####

{article}

#####

Task: Summarize the main points of the news article.

Instruction: List short takeaway points that readers are likely to remember from the article.

Key: "takeaways"

Value: A 3-4 sentence summary.

Figure A.4: **Article takeaways prompt.** To summarize the article for use in later prompts and to aid manual review of the topic and subtopic labels, we use an LLM to generate a short summary of the main points of the article. Model: GPT-4o-mini.

Article Lean

The following is a news article. Read it and perform the task that follows.
Respond with a JSON object of key-value pairs.

#####

{article}

#####

Task: Determine the political lean of this article within the U.S. political context. Is it supporting the Democrat party or the Republican party? Supporting a party can mean supporting its viewpoints, politicians, or policies. Provide reasoning for your answer.

1. Instruction: Give a short paragraph summarizing in what ways the article supports the Democrat party or the Republican party.

Key: "reason"

Value: A paragraph of text.

2. Instruction: Give a number from -5 to 5, with -5 indicating strong support for Democrats and 5 indicating strong support for Republicans. A value of 0 indicates that the article has no clear political lean towards either side.

Key: "lean"

Value: An integer number from -5 to 5.

Do not return anything except the JSON object of key-value pairs as output.

Figure A.5: **Article lean labeling prompt.** To classify an article's political lean, we instruct the LLM to first analyze the article and provide a summary of the main features which give it a certain lean. We then instruct the LLM to assign the article a numerical score on an 11-point Likert scale that accurately reflects the above analysis. Model: GPT-4o.

Article Tone

The following is a news article. Read it and perform the task that follows.
Respond with a JSON object of key-value pairs.

#####

{article}

#####

Task: Determine the overall tone of the article. Is it negative, positive, or neutral?

1. Instruction: Provide a short paragraph summarizing in what ways the article has a negative or positive tone.

Key: "reason"

Value: A paragraph of text.

2. Instruction: Provide a number from -5 to 5, with -5 indicating a very negative tone and 5 indicating a very positive tone. A value of 0 indicates that the article has a neutral tone.

Key: "tone"

Value: An integer number from -5 to 5.

Do not return anything except the JSON object of key-value pairs as output.

Figure A.6: **Article tone labeling prompt.** To classify an article's tone, we instruct the LLM to first analyze the article and provide a summary of the main features which give it a certain tone. We then instruct the LLM to assign the article a numerical score on an 11-point Likert scale that accurately reflects the above analysis. Model: GPT-4o.

Headline Lean

The following is the title of a news article on the topic of {topic} ({subtopic}) in the {category} news category. Read it and perform the task that follows.

Respond with a JSON object of key-value pairs.

#####

{article_title}

#####

Task: Determine the political lean of this article title within the U.S. political context. Is it supporting the Democrat party or the Republican party? Supporting a party can mean supporting its viewpoints, politicians, or policies. Provide reasoning for your answer.

1. Instruction: Give a short paragraph summarizing in what ways the article title supports the Democrat party or the Republican party.

Key: "reason"

Value: A paragraph of text.

2. Instruction: Give a number from -5 to 5, with -5 indicating strong support for Democrats and 5 indicating strong support for Republicans. A value of 0 indicates that the article title has no clear political lean towards either side.

Key: "lean"

Value: An integer number from -5 to 5.

Do not return anything except the JSON object of key-value pairs as output.

Figure A.7: **Headline lean labeling prompt.** We use the same prompt structure to label headline lean as we do for the article text, with the addition of explicitly mentioning the topic of the article to give the model context that the headline may lack. Model: GPT-4o.

Headline Tone

The following is the title of a news article on the topic of {topic} ({subtopic}) in the {category} news category. Read it and perform the task that follows. Respond with a JSON object of key-value pairs.

#####

{article_title}

#####

Task: Determine the overall tone of the article title. Is it negative, positive, or neutral?

1. Instruction: Provide a short paragraph summarizing in what ways the article title has a negative or positive tone.

Key: "reason"

Value: A paragraph of text.

2. Instruction: Provide a number from -5 to 5, with -5 indicating a very negative tone and 5 indicating a very positive tone. A value of 0 indicates that the article has a neutral tone.

Key: "tone"

Value: An integer number from -5 to 5.

Do not return anything except the JSON object of key-value pairs as output.

Figure A.8: **Headline tone labeling prompt.** For consistency, we use the same prompt to label headline tone as we do for the article text, except that we explicitly mention the topic of the article to give the model context that the headline may lack. Model: GPT-4o.

Article Type

The following is a news article. Read it and perform the task that follows. Respond with a JSON object of key-value pairs.

#####

{article}

#####

Task: Determine the news type of this news article.

1. Instruction: Classify the above news article into one of three categories: news report, news analysis, or opinion. Each category has distinct characteristics:

- News Report: Objective reporting on recent events, focusing on verified facts without the writer's personal views.
- News Analysis: In-depth examination and interpretation of news events, providing context and explaining significance, while maintaining a degree of objectivity.
- Opinion: Articles reflecting personal views, arguments, or beliefs on current issues, often persuasive and subjective.

Consider criteria such as language objectivity, focus on facts versus interpretation, author's intent, and article structure.

Key: "news_type"

Value: One of "news report" or "news analysis" or "opinion".

2. Instruction: Provide a short paragraph to justify your classification, citing specific elements from the text.

Key: "justification"

Value: A paragraph of text.

Do not return anything except the JSON object of key-value pairs as output.

Figure A.9: **Article type labeling prompt.** To classify the type of the article, we give the LLM descriptions of the three primary news types (report, analysis, opinion) and ask it to choose the most appropriate one. Model: GPT-4o.

Sentence Labeling

The following is a numbered list of sentences from a news article. Read them all and perform the task that follows:

#####

{sentences}

#####

Task: For each sentence above, perform the following analysis. Return the output as a list of JSON objects of key-value pairs, one for each sentence, where the keys are:

1. Instruction: Output the number of the current sentence being analyzed.

Key: "sentence"

Value: The sentence number.

2. Instruction: Determine whether the sentence is a "fact", "opinion", "borderline", "quote", or "other" type of sentence. A "fact" is something that's capable of being proved or disproved by objective evidence. An "opinion" reflects the beliefs and values of whoever expressed it, but should not be a quote. A "borderline" sentence is one that is not entirely a fact or opinion. A "quote" is a passage from another person or source that comes with quotation marks. If a quote is composed of multiple sentences, they should all be labeled as "quote".

Key: "type"

Value: One of "fact", "opinion", "borderline", "quote", "other". Select "other" if the sentence does not fit the possible values.

3. Instruction: Classify the tone of the sentence as "positive", "negative", or "neutral".

Key: "tone"

Value: One of "positive", "negative", "neutral".

4. Instruction: Classify the focus of the sentence as being "democrat", "republican", or "neither" within the U.S. political context. If the sentence is referring to the Democratic Party, including its policies or politicians, label it as "democrat". If it is referring to the Republican Party, including its policies or politicians, label it as "republican". If it is referring to both, label it as "both". Otherwise, label it as "neither".

Key: "focus"

Value: One of "democrat", "republican", "both", "neither".

Do not return anything except the list of JSON objects of key-value pairs as output. Do not skip analyzing any sentences. Do not return any key values except those specified.

Figure A.10: **Sentence labeling prompt.** To label every sentence in an article, we provide the entire article in the prompt since classifying a sentence requires knowing its neighboring context. We format the article as a list of enumerated sentences, one per line, and ask the LLM to classify the tone, type, and focus of each one. While tone is self-explanatory, we provide definitions for type and focus to serve as guidelines. Model: GPT-4o.

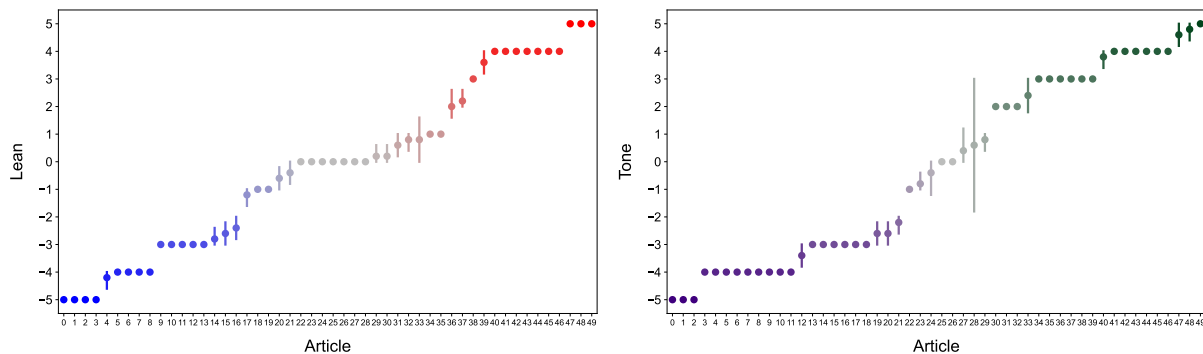


Figure A.11: **Evaluating the robustness of lean and tone ratings.** We ran each prompt five times on a set of 50 articles spanning diverse topics and the full range of lean and tone labels. The average Pearson's r was 0.99 for lean and 0.97 for tone. Of the 50 articles, the lean and tone ratings were identical across all runs for 35 and 37 articles, respectively.

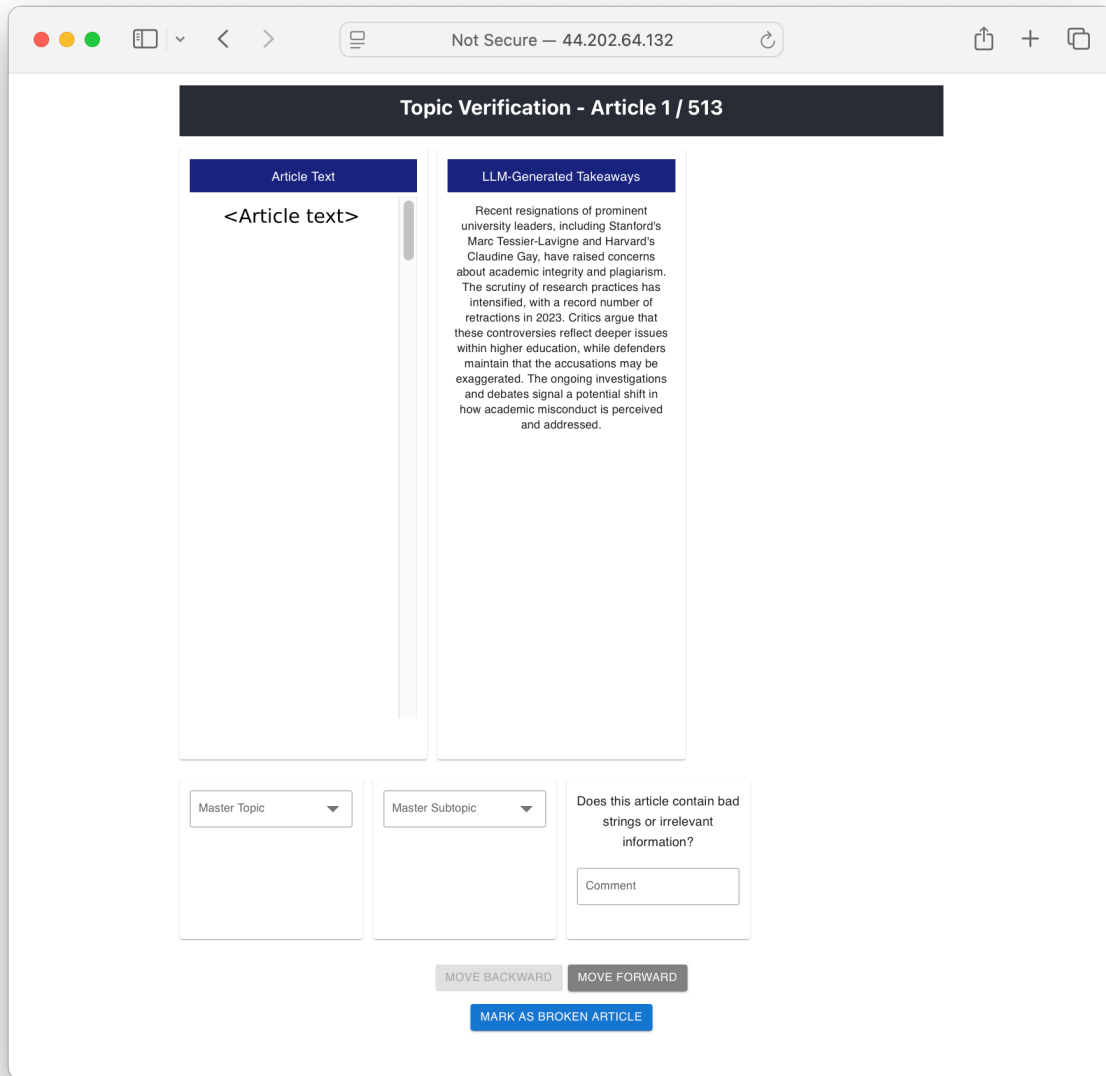


Figure A.12: **Article topic annotation interface.** Human annotators are shown the article text (left) along with a summary generated by the LLM to speed up the annotation process (center) and asked to select an appropriate topic or subtopic from our list. If the article doesn't fit under any of them, the annotator can suggest a new one for inclusion into our pipeline.

The figure displays two screenshots of a web application interface for article annotation. Both screenshots show a browser window with the address bar indicating 'Not Secure — 44.202.64.132'.

Top Screenshot: Q1: Political lean - Human Ratings

- Article Section:** A large text area labeled 'Article' containing the placeholder text '<Article text>'. Below this area is a blue button labeled 'NEXT QUESTION'.
- Q1: Political lean - Human Ratings Section:** A section titled 'Q1: Political lean - Human Ratings' with the instruction 'Please provide your own score for the article's political lean:'. Below the instruction is a horizontal scale from -5 to 5. The scale is marked with radio buttons at intervals of 1. The scale is color-coded: blue for negative values (-5 to -1), white for 0, and red for positive values (1 to 5). Labels 'Democrat' are positioned below the negative end, 'Neutral' below 0, and 'Republican' below the positive end.

Bottom Screenshot: Q2: Tone - Human Ratings

- Article Section:** A large text area labeled 'Article' containing the placeholder text '<Article text>'. Below this area is a blue button labeled 'NEXT QUESTION'.
- Q2: Tone - Human Ratings Section:** A section titled 'Q2: Tone - Human Ratings' with the instruction 'What tone rating would you assign to this article in the range [-5, +5]'. Below the instruction is a horizontal scale from -5 to 5. The scale is marked with radio buttons at intervals of 1. The scale is color-coded: purple for negative values (-5 to -1), white for 0, and green for positive values (1 to 5). Labels 'Very Negative' are positioned below the negative end, 'Neutral' below 0, and 'Very Positive' below the positive end.

Figure A.13: **Article lean and tone annotation interface.** Annotators are shown the full article text and asked to rate its political lean and tone on a scale from -5 to +5.

Article

<Article text>

Q3: News Type - Human Label

- **News Report:** Objective reporting on recent events, focusing on verified facts without the writer's personal views.
- **News Analysis:** In-depth examination and interpretation of news events, providing context and explaining significance, while maintaining a degree of objectivity.
- **Opinion:** Articles reflecting personal views, arguments, or beliefs on current issues, often persuasive and subjective.

What label would you assign to it

☐ News Report ☐ News Analysis ☐ Opinion

SUBMIT

Figure A.14: **Article type annotation interface.** Annotators are shown the full article text and asked to classify it as a news report, news analysis, or news opinion piece.

Event List

- ☐ High-Stakes Vice Presidential Debate: Vance vs. Walz
- ☐ Hurricane Helene: A Devastating Toll on North Carolina
- ☐ Escalation of Conflict: Israel and Iran's Military Engagements in Lebanon
- ☐ The Legacy and Controversy of Pete Rose: Baseball's All-Time Hits Leader Passes Away
- ☐ Nationwide Port Workers Strike Threatens Supply Chains and Economy
- ☐ Escalating Conflicts and Political Debates Amid Natural Disasters
- ☐ Sean 'Diddy' Combs Faces 120 Sexual Abuse Allegations
- ☐ Celebrating Jimmy Carter's Historic 100th Birthday
- ☐ Tributes to John Amos: Celebrated Actor of 'Good Times' and 'Roots' Dies at 84
- ☐ Political Fallout from Hurricane Helene: Claims and Responses
- ☐ Tim Walz's Controversial Claims on China and Tiananmen Square
- ☐ Frank Fritz, 'American Pickers' Star, Passes Away at 60
- ☐ Georgia's Abortion Ban Ruled Unconstitutional
- ☐ Claudia Sheinbaum: Mexico's First Female President
- ☐ Trump Withdraws from CBS '60 Minutes' Interview
- ☐ Close Encounter Between Russian Fighter Jet and US F-16 Near Alaska
- ☐ Trump's GoFundMe Campaign for Hurricane Helene Victims Raises Over \$2 Million
- ☐ Julian Assange's Plea for Freedom Through Journalism
- ☐ Fatal Shooting of Kentucky Judge by Sheriff
- ☐ Morgan Wallen's Generous Donation for Hurricane Helene Relief
- ☐ NBA Legend Dikembe Mutombo Passes Away at 58
- ☐ Major Verizon Service Outage Affects Customers Nationwide
- ☐ None of the above

Headline

Does the following article fall under any of the events listed? If not, please select "None of the above" at the end of the list.

Harris 2024 campaign senior adviser previews upcoming vice presidential debate: 'Two very different visions'

[PREV](#) [NEXT](#) [READ WHOLE ARTICLE](#)

Figure A.15: **Event cluster annotation interface.** To validate the precision and recall of our event clustering approach, we ask the human annotator to review the headline of an article (right) and assign it to an appropriate event theme from that day (left).

Sentence	Type	Tone	Focus
<Sentence 1>	☐ fact ☐ quote ☐ opinion ☐ other ☐ borderline	☐ negative ☐ positive ☐ neutral	☐ democrat ☐ both ☐ republican ☐ neither
<Sentence 2>	☐ fact ☐ quote ☐ opinion ☐ other ☐ borderline	☐ negative ☐ positive ☐ neutral	☐ democrat ☐ both ☐ republican ☐ neither
<Sentence 3>	☐ fact ☐ quote ☐ opinion ☐ other ☐ borderline	☐ negative ☐ positive ☐ neutral	☐ democrat ☐ both ☐ republican ☐ neither
<Sentence 4>	☐ fact ☐ quote ☐ opinion ☐ other ☐ borderline	☐ negative ☐ positive ☐ neutral	☐ democrat ☐ both ☐ republican ☐ neither
<Sentence 5>	☐ fact ☐ quote ☐ opinion ☐ other ☐ borderline	☐ negative ☐ positive ☐ neutral	☐ democrat ☐ both ☐ republican ☐ neither

Figure A.16: **Sentence annotation interface.** The human annotator reads one article at a time and reviews and labels the type, tone, and focus of each sentence in it.

Text S1: Sample Labeled Data

The following are ten examples of labeled articles generated by our system. You can view more examples in a spreadsheet at the following link: <https://tinyurl.com/media-bias-detector>

1. **URL:** <https://www.foxnews.com/opinion/liberal-media-outlet-inadvertently-explains-benefits-trumps-economy>

Category: Business and Economy

Topic: Economy

Subtopic: Inflation

Takeaways: The article contrasts the economic approaches of Trump and Harris, highlighting Trump's focus on macroeconomic issues like inflation and growth, while criticizing Harris for her lack of a comprehensive economic plan. It argues that Harris resorts to targeted microeconomic policies that fail to address broader economic challenges. The piece suggests that voters will demand a more effective economic strategy in the upcoming election, emphasizing that Trump's tax cuts and deregulation could benefit all Americans.

News Type: opinion

Reason News Type: The article is classified as an opinion piece due to its subjective language and persuasive tone. It presents a clear stance against Kamala Harris' economic policies while favoring Donald Trump's approach, using phrases like 'failed big government hodgepodge' and 'rising tide economic plan.' The article's intent is to argue and persuade rather than objectively report or analyze facts.

Article Lean: 5

Reason Article Lean: The article strongly supports the Republican party by highlighting former President Trump's macroeconomic policies as beneficial and criticizing the Biden/Harris administration's economic strategies. It portrays Trump's approach as comprehensive and effective, while depicting Harris's policies as inadequate and pandering to specific voter groups. The article also defends Trump's tax and tariff policies, suggesting they are economically sound and politically advantageous.

Article Tone: -3

Reason Article Tone: The article has a predominantly negative tone towards Kamala Harris and the Biden/Harris administration's economic policies. It criticizes Harris for lacking a comprehensive macroeconomic plan and accuses her of political pandering with microeconomic policies. The article contrasts this with a more favorable view of Trump's macroeconomic approach, highlighting his focus on broad economic growth strategies.

Title Lean: 4

Reason Title Lean: The article title suggests that a liberal media outlet, typically associated with Democratic viewpoints, is inadvertently highlighting the benefits of Trump's economic plan. This implies a positive portrayal of Trump's policies, which are aligned with the Republican party. By framing the situation as an unexpected endorsement from a typically opposing viewpoint, the title leans towards supporting Republican economic policies.

Title Tone: 3

Reason Title Tone: The article title suggests a positive tone towards Trump's economic plan by highlighting that even a liberal media outlet, which might typically be critical, has inadvertently acknowledged its benefits. This implies that the plan has merits that transcend political biases, casting it in a favorable light.

2. **URL:** <https://www.breitbart.com/entertainment/2024/09/23/trump-tribute-song-fighter-tops-itunes-chart->

Category: Culture and Lifestyle

Topic: Arts and Entertainment

Subtopic: Music Industry

Takeaways: Jon Kahn's new song 'Fighter,' which pays tribute to former President Donald Trump, has topped the iTunes chart for five consecutive days. Trump endorsed the song, calling it an 'unexpected honor.' The song, co-written by Chris Wallin, aims to portray Trump positively against what Kahn describes as a misleading narrative from the left. 'Fighter' has also gained over 10 million views on social media.

News Type: opinion

Reason News Type: The article reflects personal views and arguments, particularly in its portrayal of Donald Trump and criticism of the 'left's mission.' The language is subjective,

with phrases like 'gaslighting left machine' and 'threat to the left's mission,' indicating a persuasive and opinionated stance rather than objective reporting or analysis.

Article Lean: 5

Reason Article Lean: The article supports the Republican party by highlighting a song that pays tribute to former President Donald Trump, portraying him positively and countering negative narratives from the left. It emphasizes Trump's endorsement of the song and criticizes the 'Democrat media complex' for misrepresenting him.

Article Tone: 4

Reason Article Tone: The article has a positive tone as it highlights the success of Jon Kahn's song 'Fighter' in tribute to Donald Trump, emphasizing its popularity and Trump's endorsement. It portrays Trump in a favorable light, countering negative perceptions and celebrating his resilience.

Title Lean: 4

Reason Title Lean: The article title likely supports the Republican party as it references a tribute song to Donald Trump, a prominent Republican figure. The success of the song on the iTunes chart suggests a positive reception, which can be interpreted as support for Trump and, by extension, the Republican party.

Title Tone: 4

Reason Title Tone: The article title has a positive tone as it highlights the success of a song, 'Fighter,' which has topped the iTunes chart for five consecutive days. This suggests a significant achievement and popularity, which is generally perceived positively.

3. **URL:** <https://www.breitbart.com/politics/2024/10/01/poll-trump-edges-out-harris-in-four-battleground-states/>

Category: Politics

Topic: Elections

Subtopic: Presidential Horse Race

Takeaways: Former President Donald Trump is leading Vice President Kamala Harris in four key battleground states: Arizona, Nevada, North Carolina, and Georgia, according to an InsiderAdvantage survey. The margins are tight, with Trump holding a one-point lead in

Arizona and Nevada, and a slight edge in North Carolina, while Georgia is tied. The survey indicates Trump may be gaining momentum, especially as he provides relief efforts in areas affected by Hurricane Helene, contrasting with Harris's response to the disaster.

News Type: news report

Reason News Type: The article is classified as a news report because it objectively presents the results of a survey regarding the political standings of Donald Trump and Kamala Harris in key battleground states. The article focuses on factual data, such as percentages and margins of error, without delving into personal opinions or interpretations. It maintains a neutral tone, reporting on events like Trump's visit to Georgia and Harris's scrutiny over Hurricane Helene, without offering subjective commentary.

Article Lean: 3

Reason Article Lean: The article leans towards supporting the Republican party by highlighting Donald Trump's lead over Kamala Harris in key battleground states and emphasizing his actions in response to Hurricane Helene. It portrays Trump positively, mentioning his provision of relief items and a moment of silence for hurricane victims, while also noting Harris's scrutiny over her response to the disaster. The inclusion of a prayer for Trump by a prominent Christian figure further underscores the article's favorable tone towards him.

Article Tone: 3

Reason Article Tone: The article presents a generally positive tone towards Donald Trump, highlighting his lead in key battleground states and his active involvement in providing relief after Hurricane Helene. It emphasizes his momentum in the polls and includes a supportive prayer from a prominent religious figure, which adds to the positive portrayal. However, it also mentions the tightness of the races and scrutiny faced by Kamala Harris, which adds a slight neutral balance.

Title Lean: 1

Reason Title Lean: The article title suggests a competitive edge for Donald Trump, a prominent Republican figure, over Kamala Harris, a leading Democrat, in key battleground states. This framing can be perceived as supportive of the Republican party by highlighting

Trump's potential electoral strength. However, the title itself is neutral in tone and does not explicitly endorse or criticize either candidate, focusing instead on the poll results.

Title Tone: 1

Reason Title Tone: The article title "Poll: Trump Edges out Harris in Four Battleground States" carries a slightly positive tone for Trump supporters, as it highlights his lead over Harris in crucial states. This could be seen as a favorable development for Trump's campaign, suggesting momentum or strength in key areas. However, the tone is not overtly positive or negative, as it simply reports the results of a poll without additional context or emotional language.

4. **URL:** <https://www.breitbart.com/politics/2024/09/02/mike-johnson-may-attach-proof-citizenship-voting-la>

Category: Politics

Topic: Elections

Subtopic: Voter Concerns

Takeaways: Speaker Mike Johnson and House Republican leaders are considering attaching the SAVE Act, which requires proof of citizenship to vote, to a short-term spending bill to prevent a government shutdown. The bill aims to address concerns about non-citizen voting, with support from a significant majority of Americans. The Biden Administration opposes the measure, which Republicans argue is essential for election integrity. The proposed legislation is seen as a key issue leading up to the elections on November 5.

News Type: news report

Reason News Type: The article objectively reports on the discussions among House Republican leadership regarding the attachment of the SAVE Act to a stop-gap spending bill. It focuses on verified facts, such as statements from political figures and survey results, without providing the author's personal views or in-depth analysis.

Article Lean: 5

Reason Article Lean: The article supports the Republican party by highlighting their efforts to attach the SAVE Act to a spending bill, emphasizing the need for proof of citizenship to vote. It includes statements from Republican leaders criticizing the Biden Administration

and promoting election integrity.

Article Tone: 3

Reason Article Tone: The article has a positive tone towards the SAVE Act and the efforts of Republican leaders to attach it to the spending bill. It highlights support from various Republican figures and a survey showing public backing for the measure.

Title Lean: 4

Reason Title Lean: The article title suggests that Mike Johnson, likely a Republican, is considering attaching a proof-of-citizenship voting law to a stop-gap spending bill. This aligns with Republican viewpoints on stricter voting regulations and immigration control.

Title Tone: -2

Reason Title Tone: The article title suggests a potentially contentious political move by Mike Johnson, which could be seen as controversial. The attachment of a proof-of-citizenship voting law to a stop-gap spending bill may imply a strategic maneuver that could provoke debate or opposition.

5. **URL:** <https://www.cnn.com/2024/10/22/us/5-things-to-know-for-oct-22-vulnerable-seniors-exploited-hostage/index.html>

Category: Politics

Topic: Elections

Subtopic: Voter Concerns

Takeaways: The 2024 presidential race is nearing its conclusion with Election Day on November 5. An investigation revealed deceptive fundraising tactics targeting elderly Americans, resulting in over \$6 million in donations, primarily to Republican candidates. Secretary of State Antony Blinken is in the Middle East addressing the humanitarian crisis in Gaza amid ongoing hostage negotiations. Additionally, allergy season is extending due to climate change, and a recall of frozen waffles has been issued due to potential listeria contamination.

News Type: news report

Reason News Type: The article is classified as a news report because it objectively presents a series of recent events and factual updates without delving into personal opinions or inter-

pretations. It covers various topics, such as the 2024 presidential race, deceptive fundraising tactics, international diplomacy, and health alerts, focusing on delivering verified information to the reader.

Article Lean: -2

Reason Article Lean: The article appears to lean slightly towards the Democratic party. It highlights deceptive fundraising tactics associated with former President Donald Trump and other Republican candidates, potentially painting them in a negative light. Additionally, it includes a quote from former Rep. Liz Cheney, a Republican critical of Trump, urging voters to act against him, which may resonate more with Democratic viewpoints.

Article Tone: -1

Reason Article Tone: The article presents a mix of negative and neutral tones. It highlights concerning issues such as deceptive fundraising tactics targeting the elderly, ongoing conflicts in the Middle East, and false claims by a former president, which contribute to a negative tone. However, it also includes neutral or positive elements, such as updates on the NBA season and innovative AI technology to prevent wildfires, balancing the overall tone.

Title Lean: 0

Reason Title Lean: The article title does not explicitly mention any political figures, policies, or viewpoints that are directly associated with either the Democrat or Republican parties. It covers a range of topics, including vulnerable seniors, hostage talks, hurricane aid, fall allergies, and frozen waffles, which are not inherently political or partisan issues. Therefore, the title appears to be neutral in its political leaning.

Title Tone: -1

Reason Title Tone: The article title presents a mix of topics, some of which carry negative connotations, such as 'vulnerable seniors exploited' and 'hostage talks,' suggesting issues of concern and distress. However, it also includes neutral topics like 'hurricane aid,' 'fall allergies,' and 'frozen waffles,' which do not inherently carry a negative or positive tone. Overall, the presence of both negative and neutral elements results in a slightly negative tone.

6. **URL:** <https://www.theguardian.com/commentisfree/2024/oct/30/bernie-sanders-israel-gaza-harris-trump>

Category: Politics

Topic: Elections

Subtopic: Voter Concerns

Takeaways: The article expresses strong opposition to the war in Gaza and criticizes the Biden administration's support for Israel's actions. It emphasizes the need for humanitarian aid and a change in US policy towards Israel, while also highlighting the dangers of a Trump presidency, including setbacks for women's rights, climate action, and increased inequality. The author urges voters to support Kamala Harris in the upcoming election despite disagreements on Gaza, framing it as a crucial moment for the future of the country.

News Type: opinion

Reason News Type: The article is classified as an opinion piece because it reflects personal views and arguments, particularly those of Bernie Sanders, on the US political stance towards the Gaza conflict and the upcoming election. The language is subjective, with persuasive elements aimed at influencing readers' opinions, such as highlighting the consequences of a Trump presidency and advocating for Kamala Harris.

Article Lean: -5

Reason Article Lean: The article strongly supports the Democratic party by advocating for Kamala Harris as a preferable choice over Donald Trump. It criticizes Trump's policies and actions, particularly regarding humanitarian aid to Gaza, women's rights, climate change, and economic inequality. The article emphasizes the importance of defeating Trump in the upcoming election, highlighting the potential negative consequences of his presidency.

Article Tone: -2

Reason Article Tone: The article carries a predominantly negative tone towards the actions of the Israeli government in Gaza and the potential consequences of a Trump presidency. It criticizes the humanitarian impact of the conflict and highlights the risks associated with Trump's policies on climate change, women's rights, and economic inequality. However, it also contains a positive undertone in its support for Kamala Harris as a preferable alternative, suggesting hope for policy change and humanitarian aid under her leadership.

Title Lean: 1

Reason Title Lean: The article title suggests a critical stance towards Kamala Harris, a prominent Democrat, by questioning her support for Israel's war. This aligns with a viewpoint that may be more critical of Democratic foreign policy decisions, potentially resonating with Republican perspectives that often critique Democratic leaders. However, the mention of Bernie Sanders, a Democrat, indicates an internal critique rather than an external partisan attack, suggesting a nuanced position rather than outright support for Republicans.

Title Tone: -3

Reason Title Tone: The article title suggests a negative tone as it raises a critical question about Kamala Harris's support for Israel's war, implying a potential conflict or controversy. The use of the phrase "How can I vote for Kamala Harris" indicates a sense of doubt or disapproval, which is further emphasized by the association with Bernie Sanders, who is known for his progressive stance on foreign policy issues.

7. **URL:** <https://www.nytimes.com/2024/09/05/world/middleeast/israel-raids-west-bank-palestinians.html>

Category: Politics

Topic: War and International Conflict

Subtopic: Israel-Gaza War

Takeaways: Israeli military raids in the West Bank have intensified, resulting in at least 39 deaths and 145 injuries over nine days. Residents, particularly in Jenin and Tulkarm, are trapped in their homes due to fear of violence and ongoing military operations. The situation has led to significant destruction of property and infrastructure, with reports of brutality and a lack of basic necessities. The Israeli military claims the raids target armed groups, while Palestinian sources report civilian casualties.

News Type: news report

Reason News Type: The article objectively reports on recent events in the West Bank, focusing on verified facts such as the number of casualties, the actions of the Israeli military, and the impact on Palestinian residents. It does not provide in-depth analysis or personal

opinions, maintaining a neutral tone throughout.

Article Lean: -3

Reason Article Lean: The article highlights the suffering of Palestinians due to Israeli military actions, which aligns more with the Democrat party's generally more critical stance on Israeli policies and greater emphasis on human rights issues in the region.

Article Tone: -5

Reason Article Tone: The article has a negative tone as it describes the devastating impact of Israeli military raids on Palestinian civilians, including deaths, injuries, and destruction of property. It highlights the fear and suffering of residents who are trapped in their homes and unable to access basic necessities.

Title Lean: -3

Reason Title Lean: The article title highlights the frequent Israeli raids, which can be interpreted as critical of Israeli military actions. This perspective is more commonly aligned with the Democrat party, which tends to be more critical of Israeli policies towards Palestinians compared to the Republican party.

Title Tone: -4

Reason Title Tone: The article title has a negative tone as it highlights the frequent and disruptive nature of Israeli raids on Palestinians, suggesting a sense of ongoing conflict and hardship for those affected.

8. **URL:** <https://www.wsj.com/politics/national-security/pentagon-runs-low-on-air-defense-missiles-as-demand-surge>

Category: Politics

Topic: War and International Conflict

Subtopic: Israel-Iran Conflict

Takeaways: The U.S. is facing a shortage of air-defense missiles, particularly Standard Missiles, due to high demand from ongoing conflicts in the Middle East and Europe. This shortfall raises concerns about the Pentagon's readiness for potential conflicts, especially in the Pacific. The Defense Department is struggling to increase production rates, and the heavy use of interceptors in current operations is impacting the U.S.'s ability to replenish its stockpiles.

News Type: news report

Reason News Type: The article is classified as a news report because it objectively presents recent events regarding the U.S. military's missile stockpile and its implications. It focuses on verified facts, such as the number of missiles launched and statements from officials, without delving into personal opinions or interpretations. The language remains neutral, providing a straightforward account of the situation.

Article Lean: 1

Reason Article Lean: The article primarily focuses on the challenges faced by the U.S. military in maintaining its missile stockpiles amidst ongoing conflicts in the Middle East and Europe. It highlights concerns about military readiness and the need for increased production, which aligns with Republican viewpoints on strengthening national defense and military capabilities. However, it does not explicitly criticize or support any political party, maintaining a neutral tone overall.

Article Tone: -3

Reason Article Tone: The article conveys a negative tone as it highlights the U.S. military's challenges in maintaining adequate missile stockpiles amidst ongoing conflicts in the Middle East and Europe. It underscores concerns about the Pentagon's readiness and the potential vulnerability in the Pacific due to depleted resources, emphasizing the difficulties in ramping up production and the high costs involved.

Title Lean: 3

Reason Title Lean: The article title highlights a potential shortfall in military readiness, which can be interpreted as a critique of the current administration's defense policies. This perspective often aligns with Republican viewpoints that emphasize strong military capabilities and readiness. By focusing on a shortage of air-defense missiles, the title may suggest a need for increased defense spending, a stance typically supported by Republicans.

Title Tone: -3

Reason Title Tone: The article title suggests a negative tone as it highlights a shortage of air-defense missiles at the Pentagon amidst increasing demand. This implies potential vul-

nerabilities in defense capabilities, which could be concerning given the context of the Middle Eastern crisis.

9. **URL:** <https://apnews.com/article/israel-lebanon-iran-hamas-hezbollah-haniyeh-262194a2d70273207ed1d1a5>

Category: Politics

Topic: War and International Conflict

Subtopic: Israel-Lebanon Conflict

Takeaways: Recent Israeli strikes in Beirut and Tehran have targeted high-ranking figures in Hezbollah and Hamas, raising tensions and prompting discussions of retaliation. The assassination of a Hezbollah commander and Hamas leader Ismail Haniyeh has left both groups in a difficult position, as they must respond to restore deterrence without escalating the conflict further. Analysts suggest that Hezbollah may retaliate with significant military action, while Iran's response could involve its proxies or a direct strike on Israel, complicating the already volatile situation.

News Type: news analysis

Reason News Type: The article provides an in-depth examination and interpretation of the recent strikes in Beirut and Tehran, discussing their implications for Hezbollah, Iran, and the broader regional conflict. It includes expert opinions and analysis to explain the significance and potential consequences of these events, which goes beyond mere reporting of facts.

Article Lean: 0

Reason Article Lean: The article provides a detailed account of the strikes and their implications without showing clear support for either the Democrat or Republican party. It focuses on the geopolitical consequences rather than U.S. domestic politics.

Article Tone: -4

Reason Article Tone: The article has a negative tone as it discusses violent strikes, assassinations, and the potential for further escalation and regional conflict. The focus is on the serious security breaches, the loss of lives, and the delicate balance required to avoid a wider war.

Title Lean: 0

Reason Title Lean: The article title does not explicitly support either the Democrat or Republican party. It focuses on the strategic calculations of Iran and Hezbollah in response to Israeli strikes, which is a neutral observation rather than a partisan stance.

Title Tone: -3

Reason Title Tone: The article title suggests a high-stakes situation where Iran and Hezbollah must carefully manage their responses to Israeli strikes, implying potential negative consequences if they make errors. This creates a sense of tension and urgency, contributing to a negative tone.

10. **URL:** https://www.huffpost.com/entry/over-100-lawmakers-demand-independent-probe-israel-killing-aysenur_66f45028e4b01c2b5009317f

Category: Politics

Topic: War and International Conflict

Subtopic: Other

Takeaways: Over 100 members of Congress are demanding an independent investigation into the killing of Turkish American activist Ayşenur Ezgi Eygi by Israeli forces in the West Bank. Eygi was shot while volunteering during a protest against settlement expansion, and her family, along with lawmakers, have criticized the Israeli Defense Forces' preliminary investigation as inadequate. The letter calls for accountability and a thorough examination of the circumstances surrounding her death, highlighting concerns over U.S. support for Israel amid ongoing violence in the region.

News Type: news report

Reason News Type: The article is a news report as it objectively presents recent events surrounding the killing of Ayşenur Ezgi Eygi, focusing on verified facts such as the congressional letter, the IDF's investigation, and reactions from Eygi's family. It does not delve into personal opinions or interpretations but rather reports on the demands for an independent investigation and the responses from various parties involved.

Article Lean: -2

Reason Article Lean: The article highlights the actions of more than 100 members of

Congress, led by a Democrat, demanding an independent investigation into the killing of Ayşenur Ezgi Eygi, which aligns with the Democratic party's emphasis on human rights and accountability. It also criticizes the Biden administration's response, suggesting a nuanced view rather than outright support for the Democratic party.

Article Tone: -4

Reason Article Tone: The article has a negative tone as it discusses the killing of Turkish American activist Ayşenur Ezgi Eygi by Israeli forces and the subsequent demand for an independent investigation. It highlights the dissatisfaction with the Israeli Defense Forces' investigation and the lack of accountability, as well as the broader context of violence in the region.

Title Lean: -3

Reason Title Lean: The article title suggests a demand for an independent investigation into the killing of Ayşenur Eygi by Israel, which aligns with the Democrat party's tendency to advocate for human rights and accountability in foreign policy, particularly in relation to Israel. Democrats are generally more supportive of international oversight and investigations in such matters, whereas Republicans often emphasize strong support for Israel without external interventions.

Title Tone: -3

Reason Title Tone: The article title has a negative tone as it involves a demand for an independent investigation into a killing, which suggests a serious and potentially contentious issue. The mention of over 100 lawmakers implies significant concern and possibly criticism regarding the circumstances of the incident.

Text S2: Article annotation codebook

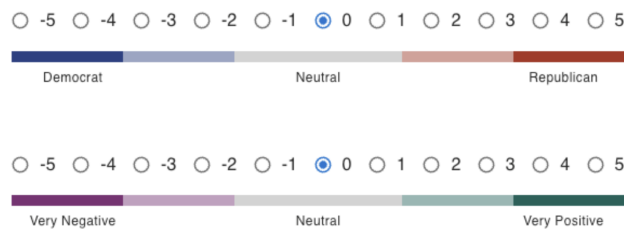
Introduction

The purpose of this exercise is to collect labels for news articles' *lean*, *tone*, and *type* in order to evaluate the performance of the **Media Bias Detector**.

You will be presented with a set of news articles and asked to label each on the following dimensions:

1. **Political Lean:** Does the article support the Democratic Party or the Republican Party?
Supporting a party can mean supporting its viewpoints, politicians, or policies.
2. **Tone:** Does the article have a positive, negative, or neutral tone?
3. **News Type:** Is the article a news report, news analysis, or opinion piece?

For *lean* and *tone*, you will be asked to rate the article on an 11-point Likert scale ranging from -5 to $+5$ as follows:



You will be asked to label a total of 50 articles that cover the entire range of lean, tone, and type categories.

Annotation Platform

- Log in to the annotation platform at this URL and select “Tone Verification”.
- Enter your Penn ID in the Submission ID field and make sure to enter it in the same way every time. You will automatically be taken to the last article you were on.

- Your responses for each article are saved after you have answered all three questions and clicked “Submit”. You will see a popup at the top of your screen indicating this.
- To update labels for a previous article, use the dropdown at the top to navigate to it, make your changes (you can leave other answers as they are), and click “Submit” again. This will save your updated responses and return you to your most recent article.

Text S3: Sentence annotation codebook

We provided the human annotators the following codebook to assist with the sentence type, tone, and focus validation task:

Introduction: The purpose of this exercise is to collect labels for sentences' news type, tone, and focus in order to evaluate the performance of the Media Bias Detector.

You will be presented with a set of news articles and asked to label each sentence in them on the following dimensions:

- **Type:** Is the sentence a “*fact*,” “*opinion*,” “*borderline*,” “*quote*,” or “*other*” type of sentence?
 - A **fact** is something capable of being proved or disproved by objective evidence.
 - An **opinion** reflects the beliefs and values of whoever expressed it but should not be a quote.
 - A **borderline** sentence is one that is not entirely a fact or opinion.
 - A **quote** is a passage from another person or source that comes with quotation marks.
If a quote is composed of multiple sentences, they should all be labeled as *quote*.
- **Tone:** Does the sentence have a “*positive*,” “*negative*,” or “*neutral*” tone?
- **Focus:** Does the sentence refer to *Democrats*, *Republicans*, *both*, or *neither*? Referring to a party includes referring to its policies or politicians. (Note that this is different from lean as we are only interested in identifying references and not labeling the sentence’s stance.)

Examples: Before starting this task, you may find it helpful to read this Pew Research Survey on facts and opinions in news articles. In particular, the 12 sentences they used to evaluate people’s ability to perform this task are as follows:

Factual statements:

- Health care costs per person in the U.S. are the highest in the developed world.
- President Barack Obama was born in the United States.
- Immigrants who are in the U.S. illegally have some rights under the Constitution.
- ISIS lost a significant portion of its territory in Iraq and Syria in 2017.
- Spending on Social Security, Medicare, and Medicaid make up the largest portion of the U.S. federal budget.

Opinion statements:

- Democracy is the greatest form of government.
- Increasing the federal minimum wage to \$15 an hour is essential for the health of the U.S. economy.
- Abortion should be legal in most cases.
- Immigrants who are in the U.S. illegally are a very big problem for the country today.
- Government is almost always wasteful and inefficient.

Borderline statements:

- Applying additional scrutiny to Muslim Americans would not reduce terrorism in the U.S.
- Voter fraud across the U.S. has undermined the results of our elections.

Annotation Platform

- Log in to the annotation platform at this URL and select “Sentence Verification”.
- Enter your Penn ID in the Submission ID field and make sure to enter it in the same way every time. You will automatically be taken to the last article you were on.

- After you have selected labels for each sentence, click on “Save Progress” to record your answers and then “Next Article” to move forward. You will see a popup at the top of your screen indicating this.
- To update labels for sentences in a previous article, click on the “Previous Article” button at the bottom to navigate to it, make your changes (you can leave other answers as they are), then click “Save Progress” and “Next Article” again. This will save your updated responses and you can then continue with the rest of the articles.

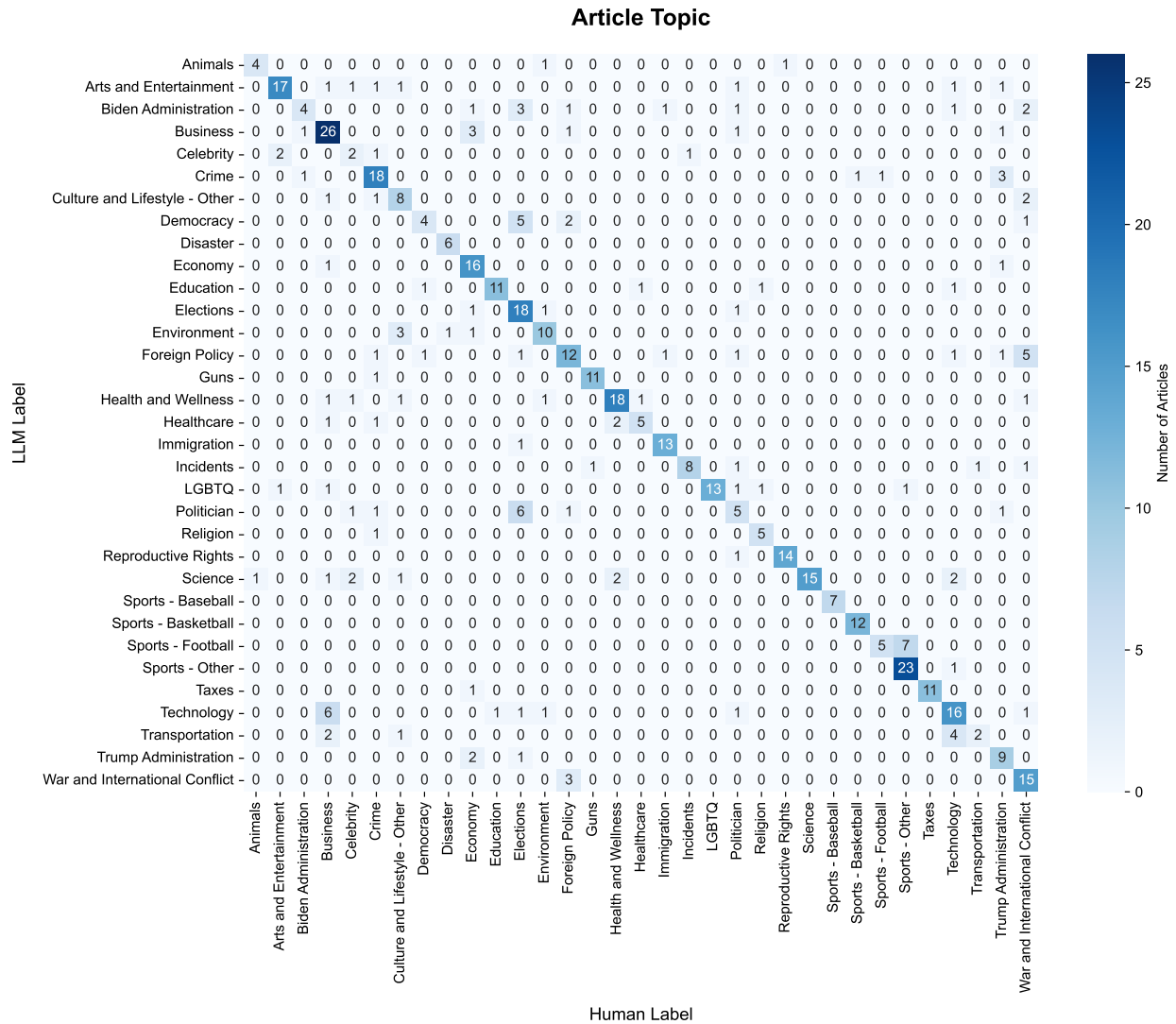


Figure A.17: **Confusion matrix for article topic classification.** The vast majority of articles are correctly classified, with confusions typically occurring for articles that touch on multiple topics and can reasonably fall under either of them, e.g., the most common correction made by human annotators is from “Politician” to “Elections”.

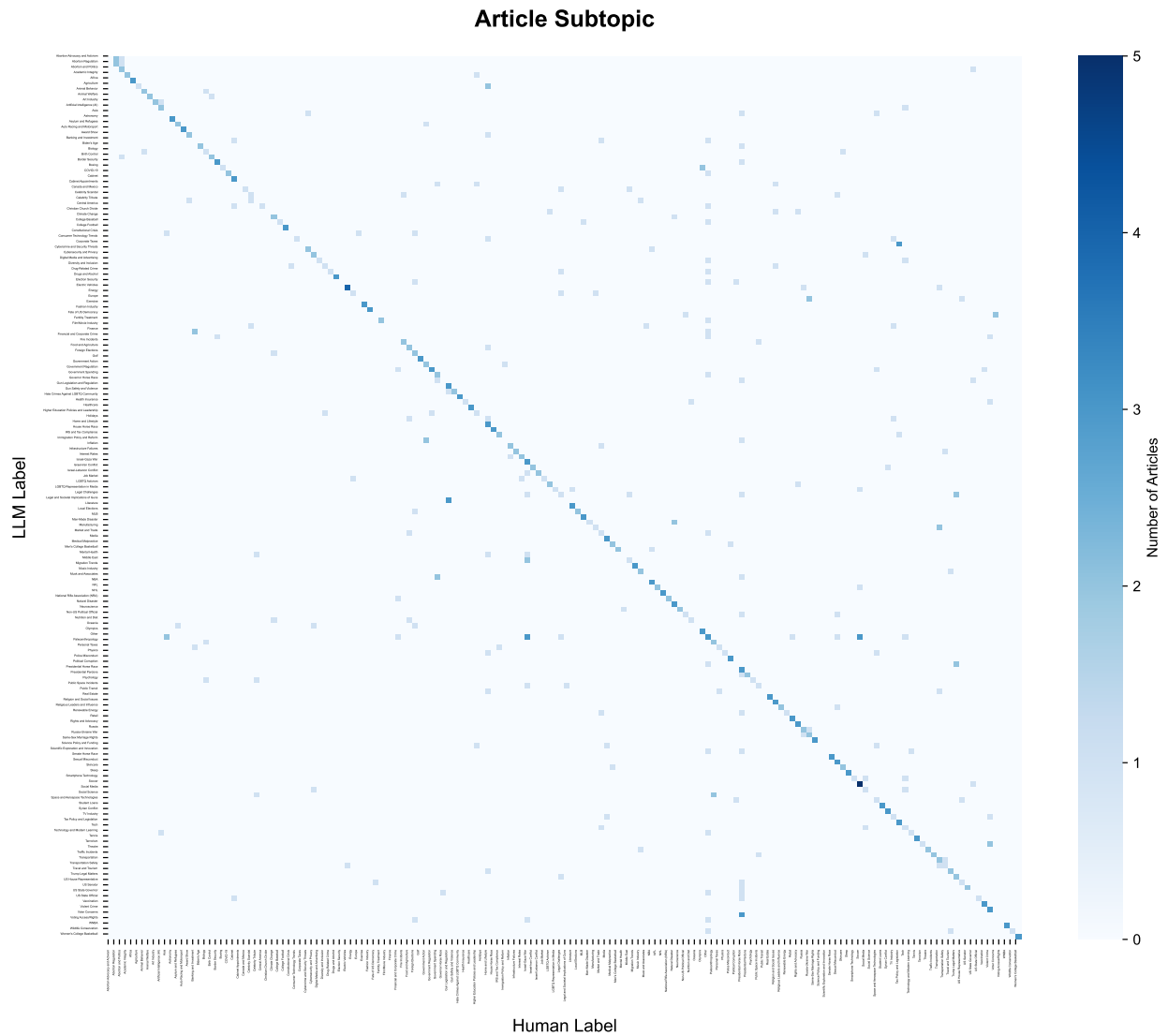


Figure A.18: **Confusion matrix for article subtopic classification.** Like in topic classification, the vast majority of articles are correctly classified, as can be seen by the density on the diagonal, with confusions typically occurring for articles that touch on multiple subtopics and can reasonably fall under either of them.

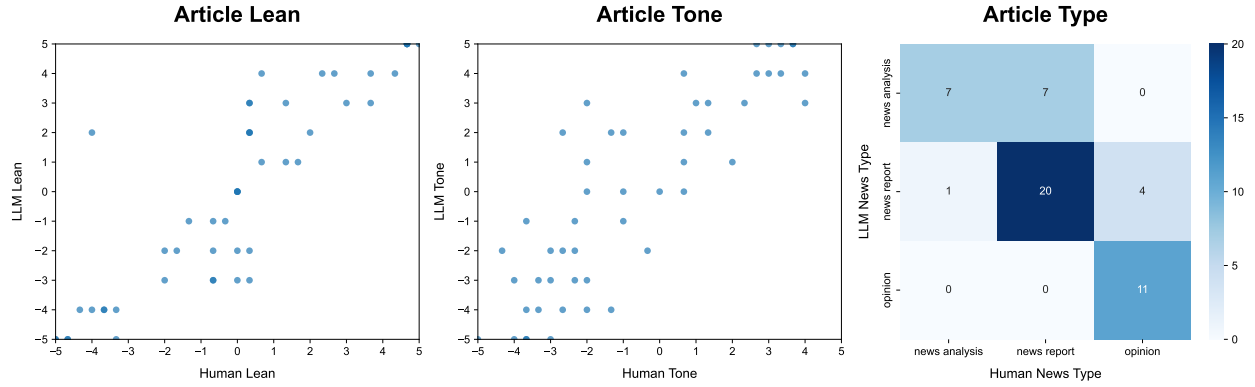


Figure A.19: **Scatter plots for article lean and tone ratings and the confusion matrix for article type labels.** For lean and tone, we average the three human ratings, and for type, we take the majority vote.

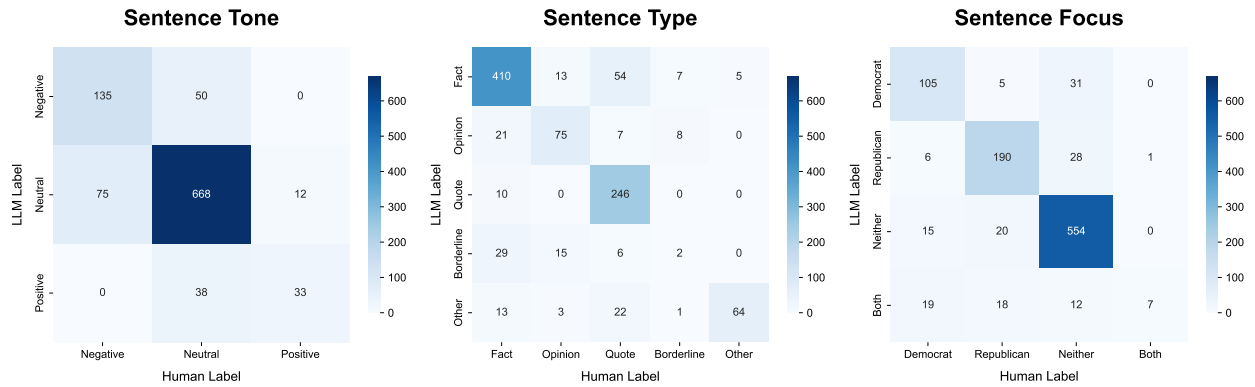


Figure A.20: **Confusion matrices for sentence tone, type, and focus.** We take the majority vote across the three human annotators to evaluate the model's labels.

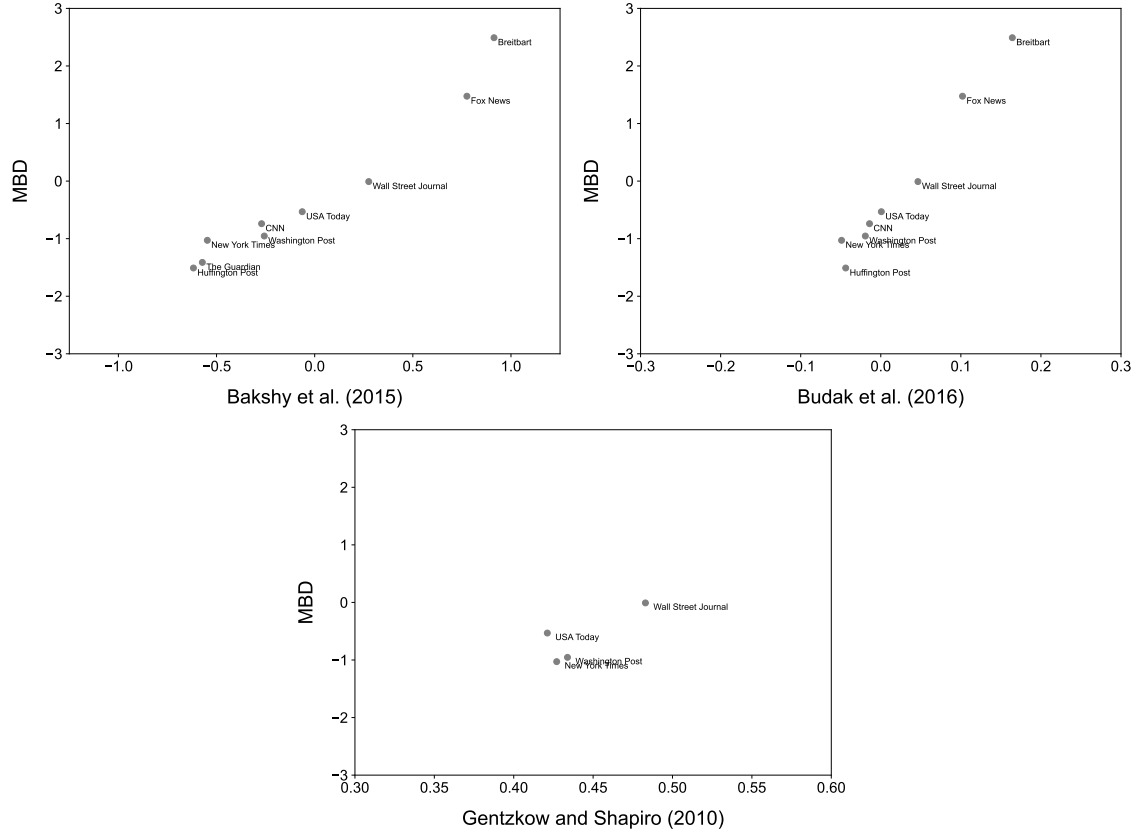


Figure A.21: **Comparison with prior work.** This figure compares our publisher-level political lean scores with those reported by (Bakshy et al., 2015; Budak et al., 2016; Gentzkow and Shapiro, 2010). Our estimates align closely with the first two studies for which most of the publishers in our dataset overlap, showing strong correlations of $R^2 = 0.946$ with Bakshy et al. and $R^2 = 0.976$ with Budak et al. We only have four publishers in common with Gentzkow and Shapiro, and the corresponding $R^2 = 0.647$.

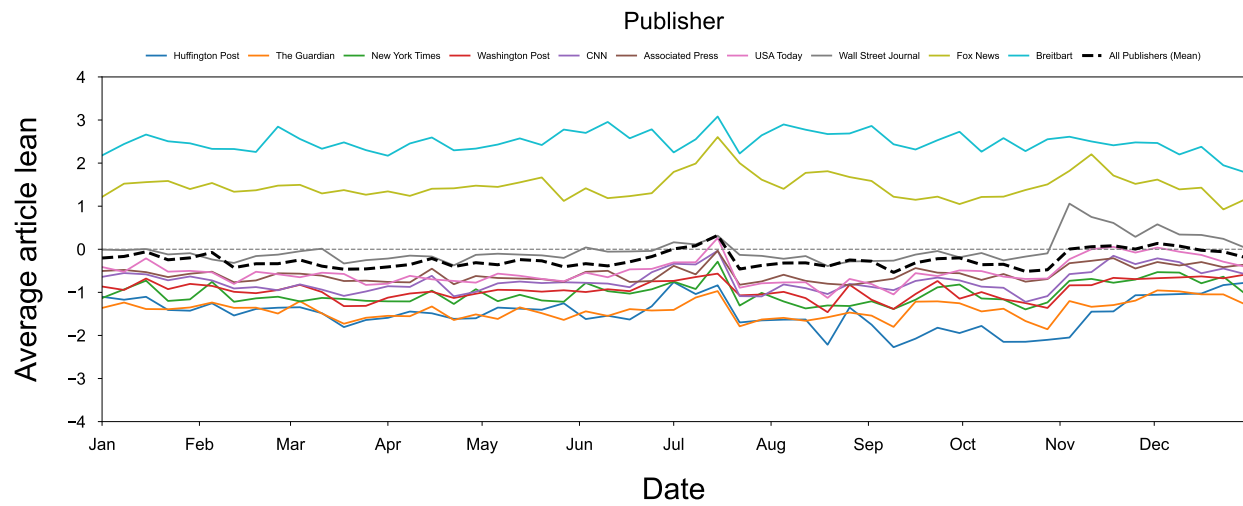


Figure A.22: **Political lean by publisher over time.** This plot shows the average political lean of all news articles in a week from each publisher. The bold dashed line shows the overall average across all publishers. We can observe how the lean shifts in response to major events, e.g., a major uptick in Republican-leaning coverage in the aftermath of the first assassination attempt on Donald Trump in July 2024.

APPENDIX B

APPENDIX FOR CHAPTER 3

Quote Extraction

The following is a news article. Read it and perform the task that follows. Respond with a list of JSON objects of key-value pairs, one for each quote, where the keys are as follows:

#####

{article}

#####

Task: Extract all direct quotes from the article above and provide a list of JSON objects of key-value pairs, one for each quote, where the keys are:

1. Instruction: Extract the full text of the quote itself. If a quote is split over multiple fragments, combine them into one.

Key: "quote"

Value: The full quote text.

2. Instruction: Extract the full name of the person being quoted.

Key: "person_name"

Value: The person's full name. Leave blank if unknown.

3. Instruction: Extract the full occupation of the person being quoted.

Key: "person_occupation"

Value: The person's full occupation. Leave blank if unknown.

4. Instruction: Extract the affiliation of the person being quoted.

Key: "person_affiliation"

Value: The entity or organization that the person is affiliated with. Leave blank if unknown.

5. Instruction: Classify the domain of the person being quoted based on their occupation and affiliation.

Key: "person_domain"

Value: One of "Politics", "Corporate", "Academia", "Legal", "Media and Arts", "Science and Technology", "Sports", "Law enforcement", "Military", "Health and Medicine", "Religion", "Community and Social Services", "Other". Select "Other" if the domain does not fit the possible values.

6. Instruction: Classify the capacity in which the person is being quoted based on the context of the article. An "illustrative anecdote" is when a person is being quoted as an example or illustration of a larger trend, a "commentary" is when a person is providing their thoughts on a topic or event, an "expert" is when a person is providing specialized knowledge or analysis, a "subject" is when a person is the main focus of the article, an "observer" is when a person is providing a first-hand account of an event, a "participant" is when the person being quoted was involved in an event, a "spokesperson" is when a person is speaking on behalf of an organization or entity.

Key: "person_capacity"

Value: One of "expert", "observer", "participant", "subject", "commentary", "illustrative anecdote", "spokesperson", "other". Select "other" if the capacity does not fit the possible values.

Do not return anything except the list of JSON objects of key-value pairs as output. Do not skip any quotes.

Figure B.1: **Quote extraction prompt.** In addition to extracting the text of the quote and the name of the person being quoted, we instruct the LLM to extract additional information like the person's occupation and affiliation as it is reported in the article. In addition to this, we also have the model classify the person's domain based on their occupation and affiliation, and their capacity based on the context in which they are being quoted in the article. Model: GPT-4o.

Emotional Tone Rating

You are an expert annotator analyzing the emotional tone of political language.

You will be given a set of quotes spoken by a single politician. Your task is to rate all the quotes overall on the following 10 emotional dimensions, using a Likert scale from 0 (not present at all) to 10 (very strongly present):

1. Anger
2. Fear
3. Uncertainty
4. Sadness
5. Hope
6. Confidence
7. Motivation
8. Compassion
9. Unity
10. Calm

Output format: Return a JSON object where each key is an emotion dimension and the value is an integer number from 0-10 in the following format:

"Anger": Value, "Fear": Value, "Uncertainty": Value, "Sadness": Value, "Hope": Value, "Confidence": Value, "Motivation": Value, "Compassion": Value, "Unity": Value, "Calm": Value
Be consistent and base ratings only on the language of the quote (not external knowledge). Do not refuse to answer.

Here are the quotes to rate, one per line:

{quotes}

Figure B.2: **Emotional tone rating prompt.** The LLM is asked to rate each politician's quotes on ten emotional dimensions from 0–10, focusing solely on linguistic cues rather than contextual knowledge. Model: GPT-4o.

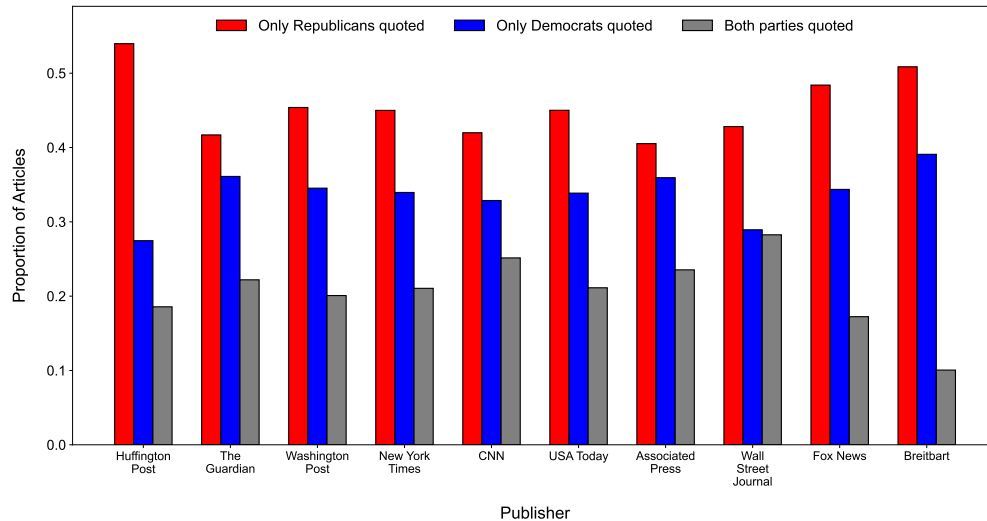


Figure B.3: **Article quote share.** Across all publishers, the vast majority of articles ($\approx 80\%$) only quote politicians from one party or the other. This ratio is the lowest for Breitbart, which quotes politicians from both sides only about 10% of the time.

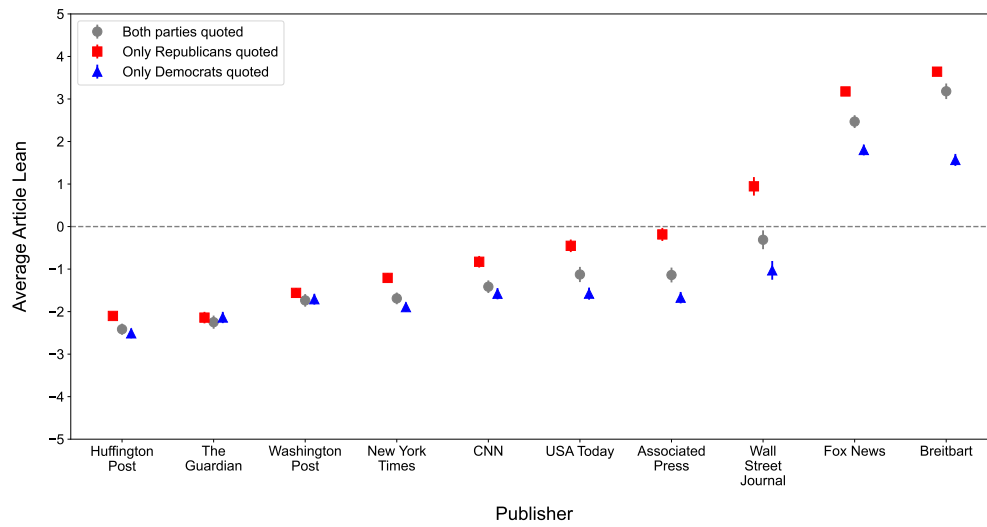


Figure B.4: **Article lean by party quoted.** Articles that exclusively quote one party are generally found to lean more in that direction compared to the base rate of articles that quote both sides.

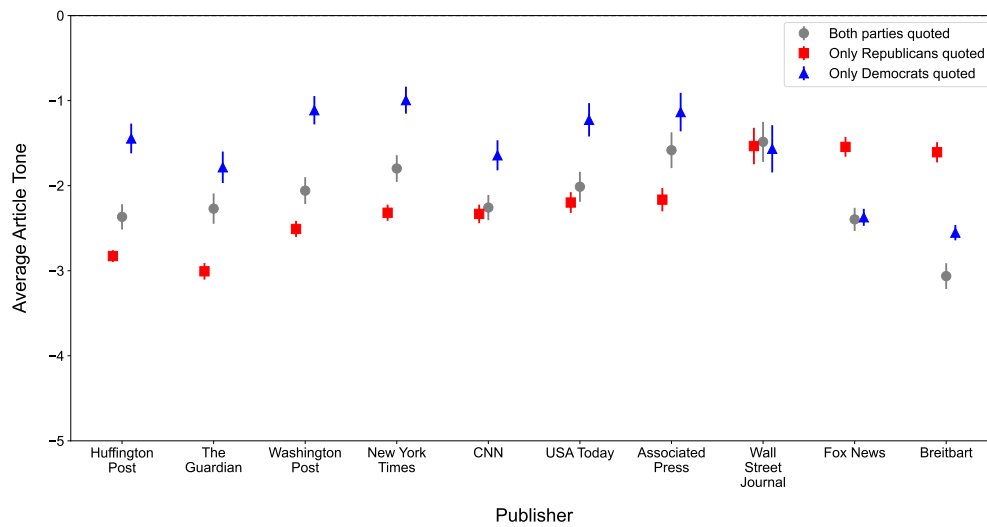


Figure B.5: **Article tone by party quoted.** While all articles are generally negative in tone, articles from right-leaning publications like Fox and Breitbart that exclusively quote Democrats are more negative on average than those that only quote Republicans. The opposite is true for center-left and left-leaning publications.

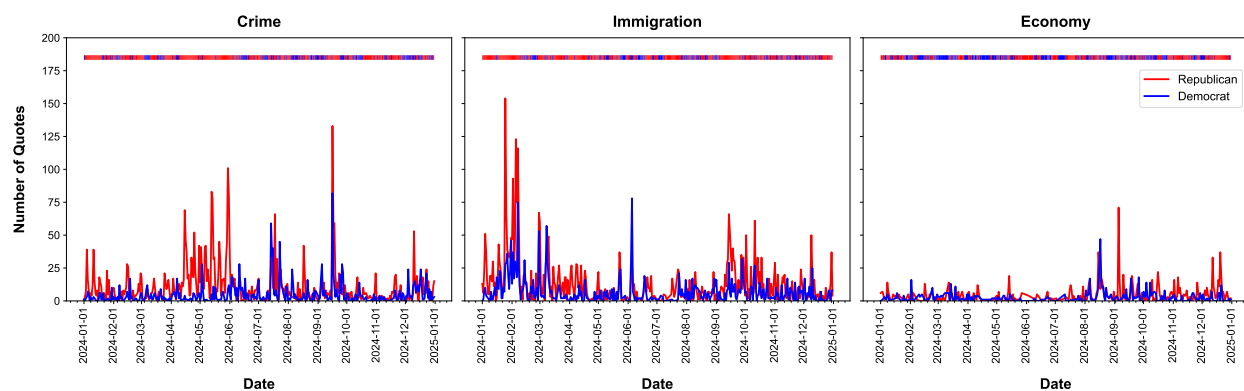


Figure B.6: **Temporal trends in topic attention.** Crime and immigration were a major focus of the Republican party during 2024, while Democrats were quoted comparatively more on the economy (before the election).

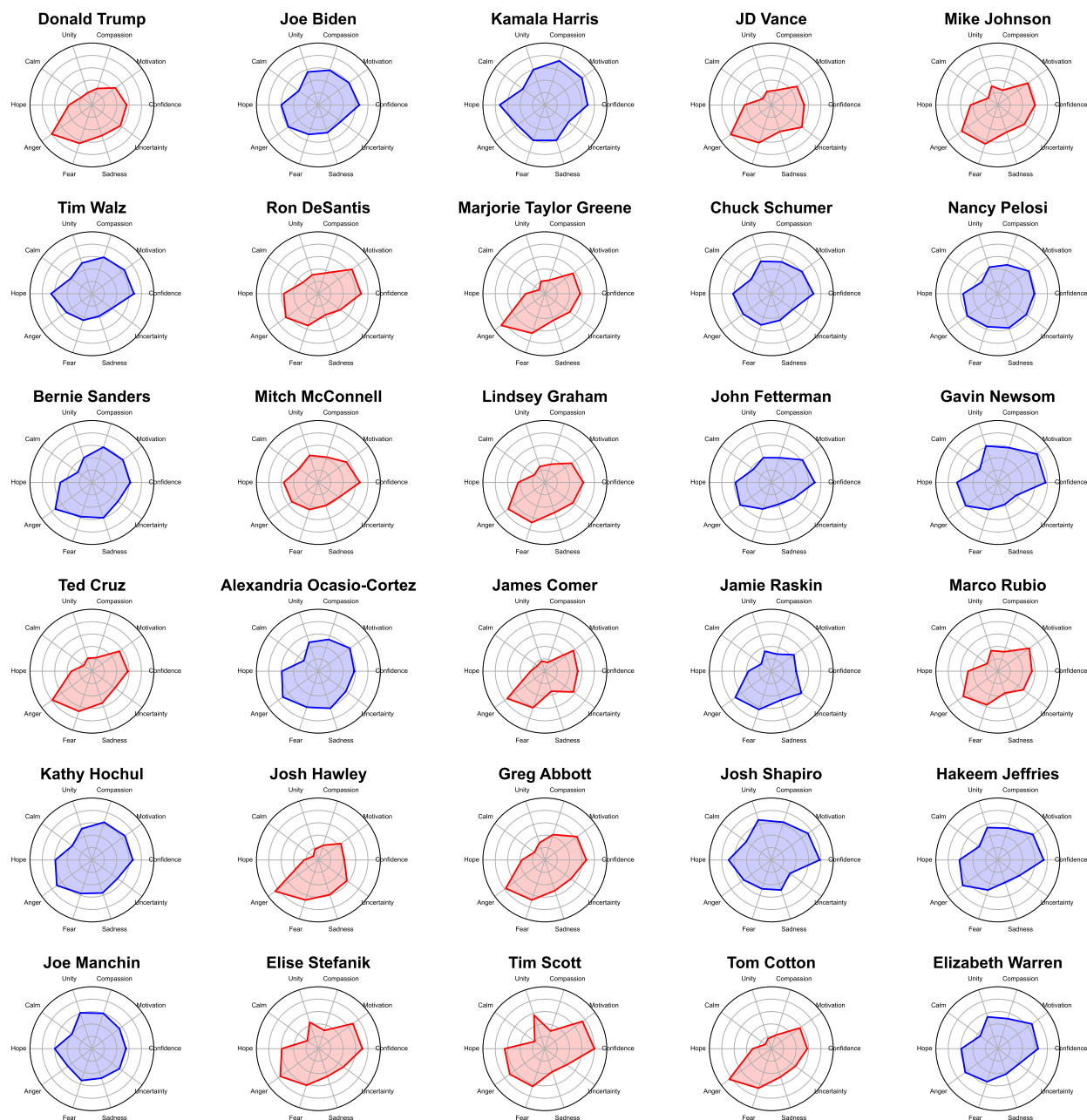


Figure B.7: **Emotional profiles of the 30 most quoted politicians.** Each radial dimension on the radar plot shows the intensity of ten emotional dimensions in their speech as identified by GPT-4o. Clear distinctions emerge between individuals, e.g. Senators Tom Cotton (R-AR) and Elizabeth Warren (D-CA).

B.1. Text S1

- <https://apnews.com/article/trump-contraception-birth-control-abortion-2024-8f73bb1b3a5864b24157f15eb2720>
 - “We’re looking at that and I’m going to have a policy on that very shortly.”
Donald Trump (Republican; Election candidate)
 - “Things really do have a lot to do with the states, and some states are going to have different policy than others.”
Donald Trump (Republican; Election candidate)
 - “I feel very strongly about it. I actually think it’s a very important issue.”
Donald Trump (Republican; Election candidate)
- <https://www.breitbart.com/entertainment/2024/05/18/sean-diddy-combs-called-monster-following-release-of-2016-cassie-ventura-assault-footage/>
 - “P. Diddy should be in jail and rejected from polite society.”
Anna Paulina Luna (Republican; Congressmember; Florida)
- <https://www.breitbart.com/politics/2024/05/17/donald-trump-heads-blue-minnesota-hopes-flipping-state/>
 - “It’s not like, in my opinion, that blue. We’re going to put a heavy move into Minnesota, I promise.”
Donald Trump (Republican; Election candidate)
 - “One of the other things I’m going to do — and I may be foolish in doing it — is I’m going to make a heavy play for New York, heavy play for New Jersey, heavy play for Virginia, heavy play for New Mexico, and a heavy play for a state that hasn’t been won in years, Minnesota. I’m going to do rallies, I’m going to do speeches, I’m going to work them. That doesn’t mean I’m going to work them as hard as I work Pennsylvania, where I’m doing very well.”

Donald Trump (Republican; Election candidate)

- “officially kick off our 2024 campaign”

Tom Emmer (Republican; Congressman; Minnesota)

- “Joe Biden’s failed policies are showing Minnesotans each day that President Trump is the only choice to lead our country forward”

Tom Emmer (Republican; Congressman; Minnesota)

- <https://www.cnn.com/2024/05/15/politics/mitt-romney-pardon-trump-biden/index.html>

- “Had I been President Biden, when the Justice Department brought on indictments, I would have immediately pardoned him. I’d have pardoned President Trump. Why? Well, because it makes me, President Biden, the big guy and the person I pardoned a little guy.”

Mitt Romney (Republican; Senator; Utah)

- “I think it’s a terrible fault for our country to see people attacking our legal system, that’s an enormous mistake.”

Mitt Romney (Republican; Senator; Utah)

- “I think it’s also demeaning for people to quite parently try and run for vice president by donning a red tie and standing outside the courthouse. It’s just, I’d have felt awkward.”

Mitt Romney (Republican; Senator; Utah)

- <https://www.foxnews.com/media/cnn-political-commentator-alice-stewart-dies-age-58>

- “Heartbreaking. Alice was wonderful and talented and a dear friend. And she loved America fiercely. She lived every day to the fullest, and she will be deeply missed. May God’s comfort and peace be upon her loved ones. RIP.”

Ted Cruz (Republican; Senator; Texas)

- <https://www.foxnews.com/politics/tuesdays-contentious-primaries-setting-up-crucial-house-senate->

- “while my opponent focuses on fighting, I’ll focus on working for you.”

Angela Alsobrooks (Democrat; Senator; Maryland)

- “If Larry Hogan wins, we lose more than the majority. We lose a woman’s right to choose
 - could lose Social Security. And with Trump knocking on the door, lose democracy itself.”

David Trone (Democrat; Congressman; Maryland)

- “The polls say I’m the only candidate who can beat Larry Hogan. And I intend to use every ounce of my energy to do just that. The stakes are just too damn high.”

David Trone (Democrat; Congressman; Maryland)

- “We like where Larry’s at. We know that’s going to be a tough race because Maryland is a blue state, but it’s a Hogan state first and foremost.”

Steve Daines (Republican; Senator; Montana)

- “The first state that we know that we’re going to win at this point is West Virginia. There’s one pickup seat right there.”

Steve Daines (Republican; Senator; Montana)

- <https://www.nytimes.com/2024/05/22/us/politics/trump-reich-contraception-trial.html>

- “I have never, and will never advocate imposing restrictions on birth control, or other contraceptives. I do not support a ban on birth control, and neither will the Republican Party!”

Donald Trump (Republican; Election candidate)

- “They were authorized to shoot me!”

Donald Trump (Republican; Election candidate)

- “A unified Reich: That’s not the language of American presidents. That’s not the lan-

guage of any Americans. It's the language of Hitler's Germany."

Joe Biden (Democrat; Election candidate)

- <https://www.theguardian.com/us-news/live/2024/may/02/trump-biden-election-campaign-politics-latest-updates>

- "It depends what happens with the supreme court."

Donald Trump (Republican; Election candidate)

- "You can't love your country only when you win."

Joe Biden (Democrat; Election candidate)

- "Dissent is essential to democracy, but dissent must never lead to disorder or to denying the rights of others, so students can finish the semester and their college education. Look, it's basically a matter of fairness. It's a matter of what's right. There's the right to protest, but not the right to cause chaos. People have the right to get an education, right to get a degree, right to walk across campus safely without fear of being attacked."

Joe Biden (Democrat; Election candidate)

- "There should be no place on any campus, no place in America for antisemitism, or threats of violence against Jewish students. There is no place for hate speech, or violence of any kind, whether it's antisemitism, Islamophobia, or discrimination against Arab Americans or Palestinian Americans. It's simply wrong. There's no place for racism in America. It's all wrong. It's not American."

Joe Biden (Democrat; Election candidate)

- "As president, I will always defend free speech. And I will always be just as strong in standing up for the rule of law. That's my responsibility to you, the American people, my obligation to the constitution."

Joe Biden (Democrat; Election candidate)

- "No"

Joe Biden (Democrat; Election candidate)

- “In America, violent protest is not protected. Peaceful protest is. It’s against the law when violence occurs. Destroying property is not a peaceful protest. It’s against the law. Vandalism, trespassing, breaking windows, shutting down campuses, forcing the cancellation of classes and graduations. None of this is a peaceful protest. Threatening people, intimidating people, instilling fear in people is not peaceful protest. It’s against the law.”

Joe Biden (Democrat; Election candidate)

- <https://www.usatoday.com/story/news/nation/2024/05/13/euclid-ohio-police-officer-killed-shooting-suspect-dead/73668208007/>

- “My prayers are with the family of Euclid Police Officer Jacob Derbin, who was killed in the line of duty. This tragic loss is a reminder of the risks that local law enforcement take every day. May God give strength to his family and the Euclid community at this time.”

Shontel Brown (Democrat; Congressmember; Ohio)

- <https://www.washingtonpost.com/politics/2024/05/26/netanyahu-embrace-republicans-strategy-israel-divisions>

- “Historically, Israel has always had a lot of support in the United States and around the world, and that’s because it was home to Jews who suffered unspeakable crimes during World War II. But what I think is clear, especially among the younger generation, is the Netanyahu government’s war against the Palestinian people and their killing of tens of thousands of people ... has significantly diminished support for Israel and especially among young people, which I think does not bode well for Israel’s future.”

Bernie Sanders (Democrat; Senator; Vermont)

- “I don’t think there’s any other way to say it: Netanyahu has been an absolute disaster

for Israel's support around the world. Here in the United States, Netanyahu made a reckless decision to integrate himself with the Republican Party, taking very clear sides in U.S. politics, and it has come with serious consequences."

Chris Murphy (Democrat; Senator; Connecticut)

– "has never wanted to listen"

Chris Murphy (Democrat; Senator; Connecticut)

– "There is no excuse for an American president to block aid to Israel."

Elise Stefanik (Republican; Congressmember; New York)

– "Let's fix this"

Joe Biden (Democrat; Election candidate)

– "black mark"

Joe Biden (Democrat; Election candidate)

APPENDIX C

APPENDIX FOR CHAPTER 4

University Normalization

You are an entity normalization assistant. Your task is to identify the official university name from a noisy input string.

Input: {input_string}

The input may contain:

- The official university name
- A common abbreviation or nickname
- A school, college, department, center, lab, hospital, or institute affiliated with a university - A possessive or informal variant, e.g., "Harvard's Kennedy School"
- Extra titles, occupations, punctuation, or location information

Return the official university name only.

Rules:

1. If the input clearly refers to a university or one of its affiliated schools/departments, return the parent university's official name.
 - "Harvard Medical School" → "Harvard University"
 - "Wharton School" → "University of Pennsylvania"
 - "MIT Media Lab" → "Massachusetts Institute of Technology"
 - "UC Berkeley Department of Sociology" → "University of California-Berkeley"
2. If the input is already an official university name, return it unchanged.
3. If the input uses a common abbreviation, expand it to the official university name.
 - "UCLA" → "University of California-Los Angeles"
 - "NYU" → "New York University"
 - "CMU" → "Carnegie Mellon University"
4. For multi-campus university systems, include the specific campus when it is part of the institution's official name if it is provided in the original input, and format it using a dash between the system name and the campus name. Do not use a comma or the word "at".
 - "UC Berkeley" → "University of California-Berkeley" - "University of Texas at Austin" → "University of Texas-Austin" - "University of Minnesota, Twin Cities" → "University of Minnesota-Twin Cities"
5. Do not guess based only on a city, state, hospital, company, think tank, or generic organization name unless the university affiliation is explicit.
6. If multiple universities are possible and the input is ambiguous, return "NOT A UNIVERSITY".
7. If the input does not refer to a university, return "NOT A UNIVERSITY".

Output format: Return only the official university name. Do not include any other text or reasoning in your response at all.

Figure C.1: **University normalization prompt.** To resolve variations in references to universities and their constituent schools or departments, we use this prompt to extract official names with consistent formatting. Model: GPT-4o-mini.

Role Classification

Your task is to classify whether the quoted person is being used as an expert source in the article, or whether the person or their institution is directly involved in the news story in a non-expert capacity.

Person name: {person_name}
Person affiliation: {clean_affiliation}

Definitions:

- "expert_source": The person is quoted for expertise, analysis, commentary, or because the article is about their research or scientific work.
- "person_involved": The article is primarily about the person themselves in a non-expert and non-research capacity, or the person is directly involved in the event, controversy, lawsuit, appointment, resignation, misconduct allegation, death, protest, investigation, political statement, or incident being reported.
- "institution_involved": The article is primarily about the person's university or affiliated institution, or the person is speaking as a representative, administrator, spokesperson, lawyer, leader, employee, student, alumnus, protester, or affected party connected to that institution.
- "ambiguous": There is not enough information to determine whether the person is an expert source or directly or institutionally involved.

Guidelines:

1. Classify as "expert_source" when the person is providing independent expertise on a broader topic.
2. Classify as "expert_source" when the article is about the person's own research.
3. Classify as "expert_source" even if the person's university is mentioned, as long as the university itself is not the main subject of the story.
4. Classify as "institution_involved" if the person's university, university hospital, research center, lab, department, or affiliated institution is the central subject of the article.
5. Classify as "institution_involved" if the person is a university president, dean, administrator, spokesperson, attorney, board member, student, staff member, or campus participant speaking about their own institution.
6. Classify as "person_involved" if the person is quoted because of their own non-research actions, experiences, statements, accusations, appointment, resignation, death, lawsuit, or controversy.
7. If the person's research is controversial or criticized, still classify as "expert_source" unless the main story is about misconduct, litigation, institutional discipline, or a personal scandal.
8. If unsure, use "ambiguous".

Article text starts here:

{text}

Article text has ended.

Respond with exactly one label:

expert_source
person_involved
institution_involved
ambiguous

Do not include any explanation or additional text.

Figure C.2: **Role identification prompt.** To identify whether an individual is being quoted for their expertise on the topic or because they or their institution are directly involved in the story for other reasons, we label the capacity in which they are being quoted by describing which kinds of roles fall in each category. Model: GPT-4o-mini.

Academic Domain Classification

Your task is to determine a person's primary academic domain based on the provided Wikipedia page.

Assign the person to exactly one of the Allowed Domains below. Choose the domain that best represents the person's primary academic discipline, based on evidence such as their academic position, field of study, research area, professional identity, or scholarly work.

Allowed Domains:

- Law
- Economics
- Political Science and International Relations
- Public Policy and Public Administration
- Business and Management
- Communication, Journalism, and Media Studies
- Sociology
- Anthropology
- Psychology and Cognitive Science
- History and Archaeology
- Philosophy and Religious Studies
- Literature, Languages, and Cultural Studies
- Education
- Medicine and Clinical Health
- Public Health
- Biology and Life Sciences
- Environmental Science and Ecology
- Earth and Planetary Sciences
- Physics and Astronomy
- Chemistry
- Mathematics and Statistics
- Computer and Information Science
- Engineering
- Agriculture and Food Science
- Geography and Urban Studies
- Architecture, Planning, and Design
- Other
- Unknown

Classification Rules:

1. Classify the person's primary academic discipline, not simply the subject matter discussed most frequently on the page.
2. Give the greatest weight to explicit descriptions of the person, such as "economist," "political scientist," "professor of sociology," or "computer scientist," as well as their academic department, appointments, education, and research specialization.
3. Use the person's institution only as supporting context. Do not infer the person's field from the institution alone.
4. If the person works across multiple fields, select the single domain that best represents their primary academic identity.
5. If a specific discipline clearly falls within one of the broader domains above, use that domain.
6. Use "Other" when the page clearly identifies the person's academic field, but that field does not reasonably fit any of the Allowed Domains.
7. Use "Unknown" when there is insufficient information to determine the person's academic domain.
8. If the Wikipedia page is clearly about a different person than the person specified in the input, return "Unknown".

Output Rules:

- Return ONLY one value exactly as written in the Allowed Domains list.
- Do not include explanations, punctuation, Markdown, or any additional text.

Input:

```
Person name: {person_name}
Person institution: {clean_affiliation}
Wikipedia page title: {wiki_name}
Wikipedia page text:
{wiki_text}
```

Wikipedia page text has ended.

Respond with exactly one value from the Allowed Domains list.

Do not include any explanation or additional text.

Figure C.3: **Academic domain classification prompt.** To identify the academic disciplines represented among quoted university scholars, we match individuals to Wikipedia pages and classify their primary academic domain based on descriptions of their academic position, field of study, research area, professional identity, and scholarly work. The model is instructed to return "Unknown" when the Wikipedia page does not match the specified individual or when there is insufficient information to determine a domain. Model: GPT-4o-mini.

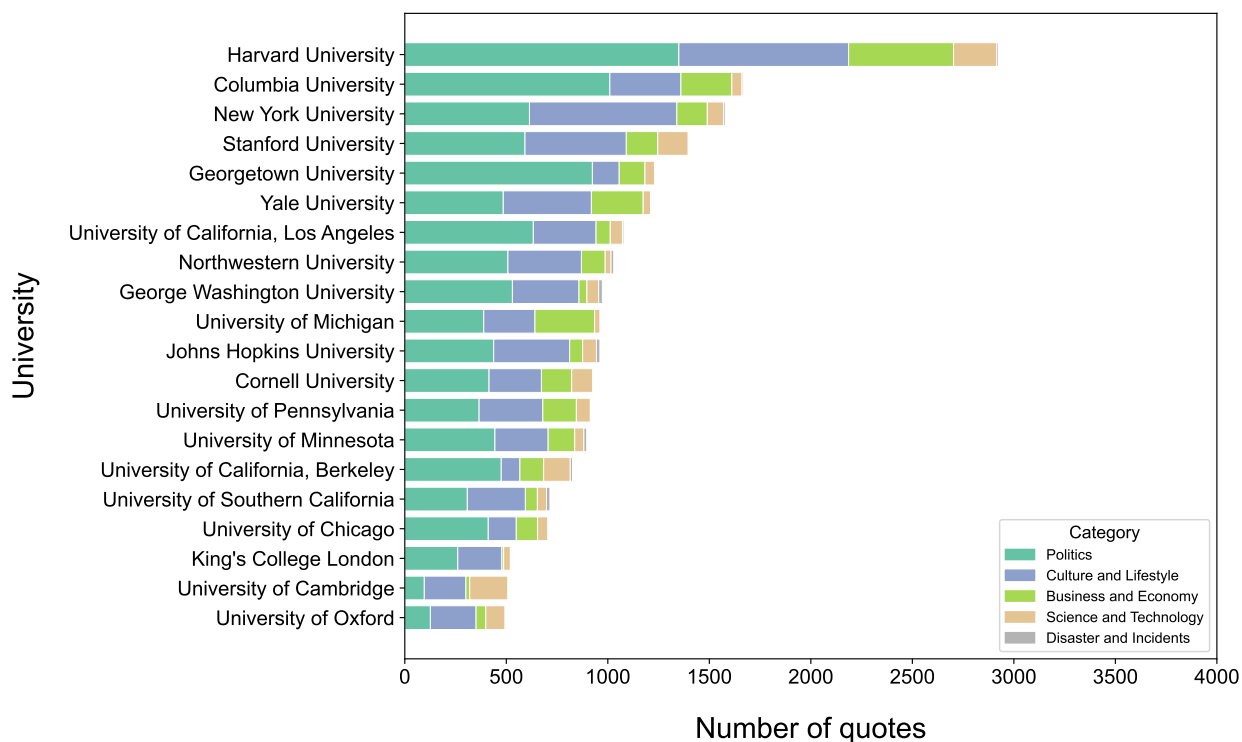


Figure C.4: **Expert quotes by news category.** Politics was the most common news category for expert quotations across all universities, followed closely by culture and lifestyle. The variation in ratios between the various categories for each institution shows where their expertise lies (according to the media). For example, Georgetown is cited substantially more on politics than on any other news topic.

APPENDIX D

APPENDIX FOR CHAPTER 5

```
Event Title

The following is a list of titles of a collection of news articles around a
particular event. Read them and perform the task that follows. Respond with a
JSON object of key-value pairs.

#####

{article_titles}

#####

Task: Summarize the titles.

1. Instruction: Analyze the titles and produce a good thematic title that best
summarizes the event.
Key: "theme"
Value: The thematic title.

2. Instruction: Analyze the titles and produce a shorter title that is not longer
than 4 to 5 words.
Key: "theme_short"
Value: The shorter title.

Do not return anything except the JSON object of key-value pairs as output.
```

Figure D.1: **Event title prompt.** To generate descriptive titles for each event, we provide the model a list of headlines of articles from a cluster and ask it to generate a suitable theme for the event based on them. We generate these themes at two levels: a short one for the high-level, multi-event view in the dashboard and a longer one for the detailed, single-event view. Model: GPT-4o-mini.

```
Fact Summarization

The following is a list of sentences about the same event. Read them and perform
the task that follows. Respond with a JSON object of key-value pairs.

#####

{sentence_list}

#####

Task: Analyze them and produce a comprehensive sentence that summarizes all of
them and is no longer than 25 words.
Key: "synthetic_sentence"
Value: The comprehensive sentence.

Do not return anything except the JSON object of key-value pairs as output.
```

Figure D.2: **Fact summarization prompt.** To generate representative sentences that capture maximum information about a given event, we use the LLM to summarize all the similar facts in a cluster into one overarching statement that covers all of them. Model: GPT-4o-mini.

Event clusters

High-Stakes Vice Presidential Debate: Vance vs. Walz

(65 articles total)

- Harris 2024 campaign senior adviser previews upcoming vice presidential debate: 'Two very different visions'
- Vance and Walz focus their attacks on the top of the ticket — not each other: VP debate takeaways
- What Vance, Walz need to do to win Tuesday's vice-presidential debate
- Experts preview Vance-Walz debate, say usually 'forgettable' VP bout 'might be different' this time
- JD Vance, Tim Walz Face Off In Vice Presidential Debate: Live Updates
- Heightened 'significance and importance' as Vance, Walz face off in VP debate
- What we can learn about Vance and Walz from past debate performances
- How to watch the Walz-Vance vice-presidential debate
- Trump Campaign Expects Tim Walz to 'Be a Chameleón' During VP Debate
- VP debate live: Vance refuses to say Trump lost 2020 election as Walz asks America to 'stand up' for democracy

Escalation of Conflict: Israel and Iran's Military Engagements in Lebanon

(75 articles total)

- Iran launches major missile attack on Israel
- Israel orders evacuation of southern Lebanon villages as ground raids begin
- Iran Attacks Israel With Missiles, as Israel Vows to Respond - WSJ
- Israel launches ground invasion in Lebanon with 'limited' raids on Hezbollah
- Israeli military says it is carrying out 'limited' ground operation targeting Hezbollah in Lebanon
- Israel launches ground invasion in Lebanon with 'limited' raids on Hezbollah
- The decapitation of Hezbollah leaves Iran weighing its options
- Iran fires at least 180 missiles into Israel as nationwide conflict grows
- Iran launches waves of missiles at Israel hours after US warning
- Israel vows to retaliate after Iran launches unprecedented missile attack

Nationwide Port Workers Strike Threatens Supply Chains and Economy

(29 articles total)

- The real reason 47,000 dockworkers are on strike
- Biden Scrambles to Contain Economic and Political Fallout of Port Strike
- Port strike union leader Harold Daggett: What to know
- Dockworkers may have the negotiating advantage in their strike against US ports
- Dockworkers' Strike Halts Commerce at Newark Port, Affecting the Supply Chain Ecosystem
- Major Longshoremen Strike Hits East Coast Ports
- How the Port Strike Could Affect Certain Supply Chains: Bananas, Coffee and More
- Port workers strike at East Coast, Gulf ports sparks fears of inflation and more shortages
- Port Strike Begins on East and Gulf Coasts
- Dockworkers at ports from Maine to Texas go on strike, a standoff risking new shortages

Hurricane Helene: A Devastating Toll on North Carolina

(60 articles total)

- More than 150 dead after Hurricane Helene dumps over 40in gallons of rain
- How Helene demolished a once-bustling North Carolina lake town
- Helene Rescue Teams Push Through Debris to Find People and Restore Power
- As death toll mounts, Helene rescue efforts underway across Southeast
- Search Crews With Cadaver Dogs Wade Through Muck Of Communities 'Wiped Off The Map' By Helene
- Helene's brutal toll: More than 100 dead; Biden to survey damage: Monday updates
- Stunning Video Shows North Carolina Lake Packed With Ruins Of Nearby Village After Storm
- How to help those affected by Hurricane Helene
- Before the floods, Asheville was called a 'climate haven': Is anywhere safe?
- Why Helene Was So Destructive in Florida, the Carolinas and Appalachia

Sentence clusters

Hurricane Helene made landfall in Florida as a Category 4 storm, causing destruction and flooding across the Southeastern U.S. with numerous fatalities.

- "Helene made landfall last Thursday night in Florida's Big Bend region as a category 4 hurricane."
- "Helene roared ashore in northern Florida late Thursday as a Category 4 hurricane and quickly moved through Georgia, the Carolinas and Tennessee with torrential rains that flooded waterways."
- "Helene left dozens of people dead and a wide path of destruction across the Southeastern U.S. after making landfall in Florida as a Category 4 hurricane."
- "Hurricane Helene made landfall late Thursday night as a category 4 hurricane in the panhandle of Florida and is working its way north, it is now considered a tropical storm."
- "Hurricane Helene made landfall Thursday night in Florida's Big Bend with winds up to 140 mph."
- "Helene made landfall in northwestern Florida late Thursday as a Category 4 storm, with winds of 140 miles per hour."

Hurricane Helene is one of the deadliest hurricanes to strike the U.S. mainland, second only to Hurricane Katrina in the last 55 years.

- "Helene is now the second-deadliest hurricane to strike the mainland U.S. in the last 55 years, topped only by Hurricane Katrina in 2005 and the most since Hurricane Camille hit the Gulf Coast in August 1969."
- "Helene is one of the deadliest hurricanes to make landfall on the U.S. mainland in modern history."
- "Helene one of the deadliest storms to hit the US Helene is considered to be one of the deadliest hurricanes to make landfall in the mainland U.S. in the modern era."

A storm caused unprecedented flooding in North Carolina, with over 2 feet of rain, marking the worst flooding in a century.

- "How some of the hardest-hit areas are coping The storm unleashed the worst flooding in a century in North Carolina, dumping more than an estimated 2 feet (61 centimeters) of rain in places."
- "According to the, the storm unleashed the worst flooding in a century in North Carolina."
- "The storm unleashed the worst flooding in a century in North Carolina, dumping more than an estimated 2 feet (61 centimeters) of rain in places."

Over 1.5 million customers in Florida, Georgia, the Carolinas, and Virginia were without power as of Tuesday, with significant outages reported since Monday.

- "By Tuesday night, more than 1.4 million electric customers remained without power in Florida, Georgia, the Carolinas and Virginia, according to poweroutage.us."
- "More than 1 million Americans were still without power in the Carolinas and Georgia as of Monday morning."
- "As of Monday morning, more than 700,000 homes in South Carolina were without power — including more than 500,000 in Georgia and 400,000 in North Carolina, according to poweroutage.us."
- "More than 1.5 million electricity customers from Florida to West Virginia were still without power on Tuesday afternoon, according to the tracking site poweroutage.us."
- "More than 1.5 million customers were still without power in the Carolinas, Georgia, Florida and Virginia early Tuesday, according to PowerOutage.us."

Figure D.3: **A sample of the data generated by our event clustering pipeline.** At the top are the major events that occurred on October 1, 2024, showing the event titles generated by GPT-4o-mini and a sample of headlines from articles that have been clustered together. The top facts about the event on Hurricane Helene are shown on the bottom. The sentence at the top of each group shows a synthetic statement generated by summarizing all the facts in the cluster while the following sentences show the original versions from those articles.

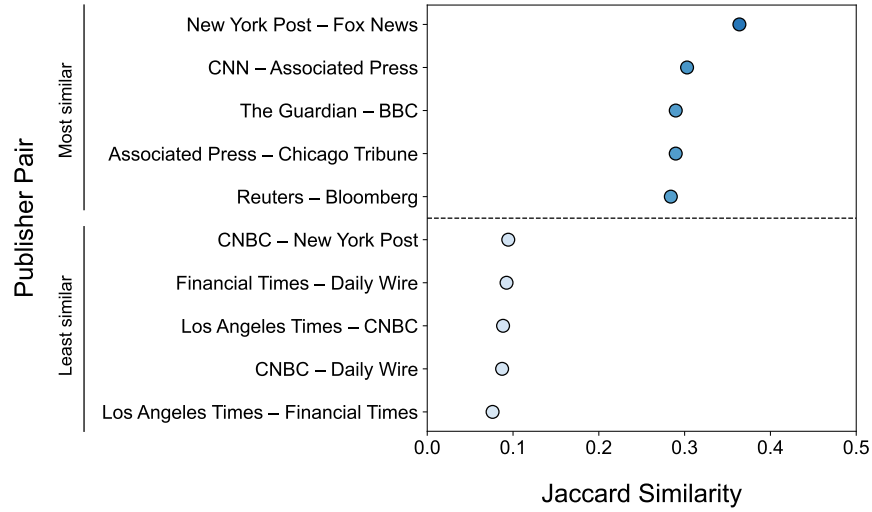


Figure D.4: **Publishers with high and low similarity in event selection.** Jaccard similarity for the top five most and least similar pairs of publishers. Similarity is based on whether they covered the same events or not, and not on how much attention they gave each event.



Figure D.5: **Least polarizing major events of 2025.** This figure shows the 20 events with the smallest standard deviation in article lean among events covered in at least 20 articles from at least 10 publishers. Each row represents an event, with each point representing an article about it and its political lean rating.

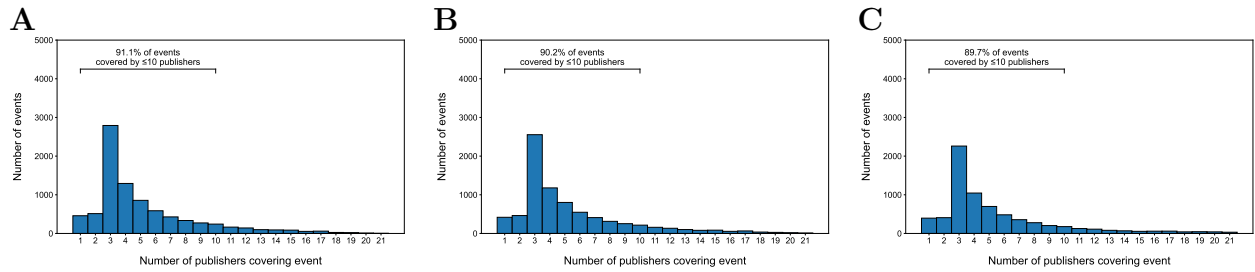


Figure D.6: **Robustness checks for event overlap.** Even after using progressively lower clustering thresholds, the skew in shared event coverage does not change substantially. The cosine similarity thresholds in Panels **A**, **B**, and **C** are 0.8, 0.75, and 0.7, respectively. The corresponding share of events covered by 10 or fewer publishers is 91.1%, 90.2%, and 89.7%.

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